

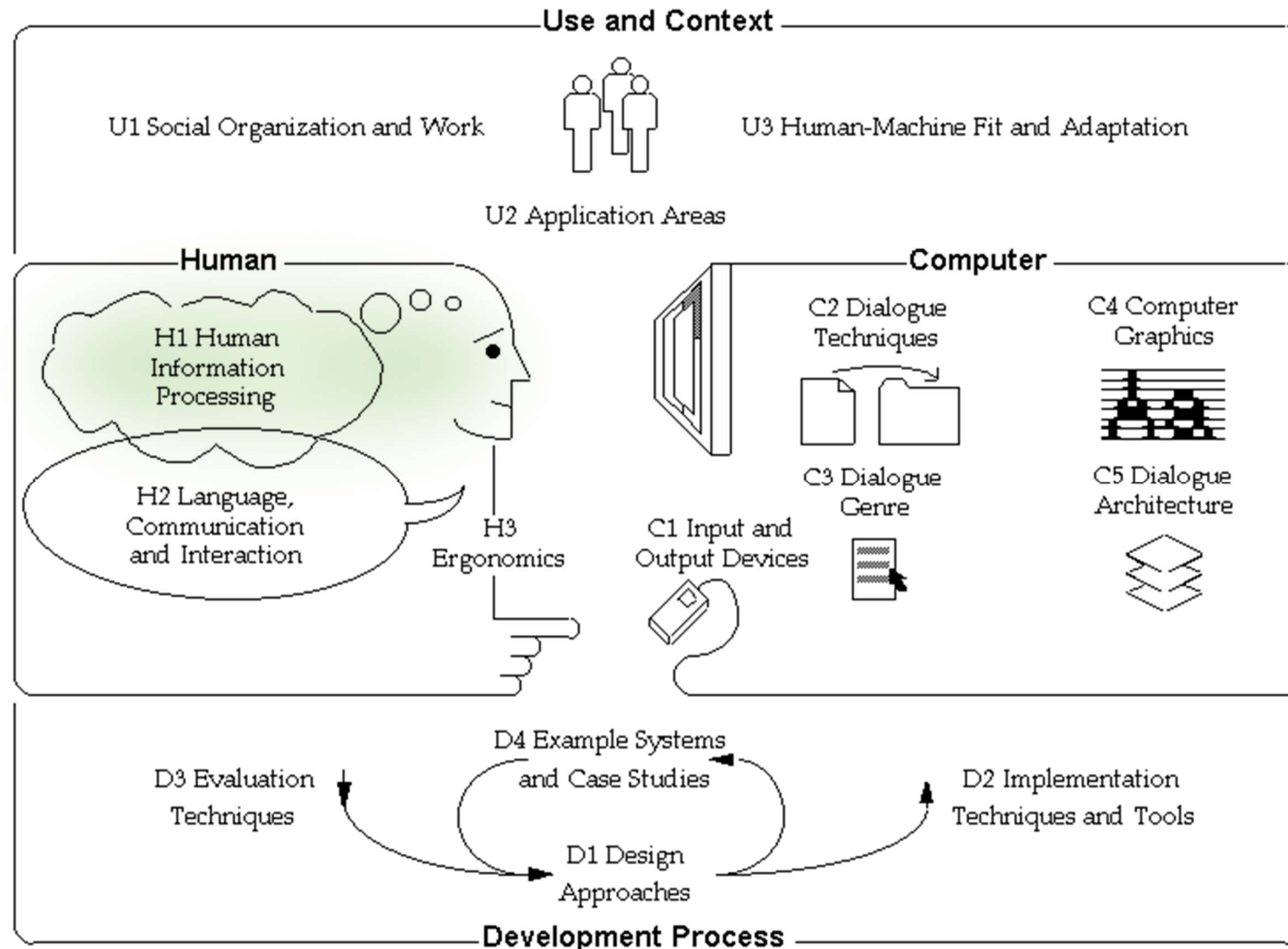
Human-Computer Interaction

Johan F. HOORN and Jeff TANG

Lecture 02: Human Information Processing, Part 1

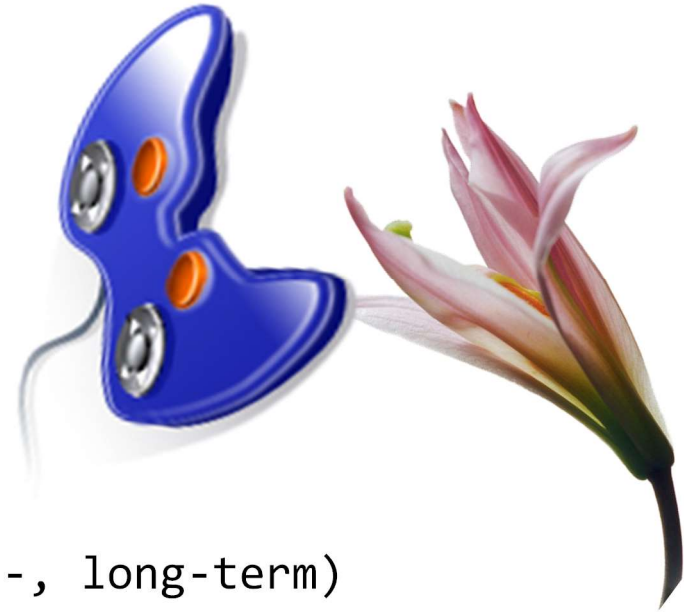


Overview



Contents

- ❑ Lecture's objectives
- ❑ The Human Information Processor
- ❑ Input: perception
- ❑ Throughput: memory (sensory, short-, long-term)
- ❑ Problem solving: reasoning, association
- ❑ Output: motor response, performance evaluation
- ❑ Summary

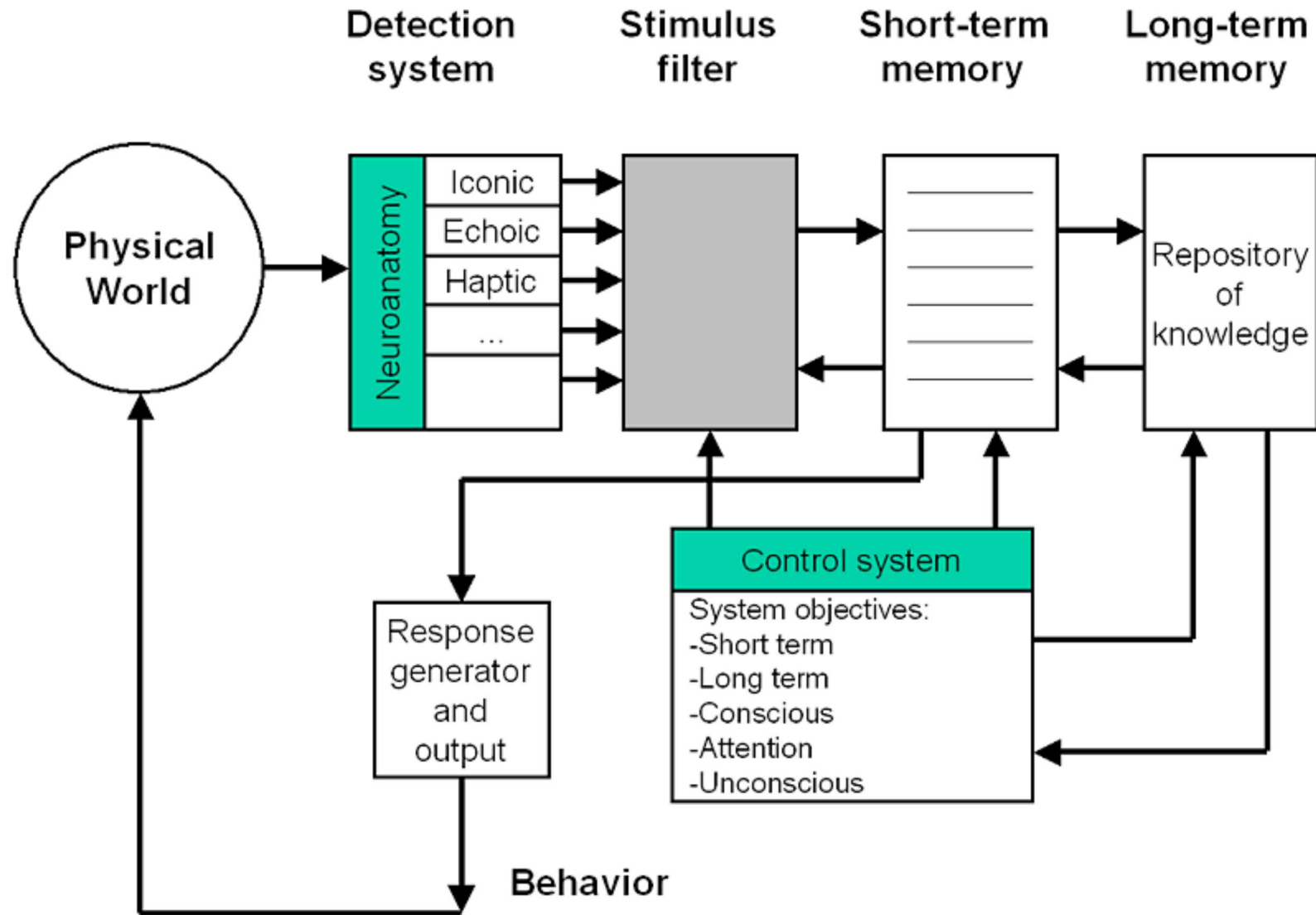


What you'll learn this afternoon

- ❑ All computer systems require human involvement
- ❑ How humans are different from computers
- ❑ Limitations and possibilities of humans
- ❑ What to watch for in system design



The Human Information Processor



Input: perception

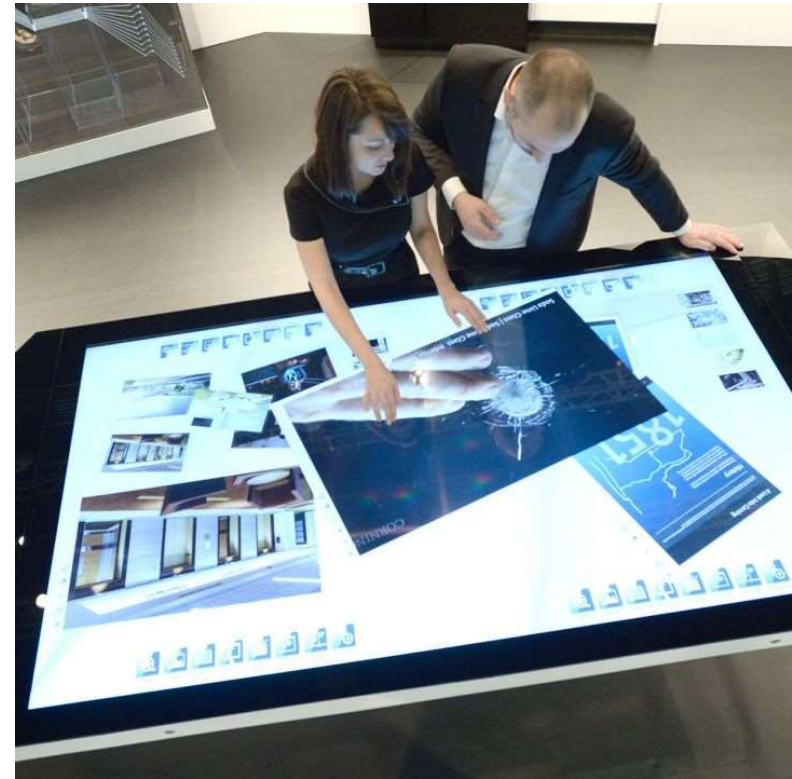
❑ Visual

- Image
 - Physical reception
 - Interpretation: 3D, color
 - Images in design
- Text
 - Physical reception
 - Interpretation: semantics, pragmatics
 - Text in design

❑ Hearing

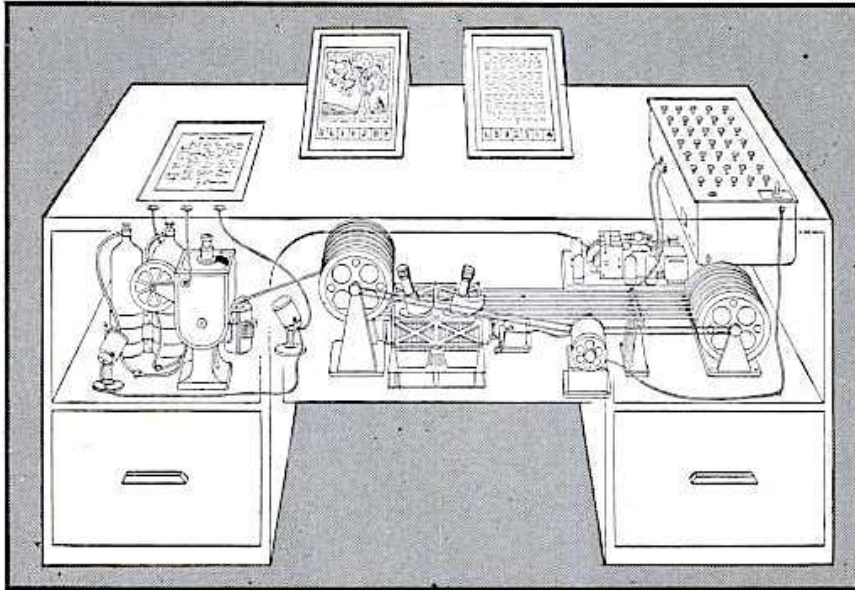
- Sound
 - Physical reception
 - Interpretation: semantics, pragmatics
 - Sonification of the interface
- Speech
 - Physical reception
 - Interpretation: semantics, pragmatics
 - Spoken word in design

❑ Touch



Collaboration Surface Connected Home Interactive Table |
Corning

Input: perception



MEMEX in the form of a desk would instantly bring files and material on any subject to the operator's fingertips. Slanting translucent viewing screens magnify supermicrofilm filed by code numbers. At left is a mechanism which automatically photographs longhand notes, pictures and letters, then files them in the desk for future reference.

AS WE MAY THINK CONTINUED



Collaboration Surface Connected Home Interactive Table |
Corning

Vannevar Bush's Memex (Memory Extender) (1945)

Throughput: processing

❑ Memory

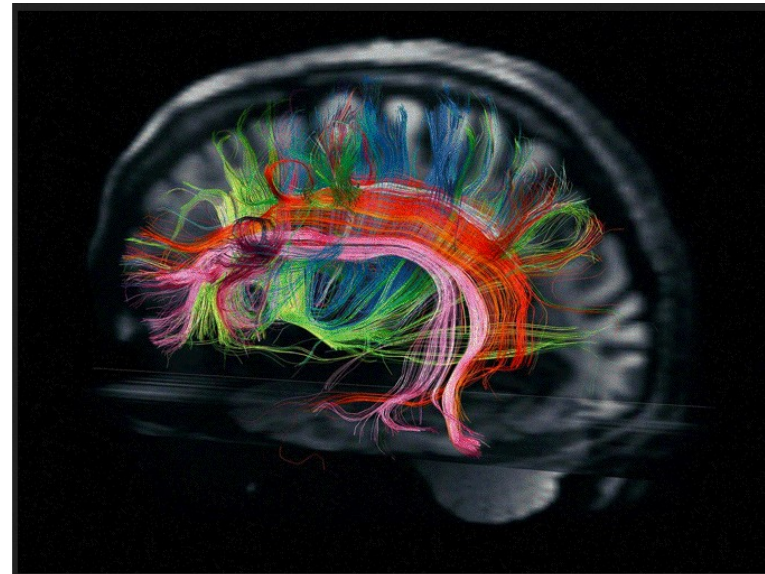
- Sensory
 - Neuroanatomy
- Short and Long Term
 - Information in the cell body

❑ Cognitive processing

- Problem solving
 - Trial and error
 - Difference reduction
 - Means-end analysis
 - Forward and backward reasoning
 - Analogical reasoning
 - Creativity

❑ Affective processing

- Goals and concerns
- Relevance
- Valence



Output: response

☐ Motor

- Speed-accuracy trade-off
 - Fitts' law
 - Shannon
- Performance evaluation
 - Index of Difficulty
 - Index of Performance
- User evaluation
 - Like?
 - Dislike?



Focus of today: Throughput

- ☐ Memory

- Sensory
 - Neuroanatomy
- Short and Long Term
 - Information in the cell body

- ☐ Cognitive processing

- ☐ Affective processing

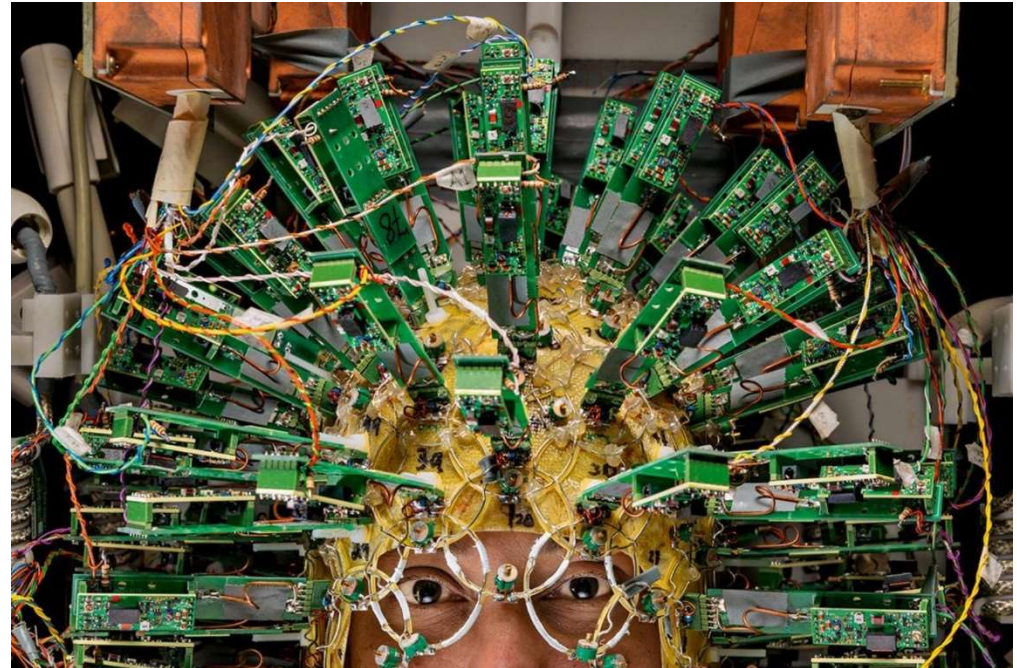
Focus of today: Throughput

- ☐ Memory
 - Sensory
 - Neuroanatomy
 - Short and Long Term
 - Information in the cell body
- ☐ Cognitive processing
- ☐ Affective processing

Latest fashion trend

Focus of today: Throughput

- ❑ Memory
 - Sensory
 - Neuroanatomy
 - Short and Long Term
 - Information in the cell body
- ❑ Cognitive processing
- ❑ Affective processing



Part of a brain scanner requiring almost as much power as a nuclear submarine. Antennas pick up signals produced when the scanner's magnetic field excites water molecules in the brain. Computers convert this data into brain maps like the one shown next (National Geographic, 2014)

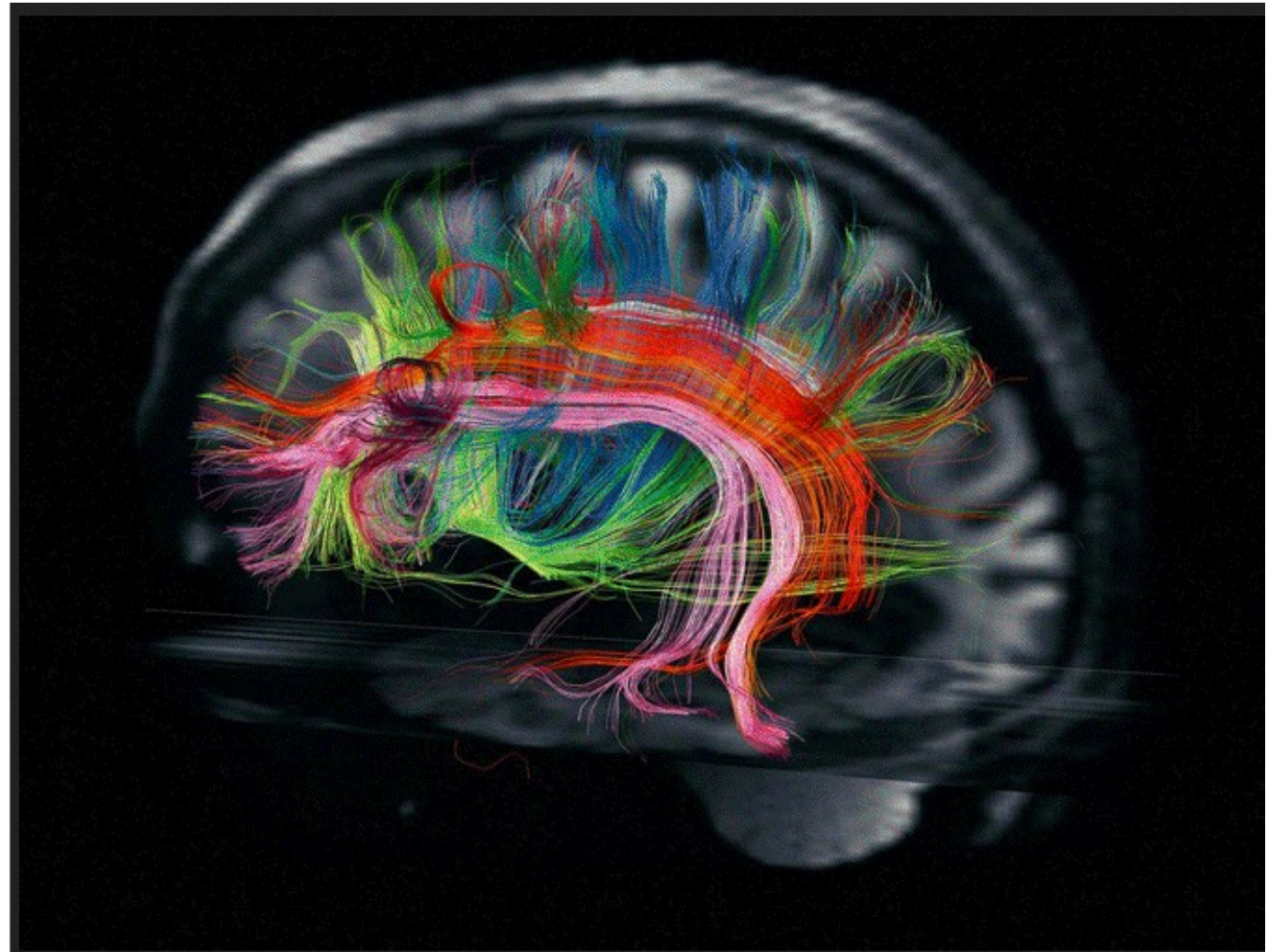
National Geographic - Secrets of the Brain. New technologies are shedding light on biology's greatest unsolved mystery: how the brain really works. By Carl Zimmer. Photographs by Robert Clark. National Geographic, February 2014. Source: <http://ngm-beta.nationalgeographic.com/2014/02/articles/secrets-of-the-brain/>

Structures and Inner Workings of the Brain

150,000 km of wiring (4 times around the earth), called 'white matter'

Weight: 1.5 kg

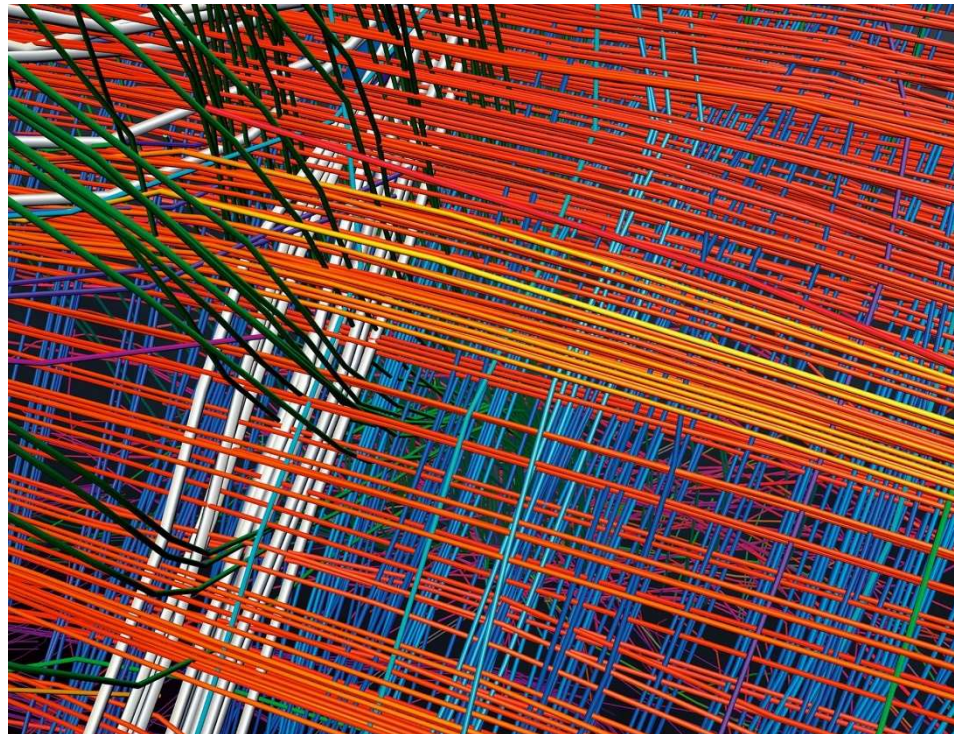
Pink and red paths (artificially colored) are involved in language processing



Structures and Inner Workings of the Brain

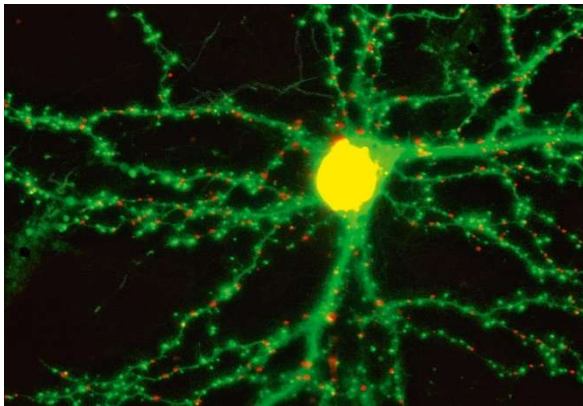
- White matter looks a bit woolly. It is not. When you zoom in, you see a frame of horizontal vs vertical lines with straight edges (Wedeen, 2012)

What we do not know yet is how information is transported over those wire frames

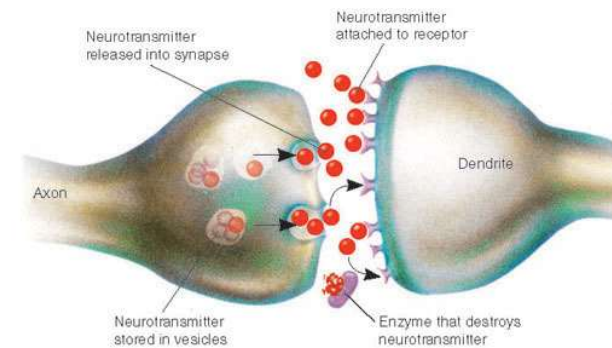
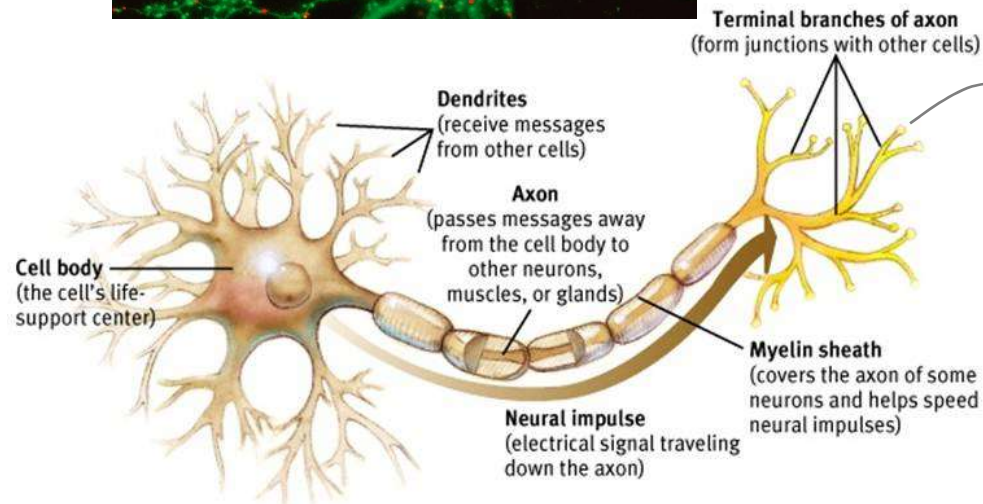


Structures and Inner Workings of the Brain

- Zooming in a little more, you see nerve cells with bumps and shoots, the 'fine structure' of the nerve cell



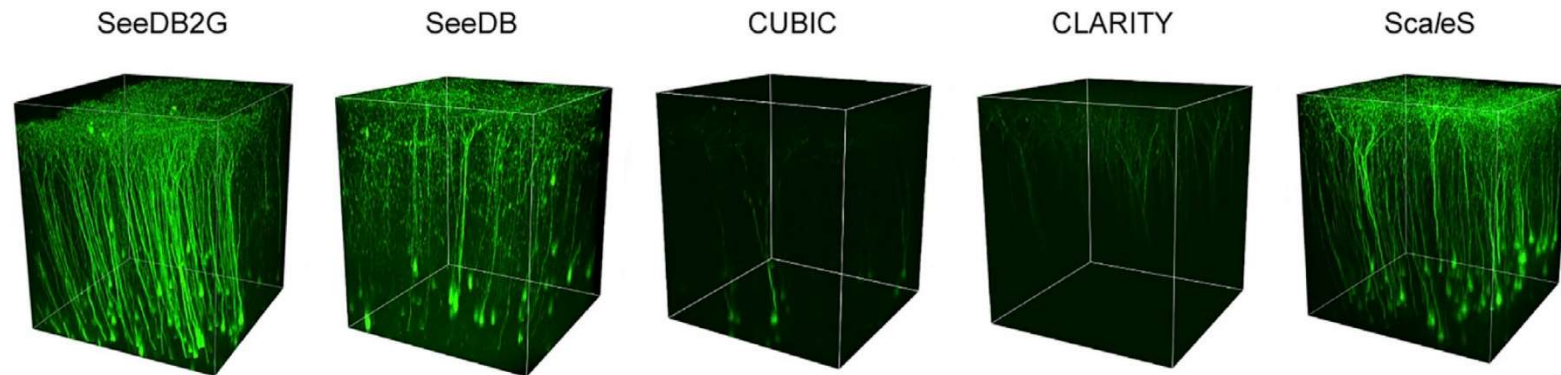
Arnold: When you remember something, things physically change in your brain. Green and red dots show contact with other cells. During the memory, new dots emerge, other fade



Synapses are electro-chemical 'switches'

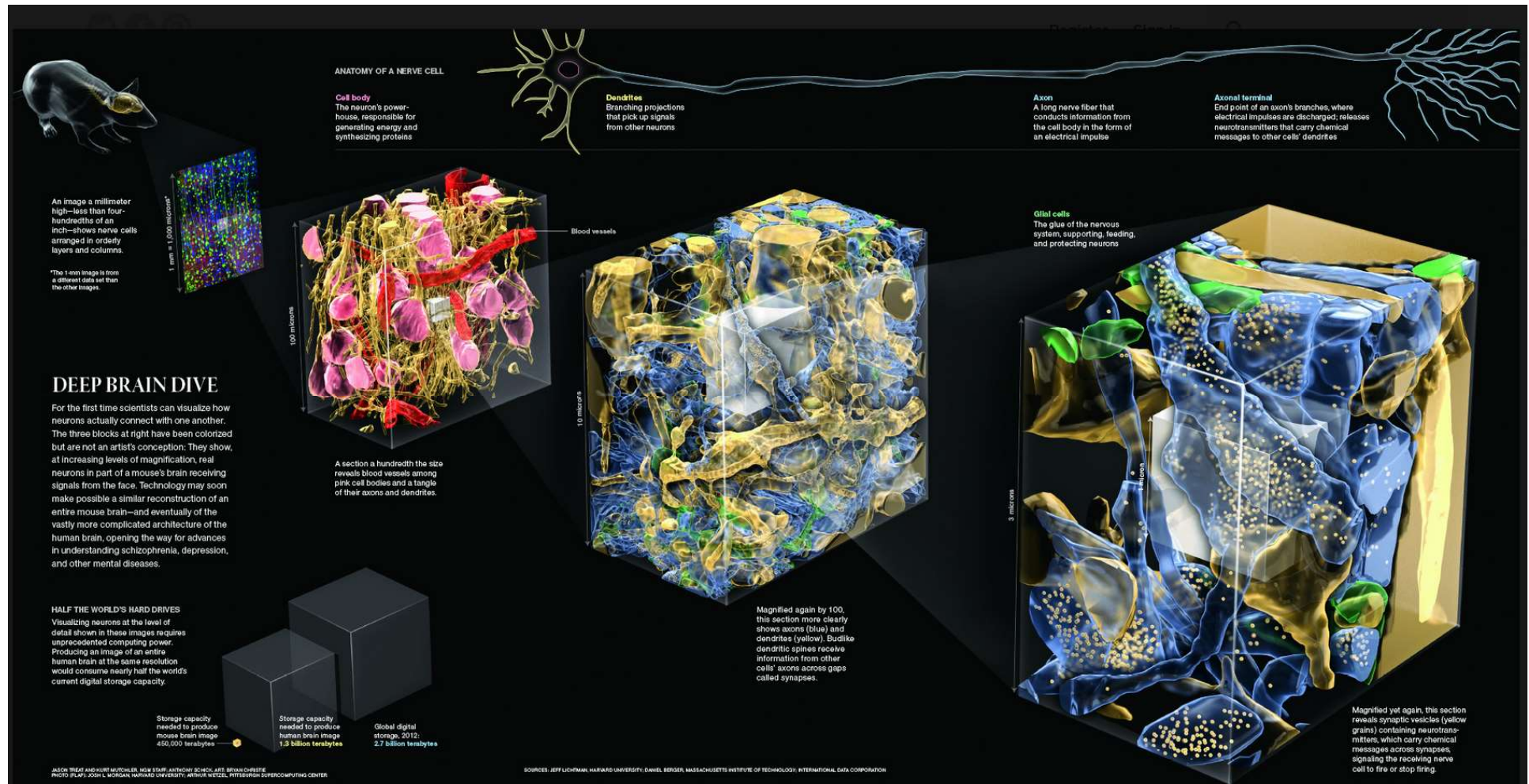
Structures and Inner Workings of the Brain

- Researchers make brain samples translucent with e.g., SeeDB2G and cut 10,000 thin slices one thousandth (.001) the size of a human hair. Then taking a picture of each with an electron microscope. Put in a row, the micro photos form a 3D model of a volume of brain tissue the size of a grain of salt

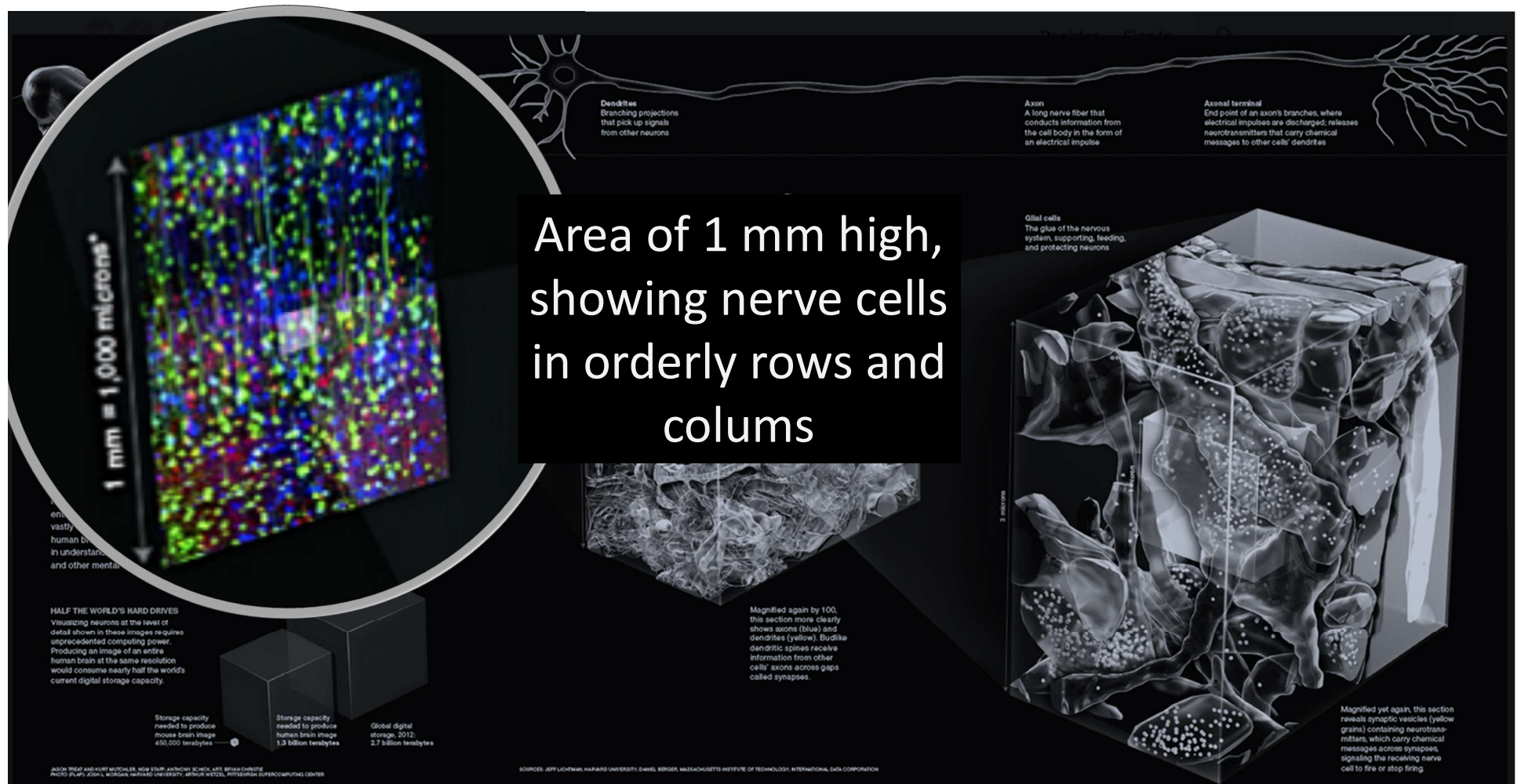


SeeDB2G provides the most detailed picture of neurons in a section of the brain

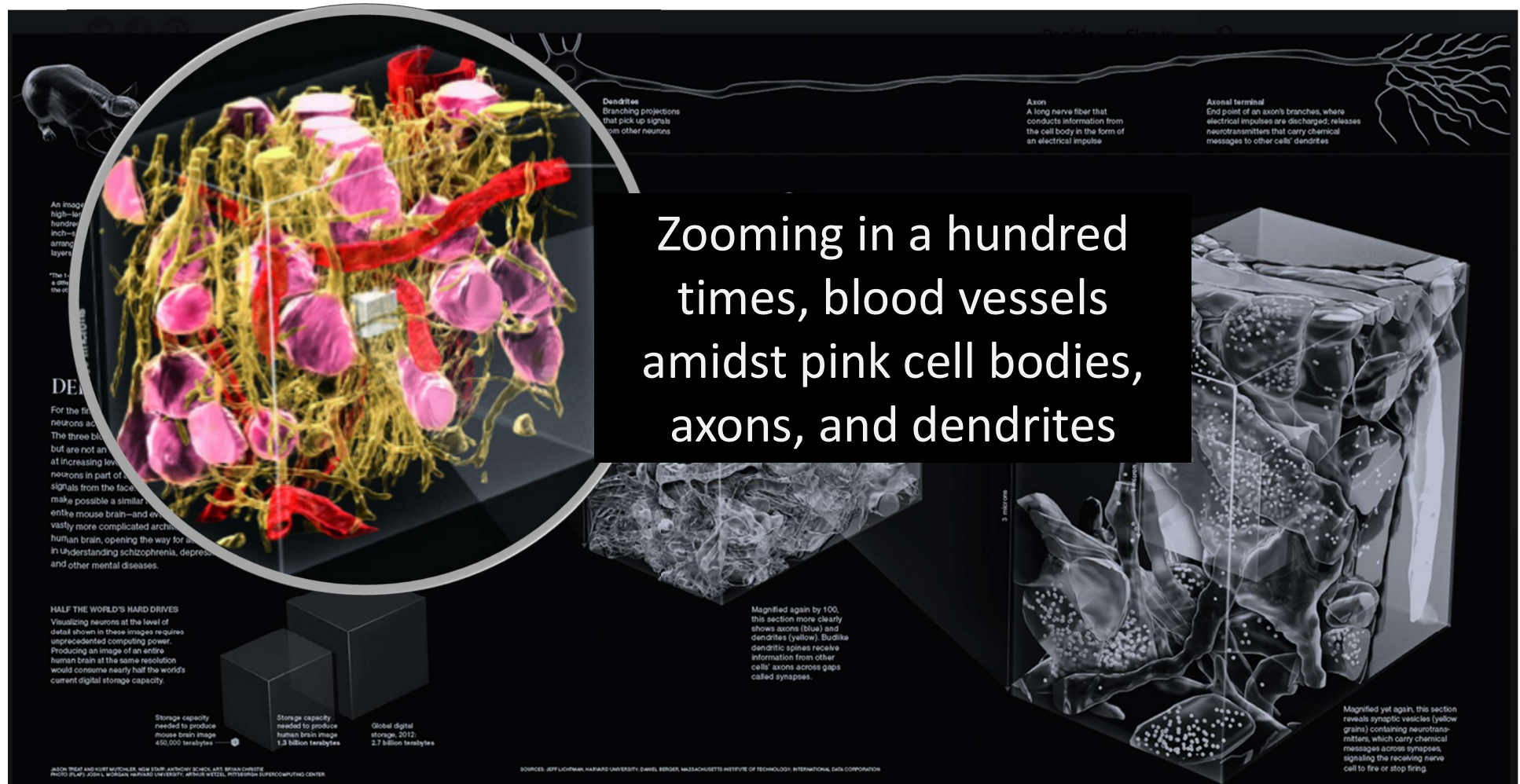
Structures and Inner Workings of the Brain



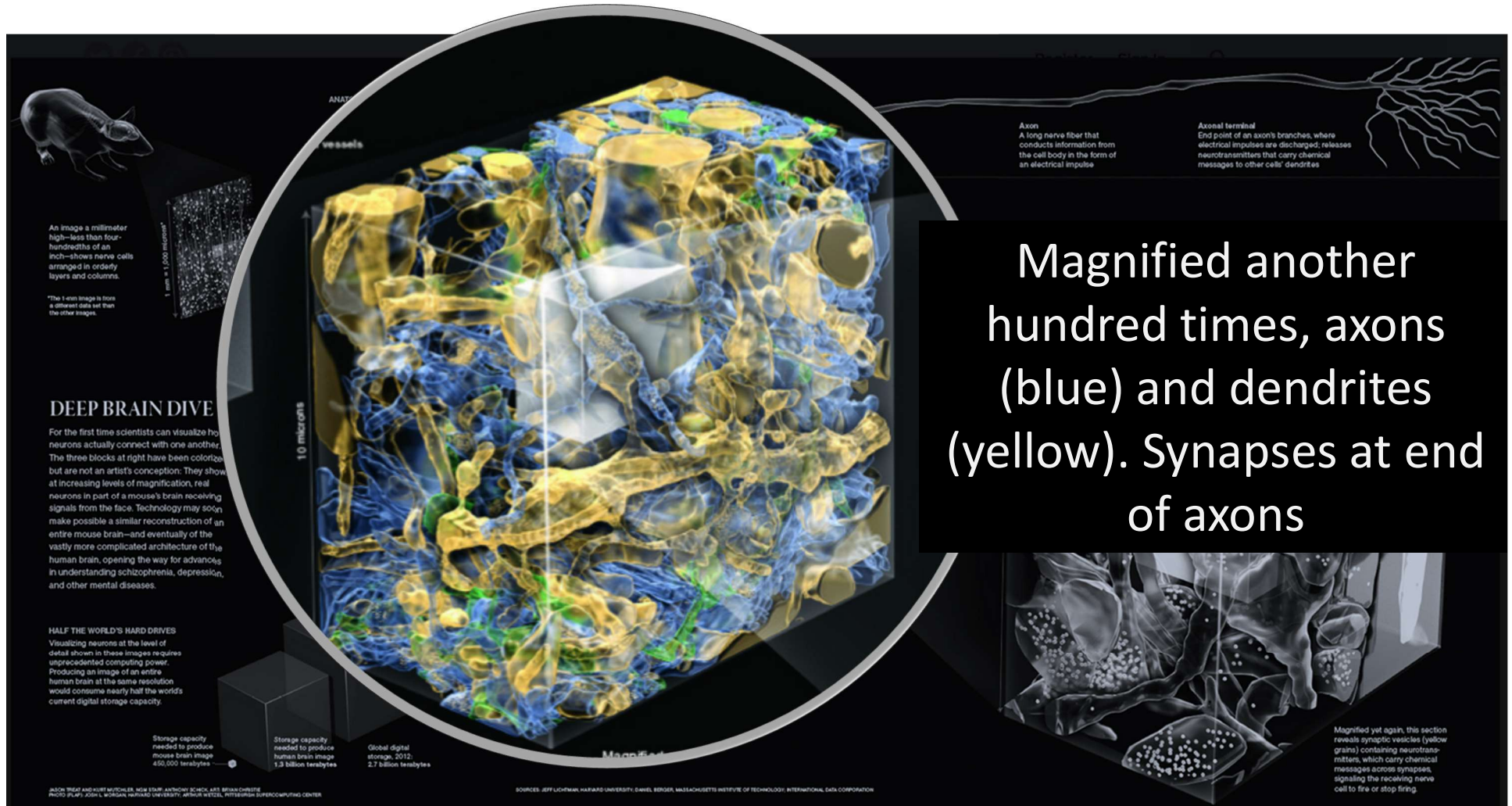
Structures and Inner Workings of the Brain



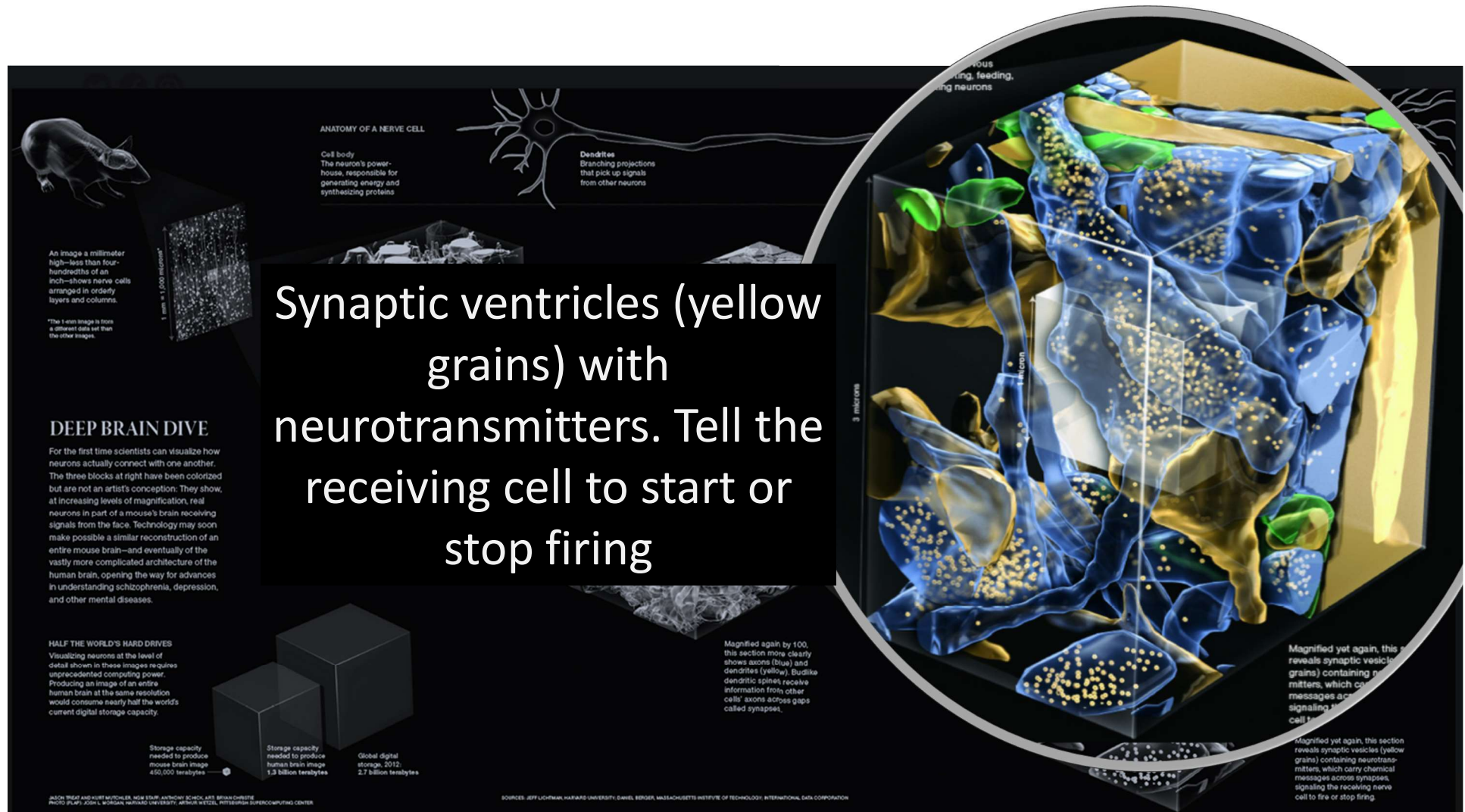
Structures and Inner Workings of the Brain



Structures and Inner Workings of the Brain



Structures and Inner Workings of the Brain



Structures and Inner Workings of the Brain

- If you take a sample of brain tissue of one thousandth (.001) micron (smaller than a grain of salt), you find about 100 axons and 80 dendrites
- Humans have about 70 billion brain cells
- That is how complex the brain is structured
- Complex, not chaotic
- Lichtman and Kasthuri discovered that each nerve cell makes contact with just one other cell, avoiding all other cells. They are very selective who to connect with

Structures and Inner Workings of the Brain

- Fried (2005) discovered the 'Jennifer Aniston neuron'
- Quian Quiroga also found brain cells for recognizing Oprah Winfrey, Golden Gate Bridge, and Eiffel Tower
- Specific information (faces, places) are stored in just one or two cells

fMRI resolution is typically around 3 x 3 x 6 mm so each sample comes from millions of neurons

Neuron 1
likes
Jennifer Aniston



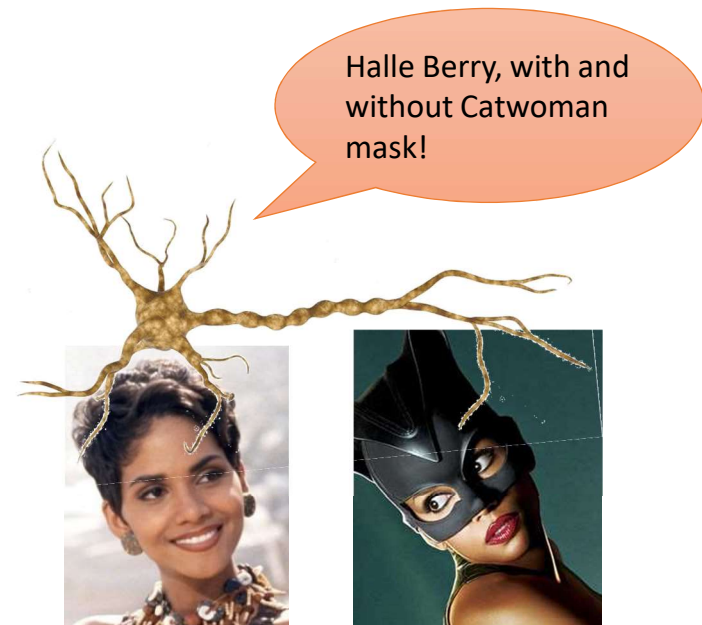
Neuron 2
likes
Julia Roberts



Neuron 3
likes
Brad Pitt



Even though there are neurons tuned to each object, the population as a whole shows no preference



Structures and Inner Workings of the Brain

VR without goggles!

Creating a False Memory in the Hippocampus

Steve Ramirez,^{1*} Xu Liu,^{1,2*} Pei-Ann Lin,¹ Junghyup Suh,¹ Michele Pignatelli,¹ Roger L. Redondo,^{1,2} Tomás J. Ryan,^{1,2} Susumu Tonegawa^{1,2†}

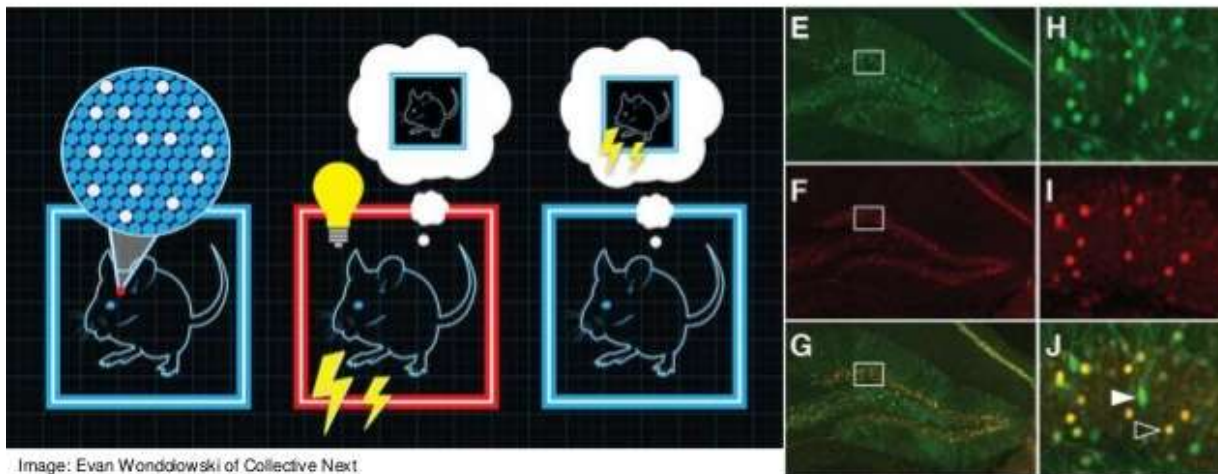


Image: Evan Wondolowski of Collective Next

Mouse in Box 1
Explores box
No fear induction

Same mouse Box 2
Shock delivered
Mouse has fear for Box 2

Same mouse Box 1
Researchers activate neurons
related to fear of Box 2
Fear response without shock

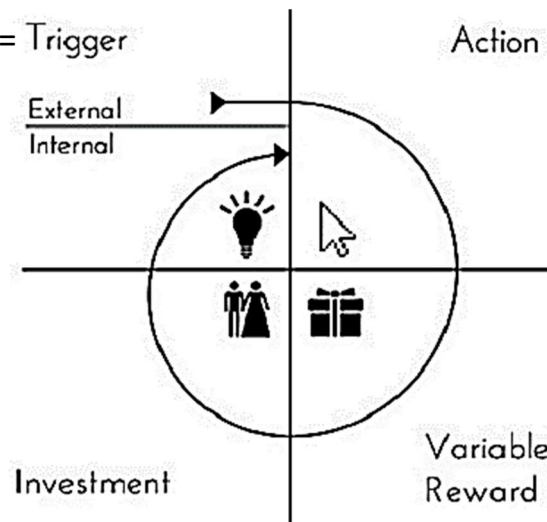
Ramirez, S., Liu, X., Lin, P. A., Suh, J., Pignatelli, M., Redondo, R. L., ... & Tonegawa, S. (2013). Creating a false memory in the hippocampus. *Science*, 341(6144), 387-391.

Structures and Inner Workings of the Brain

- Nerve cells are filled with DNA, proteins, and other molecules. About 84% of all genes in our DNA become active in our adult brains
- Every type of nerve cell uses a specific range of genes to build the molecular mechanism for its function
 - Light receptors in the eyes make proteins that can catch photons
 - Nerve cells in the *substantia nigra* make dopamine, crucial to our reward system (cf. affective processing)

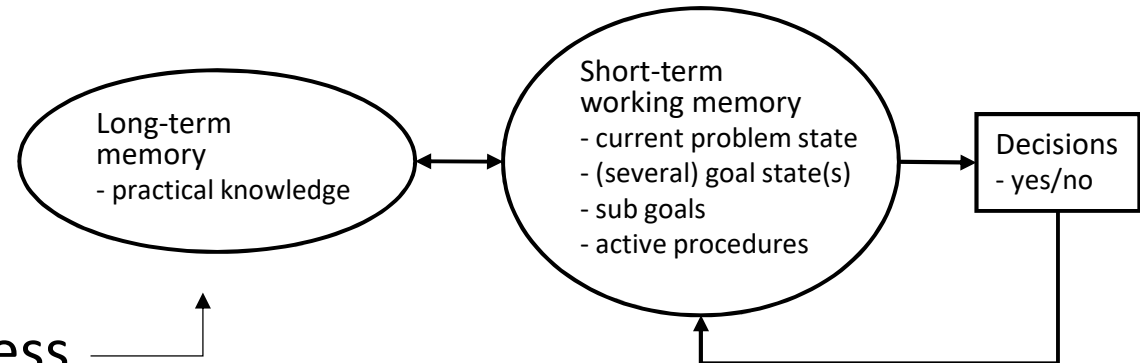
From external reward
to internal motivation
(cf. mouse in Box 1)

Your FinTech application = Trigger



What is problem solving?

- Problem
- Goal state
- Sub goals
- Problem space
- Problem-solving process



Problem solving: strategies, means, and procedures to achieve a goal

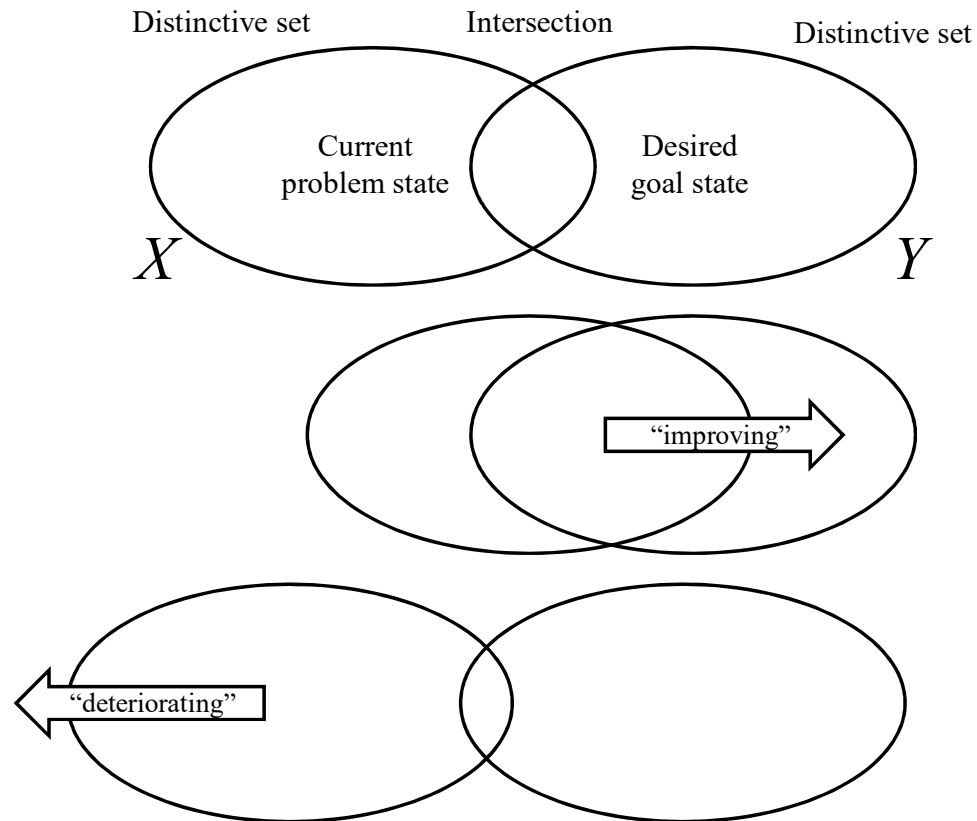
Problem: How to get what I want?

Goal state: the desired end state or solution of the problem solving activity

Sub goal: middle-term goal along the way to solving a problem

Problem space: short, middle, and long term goal states and the information needed at each step

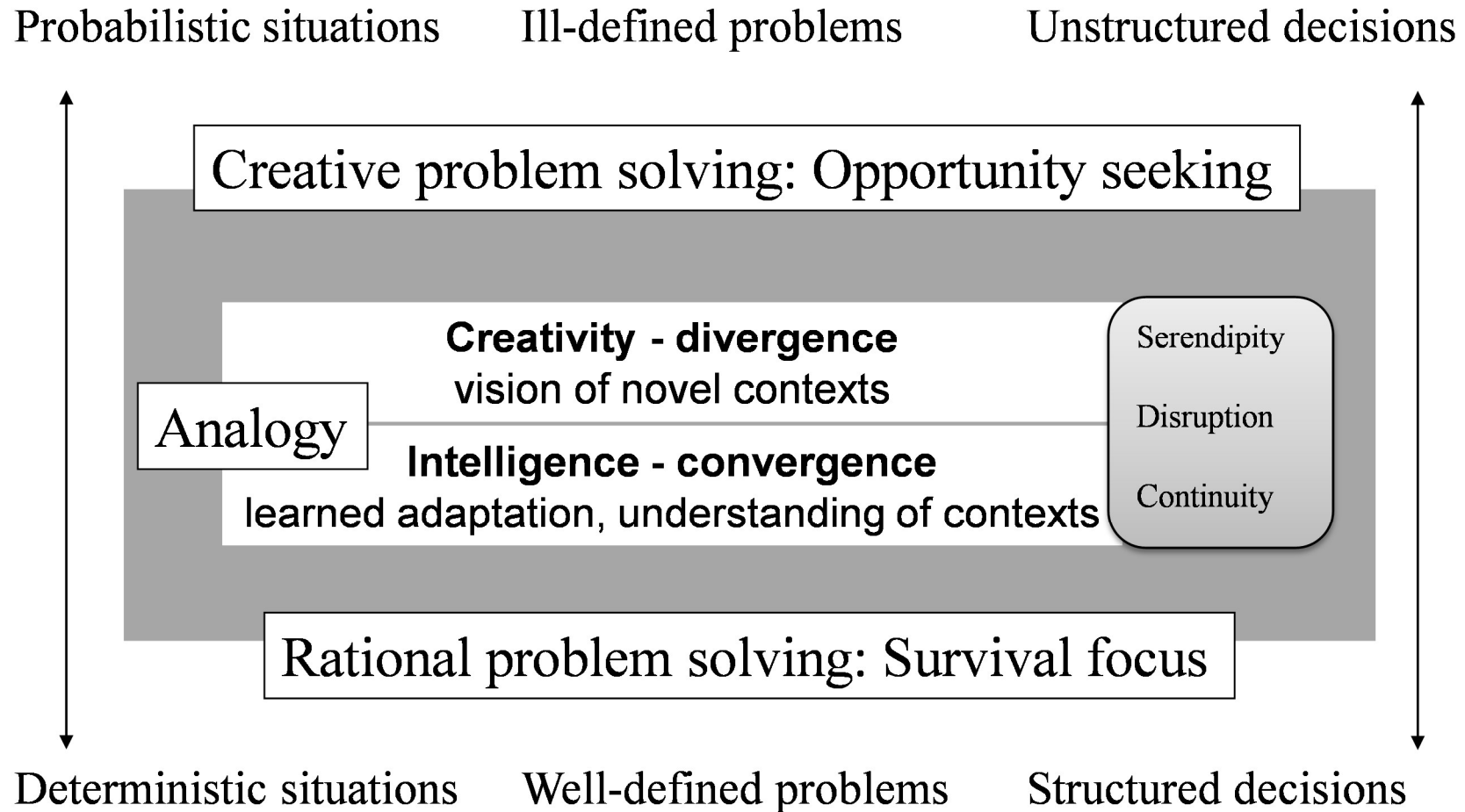
Goal states



Solving a problem: *X* should change so to overlap *Y*

Often, goal satisfaction is incomplete:
a good-enough solution

The problem continuum

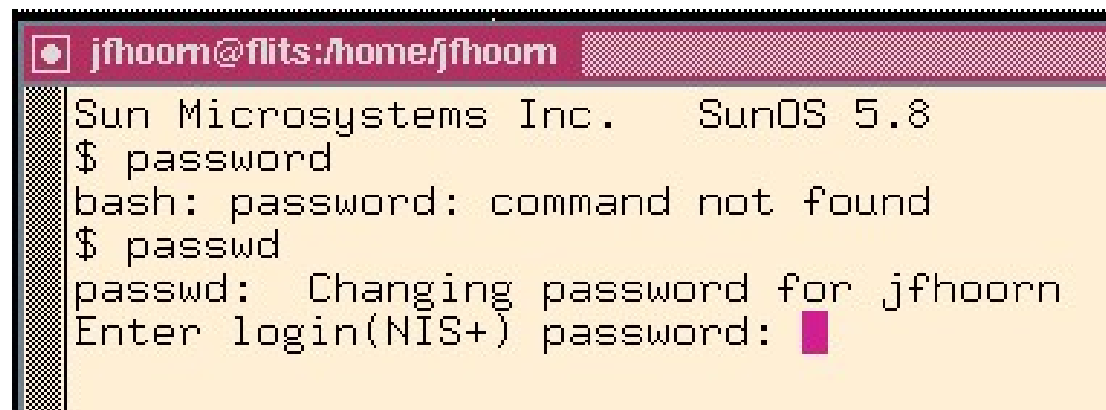


The problem continuum

Well-defined problem: full specification of the provided data, actions, strategies, and goal states, which can be represented in a problem space

Example: find the bug in Python

```
if (<boolean expression>) {  
    S1 ;  
} else {  
    S2  
}
```



```
jfhoom@flits:/home/jfhoom  
Sun Microsystems Inc. SunOS 5.8  
$ password  
bash: password: command not found  
$ passwd  
passwd: Changing password for jfhoom  
Enter login(NIS+) password: 
```

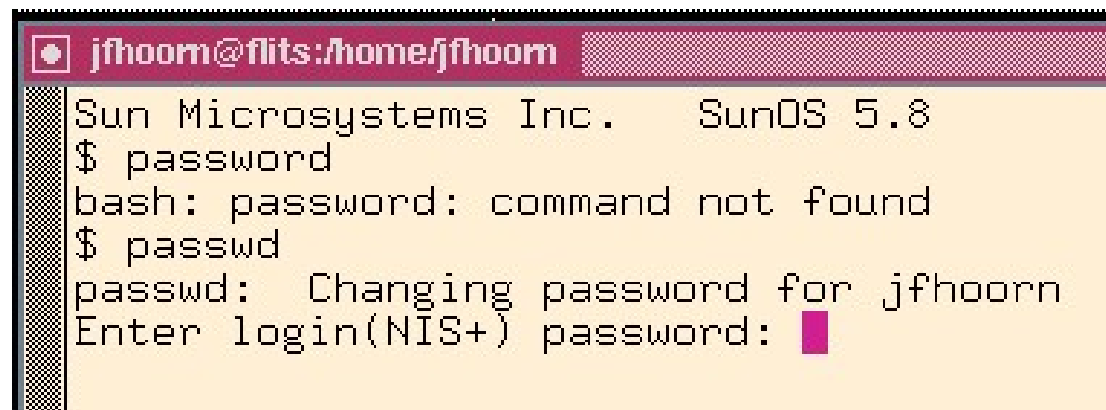
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Example: find the bug in Python

```
if (<boolean expression>) {  
    S1 ;  
} else {  
    S2 ;  
}
```

missing semicolon

A terminal window with a title bar showing the user 'jfhoorn' at host 'flits' in the directory '/home/jfhoorn'. The terminal output shows the user entering '\$ password' which results in the error 'bash: password: command not found'. The user then enters '\$ passwd', which prompts them to change their password for 'jfhoorn'. The prompt 'Enter login(NIS+) password:' is shown with a redacted password field.

```
jfhoorn@flits:/home/jfhoorn  
Sun Microsystems Inc. SunOS 5.8  
$ password  
bash: password: command not found  
$ passwd  
passwd: Changing password for jfhoorn  
Enter login(NIS+) password: 
```


ill-defined problem: partial or imprecise specification of the provided data, actions, strategies, and goal states, which cannot be (entirely) represented in a problem space

Example: mission statement

Goal decomposition:

breaking down a complex problem into smaller ones by determining sub goals

Goal state: mission



Sub goal: business goals



Sub sub goal: targets
(e.g., 5% up in two years)



Computing Services

Mission Statement

To create an online community, like no other, encompassing every facet of every municipality in the United States; to aid in online economic revitalization in partnership with local businesses and local, state and national government agencies; to publish, maintain and connect a network of community websites designed to entertain, educate and enlighten the residents of our great nation, while maintaining the highest standards of personal morals, business ethics and web etiquette.



Intermediate Summary



Human Information Processing is different from computation because perception, memory, problem solving, and (motor) responses are less 'stable' but also more adaptive

All the visual, textual, sound, and touch input become an integrated whole and recede in memory, sometimes in a single cell (e.g., faces and places)

Memory is not fixed but changes over time (neurons change their connections). Memory is context-sensitive

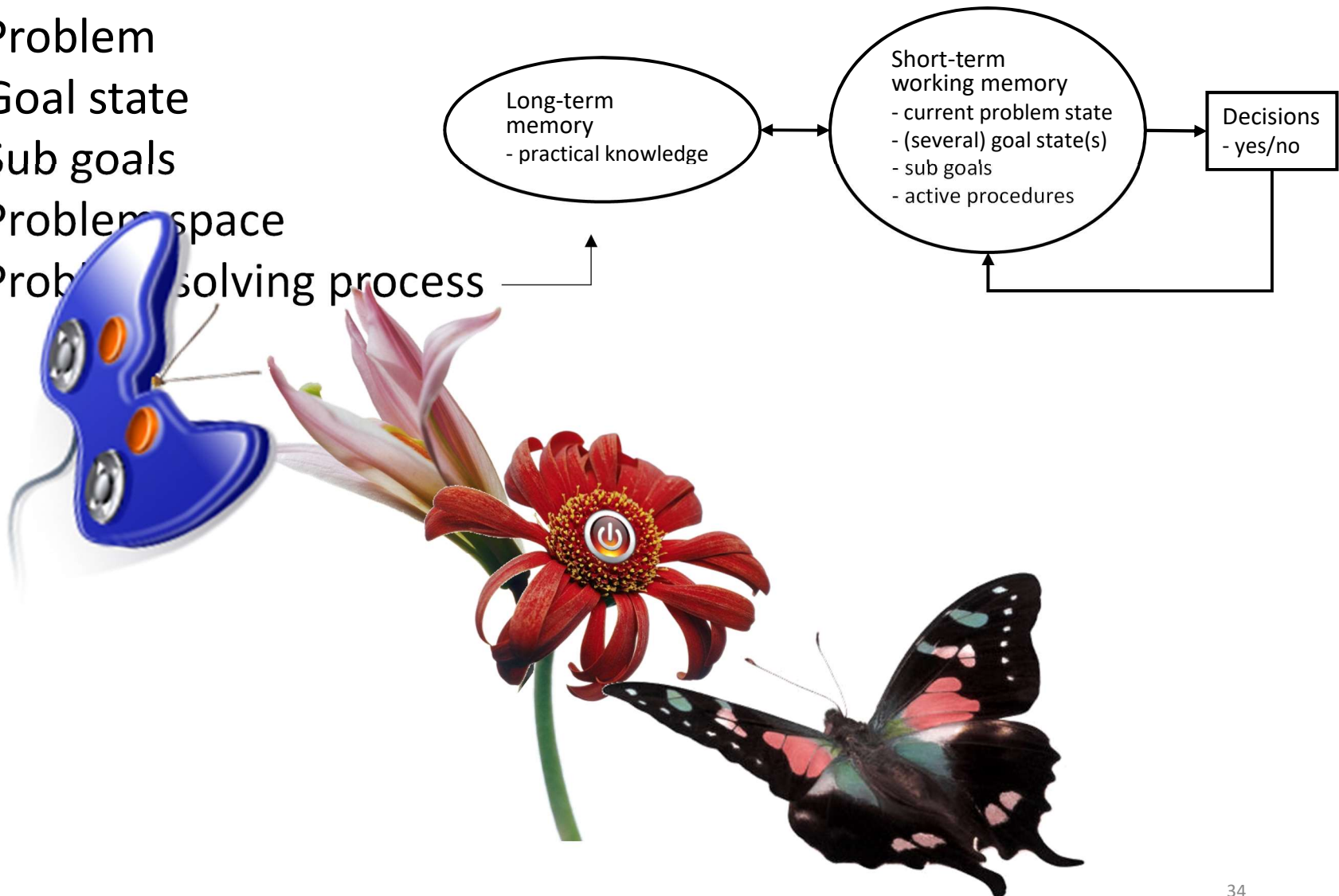
Problem-solving skills may differ. For deterministic problems, you need reasoning (e.g., means-end analysis). For underdetermined problems, you need creative solutions (see after the break)

References and recommended reading

- Anderson, J. R. (1995). *Cognitive Psychology and its Implications*. New York, NY, USA: W. H. Freeman.
- Entwistle, N. J. (2013). *Styles of Learning and teaching: An integrated outline of educational psychology for students, teachers and lecturers*. New York: Routledge.
- Felder, R. M. (1988). Learning and teaching styles in engineering education. *Engineering Education*, 78(7), 674-681.
- Hoorn, J. F. (Ed.) (2013). *Organic creativity and the physics within*. Philadelphia, PA: John Benjamins.
Available from <http://www.jbe-platform.com/content/books/9789027212030>
- Hoorn, J. F. (2014). *Creative confluence*. Philadelphia, PA: John Benjamins. Chapter 3.
- Kahneman, D., Knetsch, J. L., & Thaler, R. (1986). Fairness as a constraint on profit seeking: Entitlements in the market. *American Economic Review*, 76, 728-741.
- Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. *Econometrica*, 47, 263-291.
- Newell, A., & Simon, H. A. (1972). *Human Problem solving*. Englewood Cliffs, N.J., USA: Prentice-Hall.
- Schneider, S. L., & Lopes, L. L. (1986). Reflection in preferences under risk: Who and when may suggest why. *Journal of Experimental Psychology: Human Perception and Performance*, 12, 535-548.
- Schneider, S. L. (1992). Framing and conflict: Aspiration level contingency, the status quo, and current theories of risky choice. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 18, 1040-1057.
- Simon, H. A. (1960). *The New Science of Management Decision*. New York, NY, USA: Harper and Row.
- Tversky, A., & Kahneman, D. (1981). The framing of decisions and the psychology of choice. *Science*, 211, 453-458.

After the break: problem solving strategies

- Problem
- Goal state
- Sub goals
- Problem space
- Problem solving process



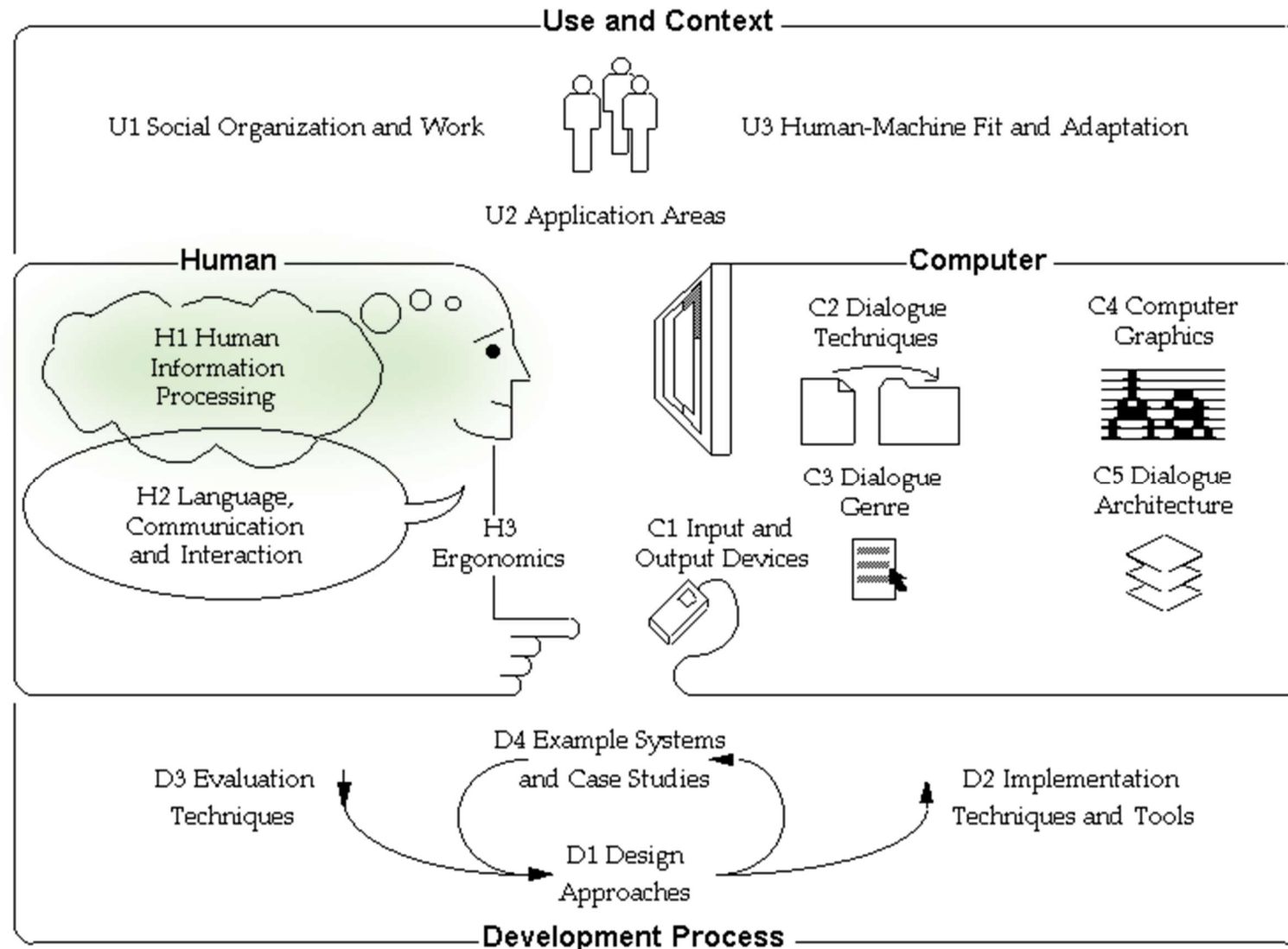
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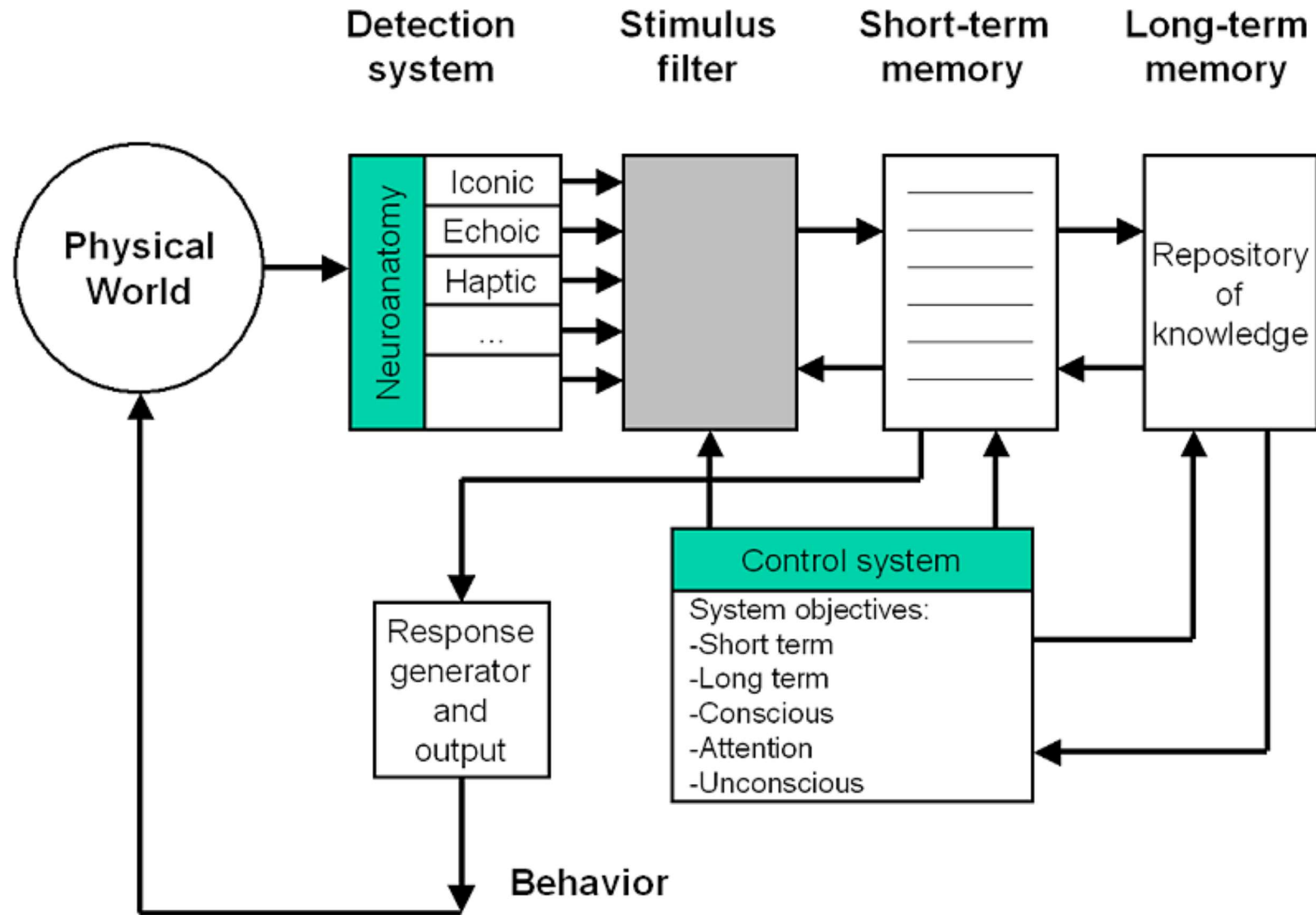
Human Information Processing, Part 2



Overview



The Human Information Processor



Problem solving strategies

Intelligence

Rule following
Difference reduction

Means-end analysis
Forward and backward reasoning

Creativity

Analogy
Random trial and error

Creative problem solving

May lead to innovation

ill-defined problem: partial or imprecise specification of the provided data, actions, strategies, and goal states, which cannot be (entirely) represented in a problem space

Example: mission statement



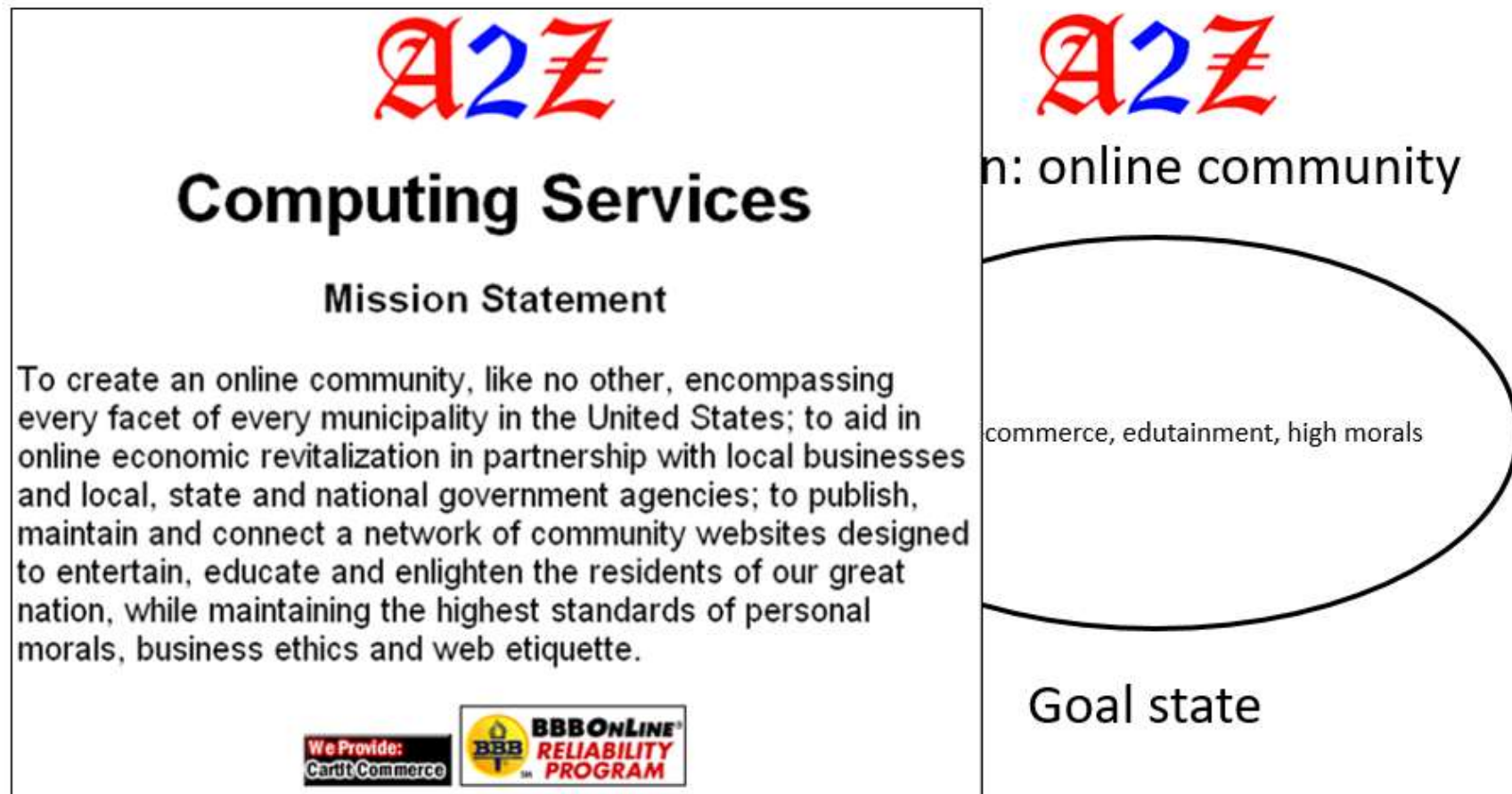
Computing Services

Mission Statement

To create an online community, like no other, encompassing every facet of every municipality in the United States; to aid in online economic revitalization in partnership with local businesses and local, state and national government agencies; to publish, maintain and connect a network of community websites designed to entertain, educate and enlighten the residents of our great nation, while maintaining the highest standards of personal morals, business ethics and web etiquette.



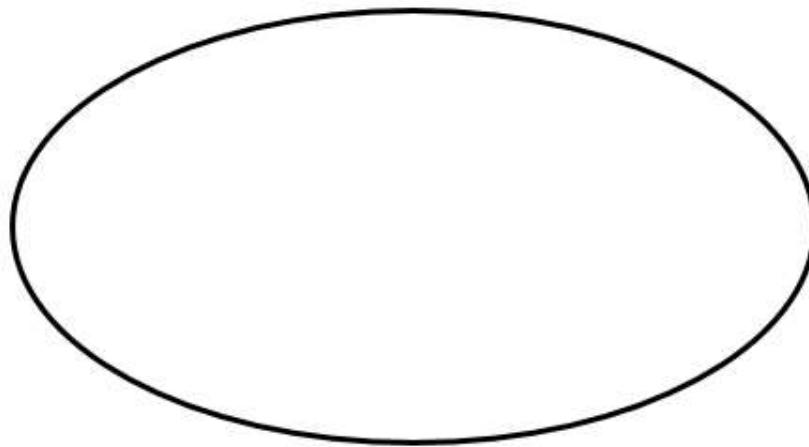
Difference reduction: each sub goal that is achieved brings the current problem state closer to the goal state



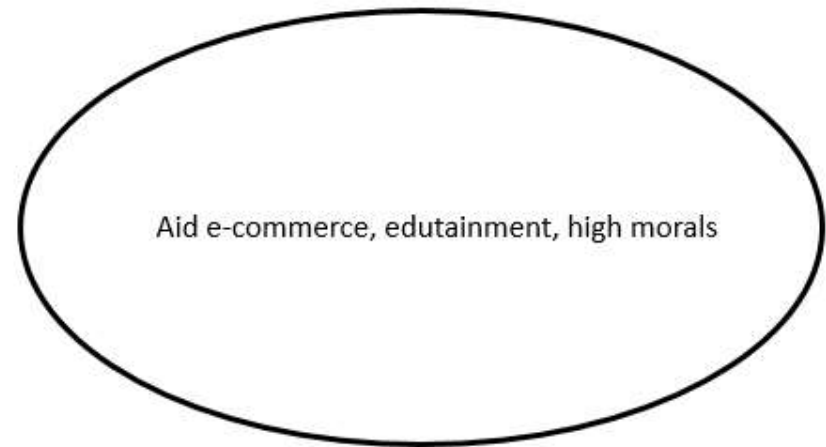
Difference reduction: each sub goal that is achieved brings the current problem state closer to the goal state

A2Z

Mission: online community



Current problem state

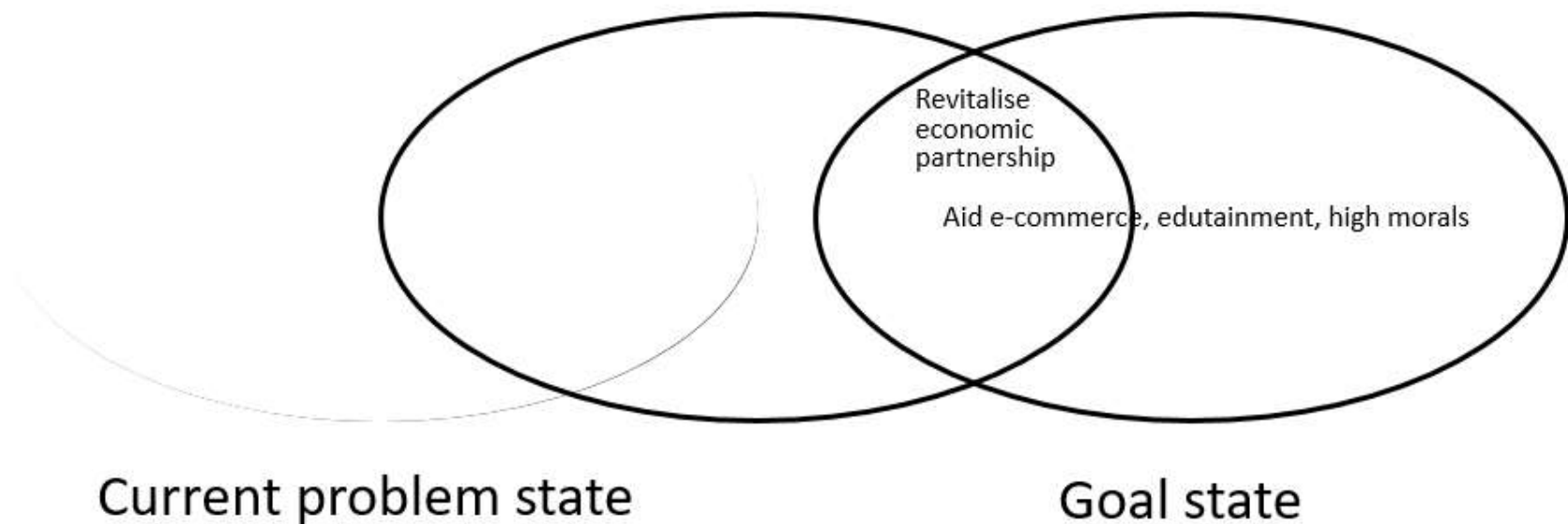


Goal state

Difference reduction: each sub goal that is achieved brings the current problem state closer to the goal state

A2Z

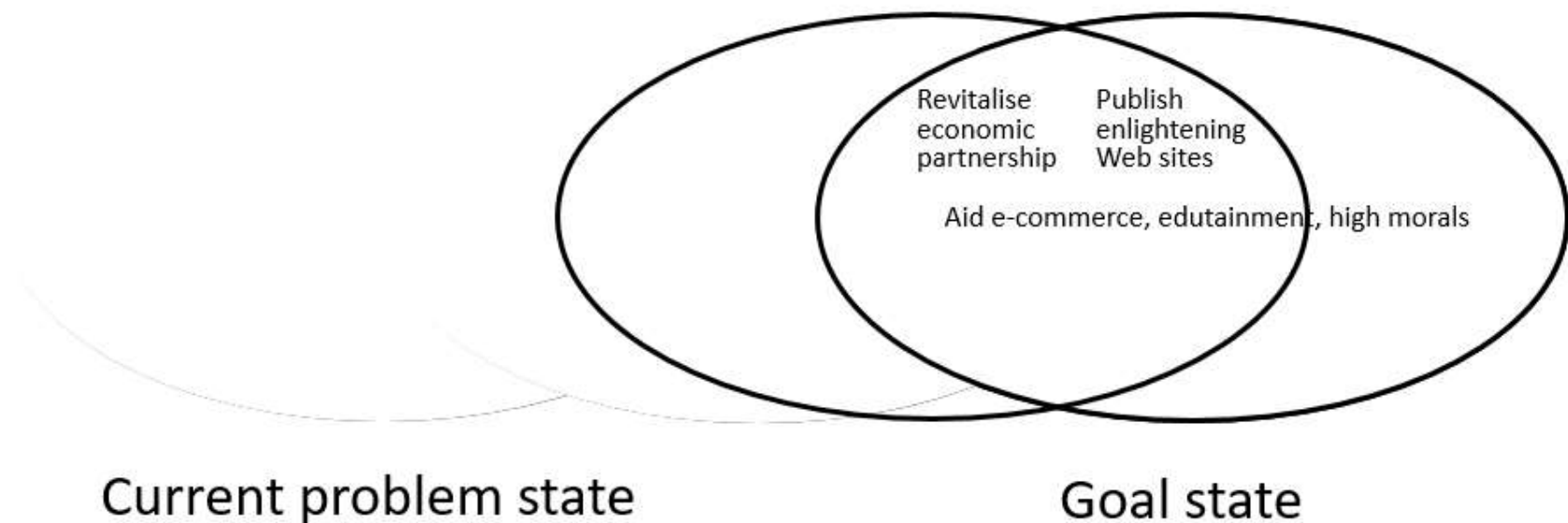
Mission: online community



Difference reduction: each sub goal that is achieved brings the current problem state closer to the goal state

A2Z

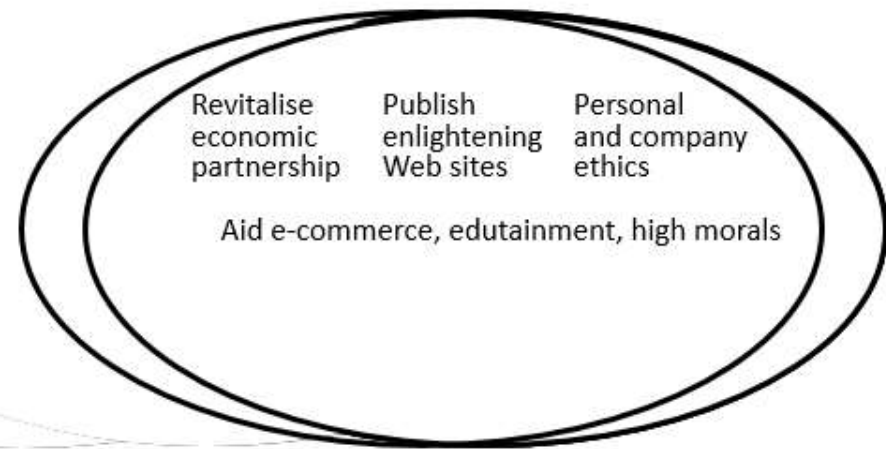
Mission: online community



Difference reduction: each sub goal that is achieved brings the current problem state closer to the goal state



Mission: online community

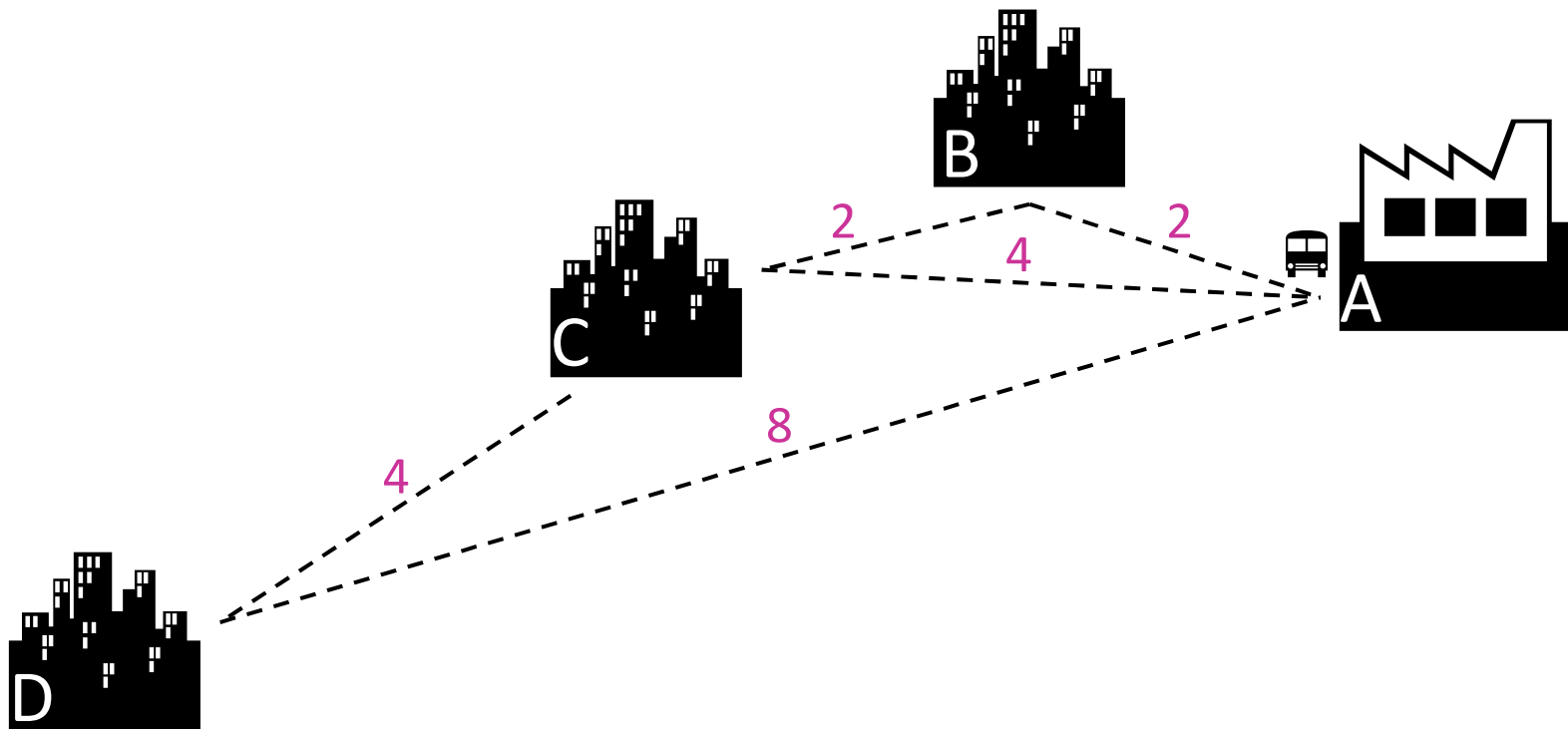


Current problem state

Goal state

Means-end analysis: for each sub goal, a (different) procedure decreases the difference between current problem state and goal state even if this temporarily leads away from the goal state

Logistic problem: Supply three different cities with one lorry, then return to factory

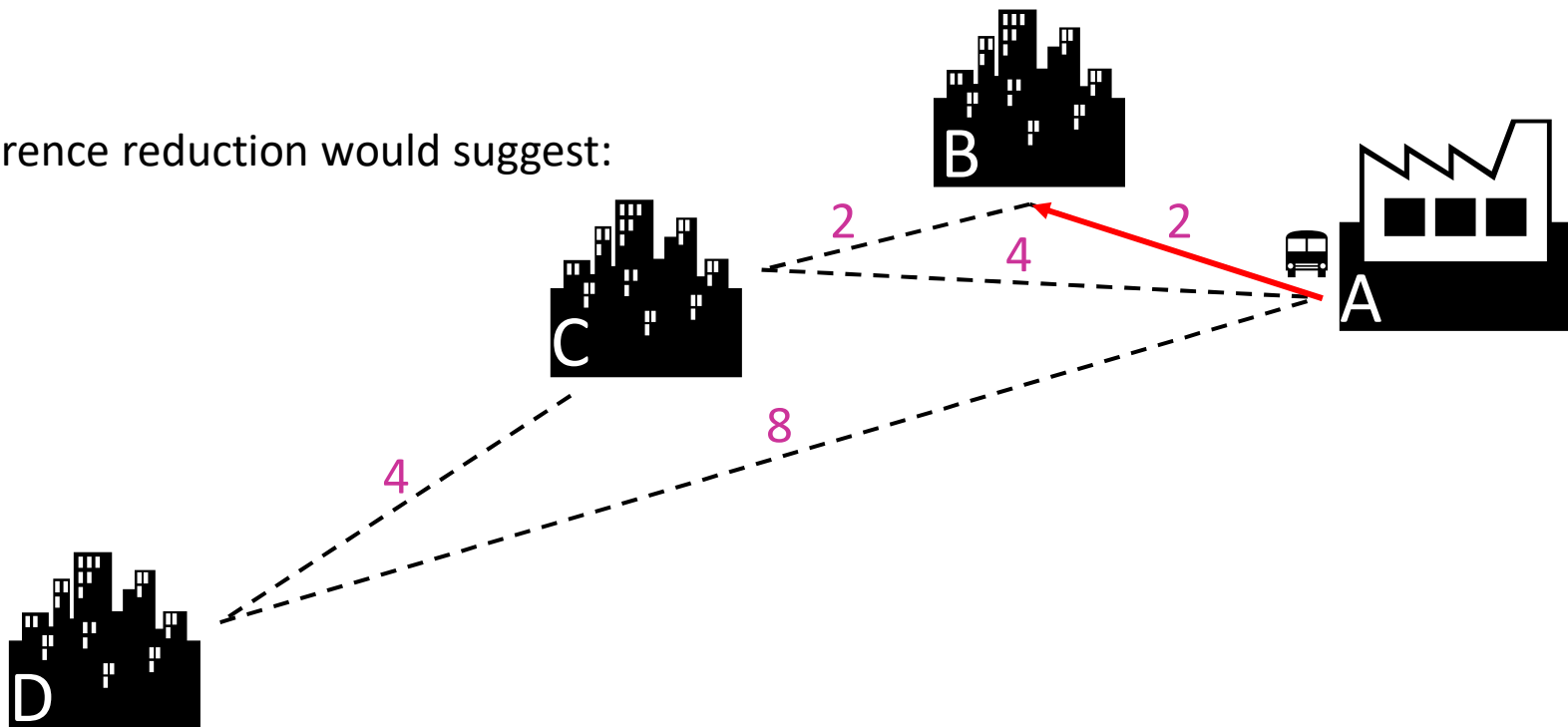


Means-end analysis: for each sub goal, a (different) procedure decreases the difference between current problem state and goal state even if this temporarily leads away from the goal state

Logistic problem: Supply three different cities with one lorry, then return to factory

Difference reduction would suggest:

2 +

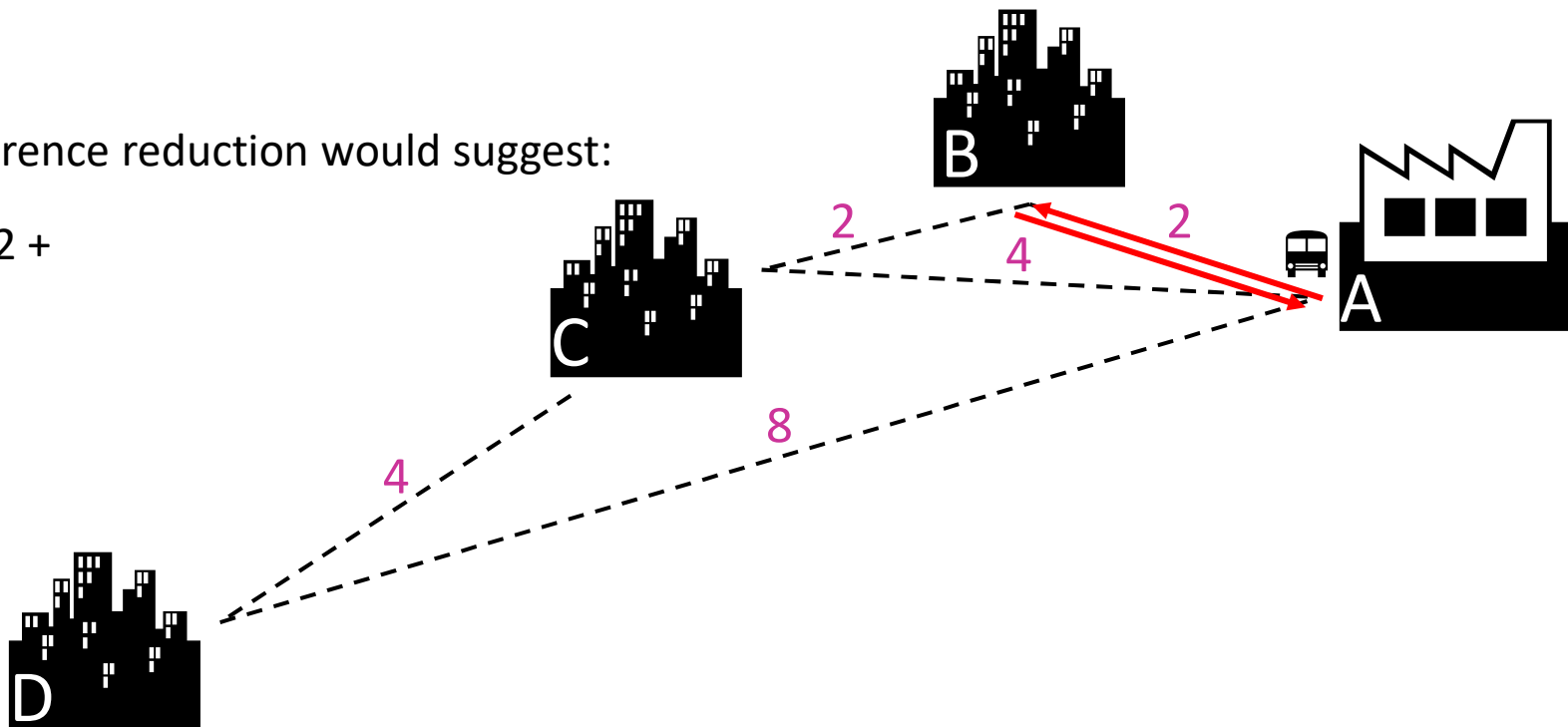


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2 + 2 +

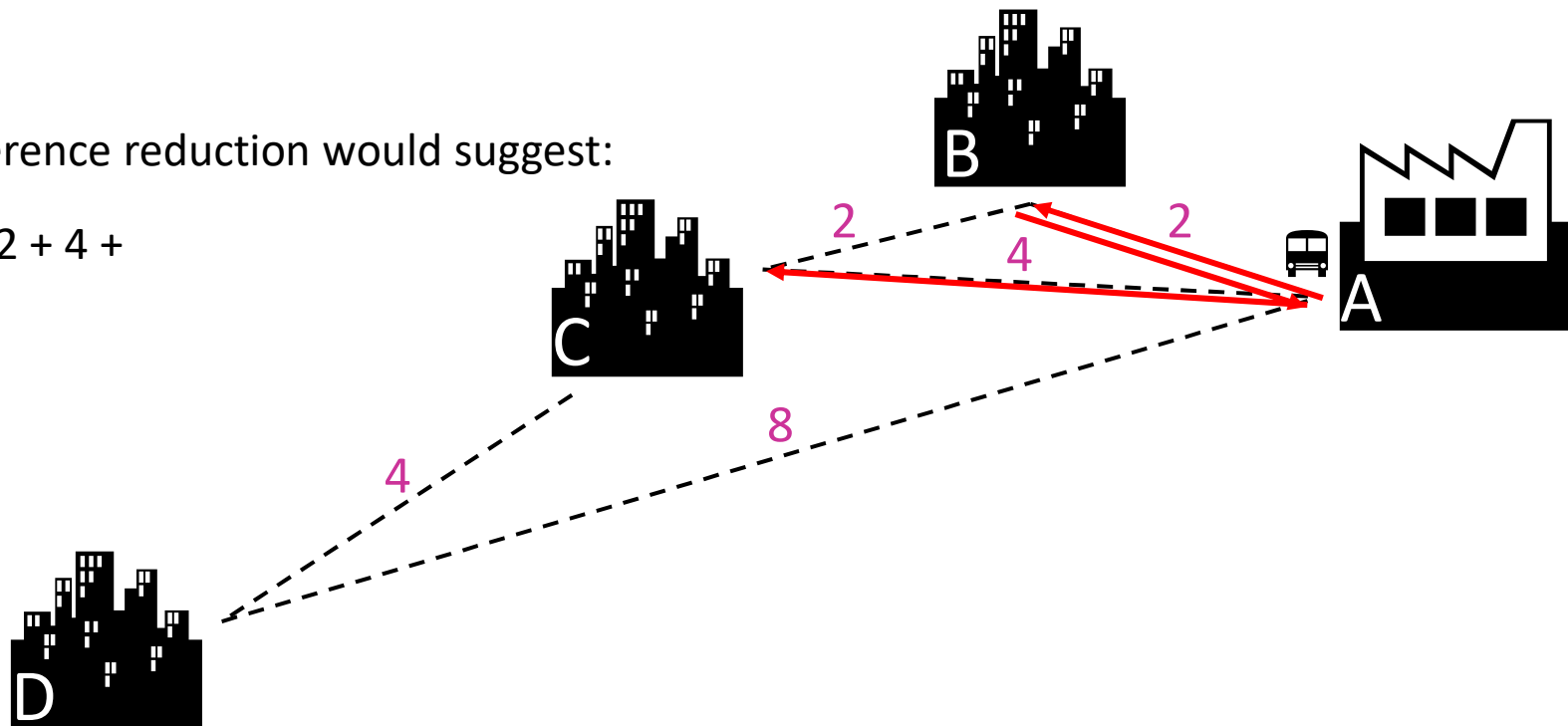


Means-end analysis: for each sub goal, a (different) procedure decreases the difference between current problem state and goal state even if this temporarily leads away from the goal state

Logistic problem: Supply three different cities with one lorry, then return to factory

Difference reduction would suggest:

$2 + 2 + 4 +$

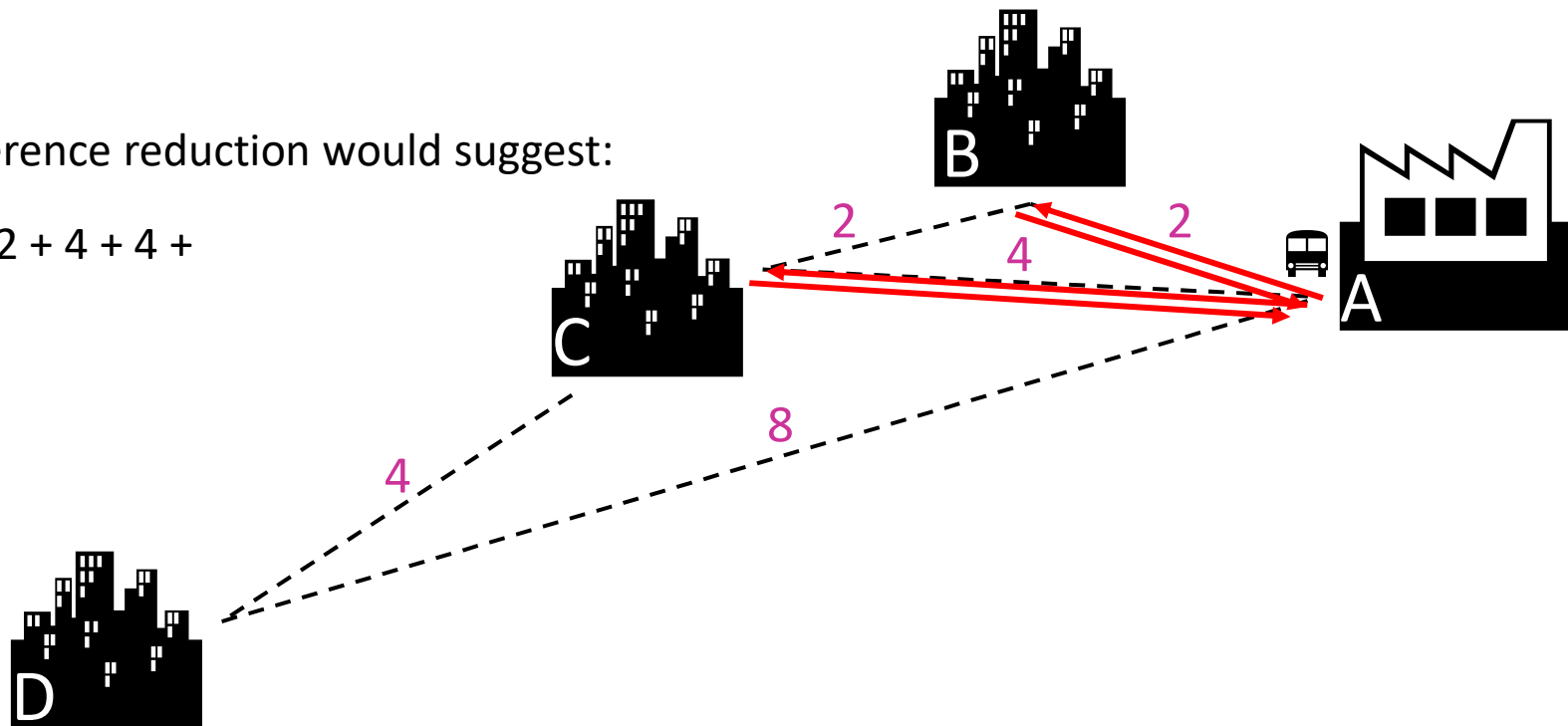


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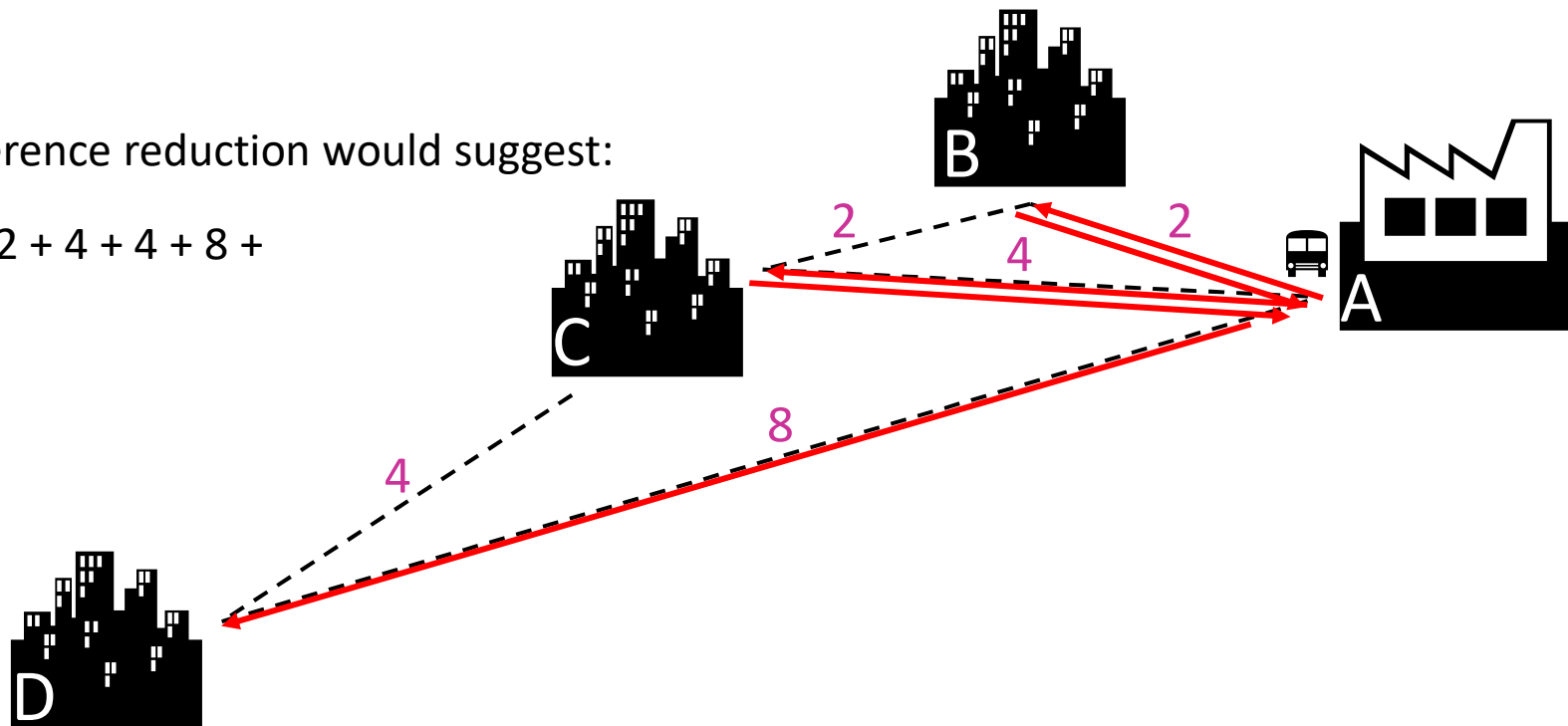


Means-end analysis: for each sub goal, a (different) procedure decreases the difference between current problem state and goal state even if this temporarily leads away from the goal state

Logistic problem: Supply three different cities with one lorry, then return to factory

Difference reduction would suggest:

$2 + 2 + 4 + 4 + 8 +$

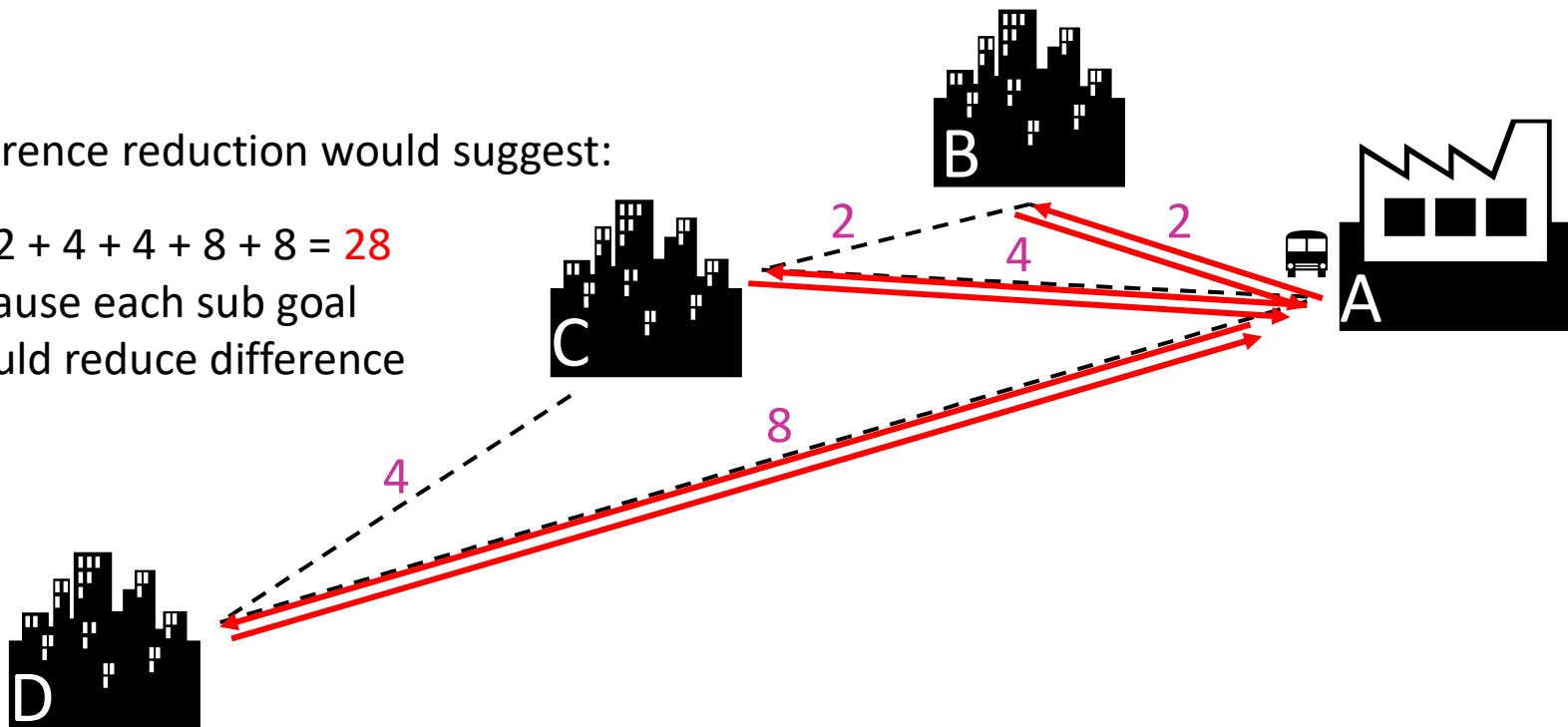


Means-end analysis: for each sub goal, a (different) procedure decreases the difference between current problem state and goal state even if this temporarily leads away from the goal state

Logistic problem: Supply three different cities with one lorry, then return to factory

Difference reduction would suggest:

$2 + 2 + 4 + 4 + 8 + 8 = 28$
because each sub goal
should reduce difference

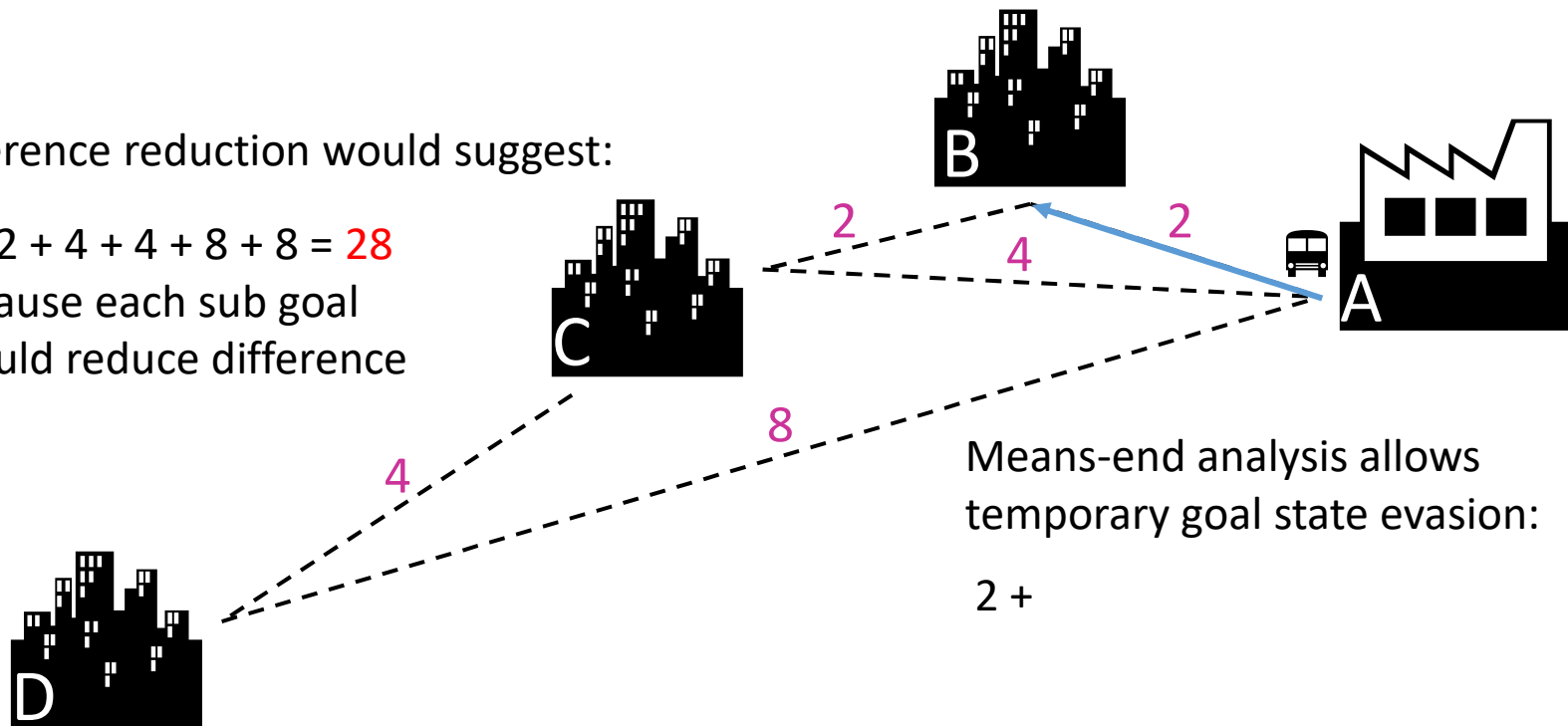


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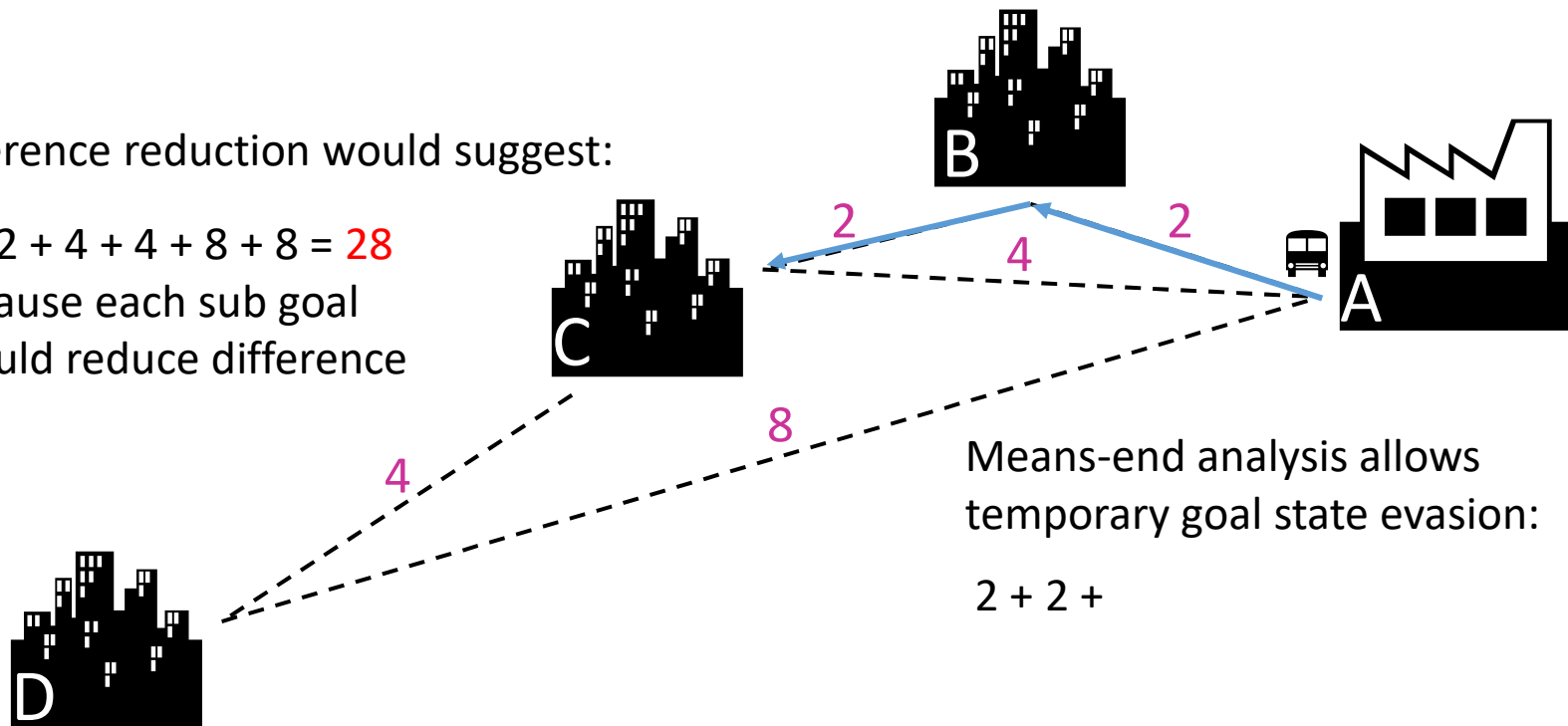


Means-end analysis: for each sub goal, a (different) procedure decreases the difference between current problem state and goal state even if this temporarily leads away from the goal state

Logistic problem: Supply three different cities with one lorry, then return to factory

Difference reduction would suggest:

$2 + 2 + 4 + 4 + 8 + 8 = 28$
because each sub goal
should reduce difference

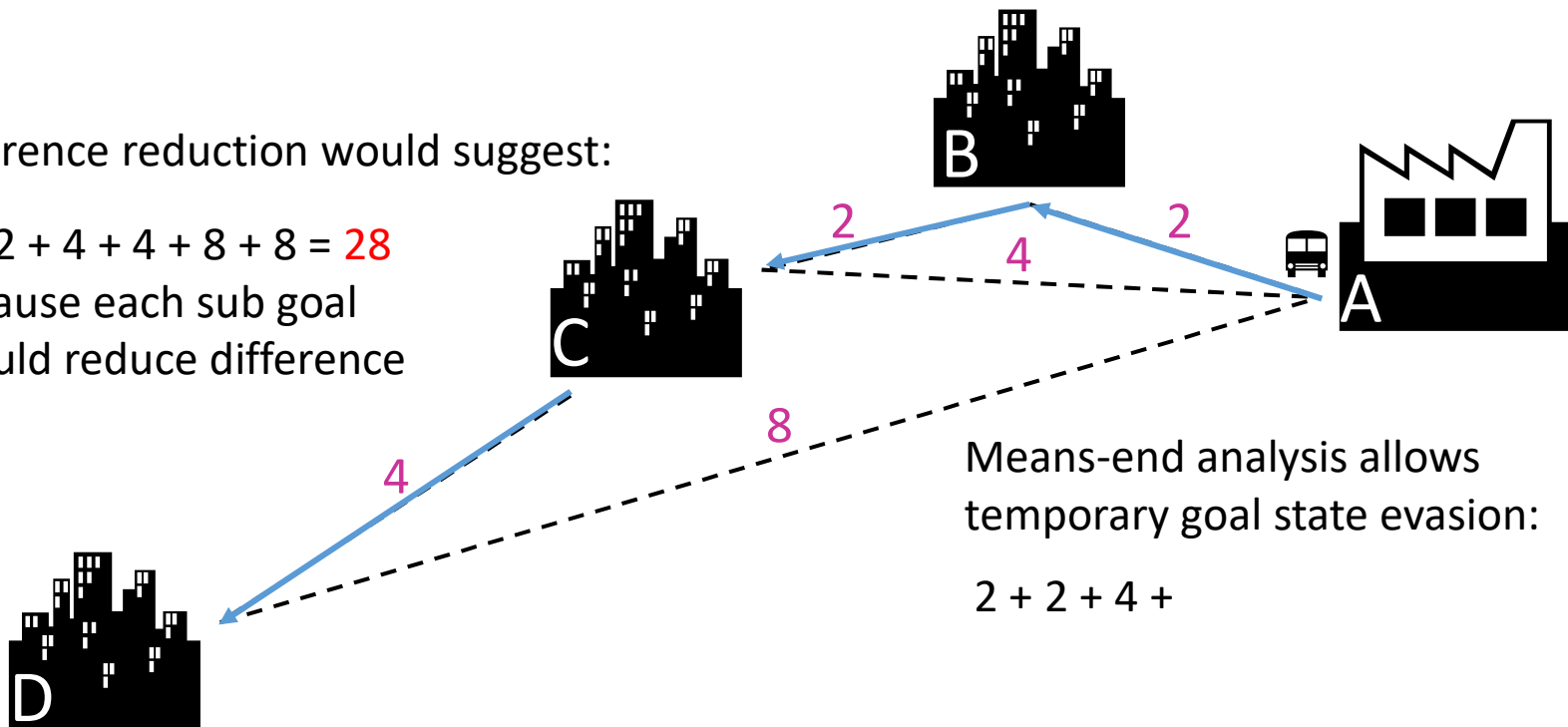


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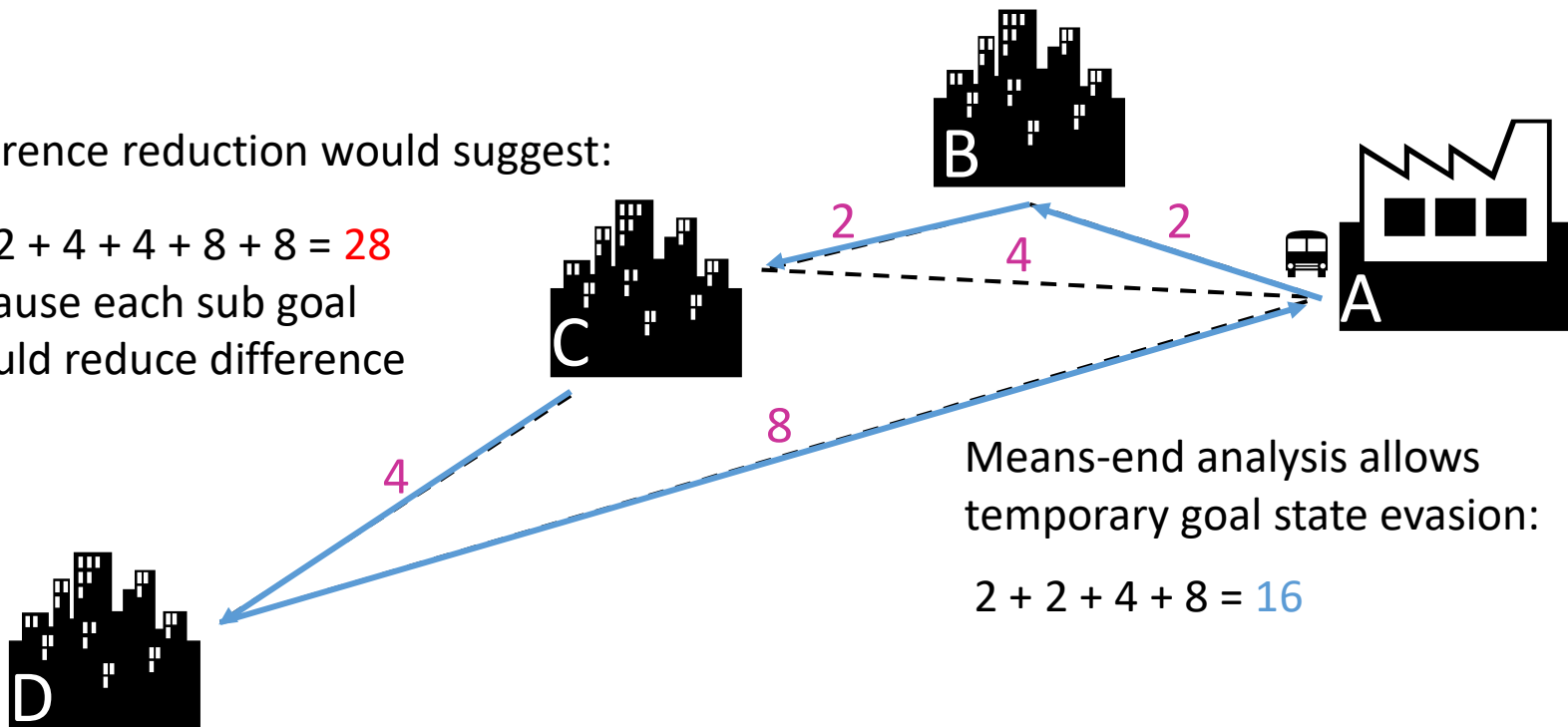


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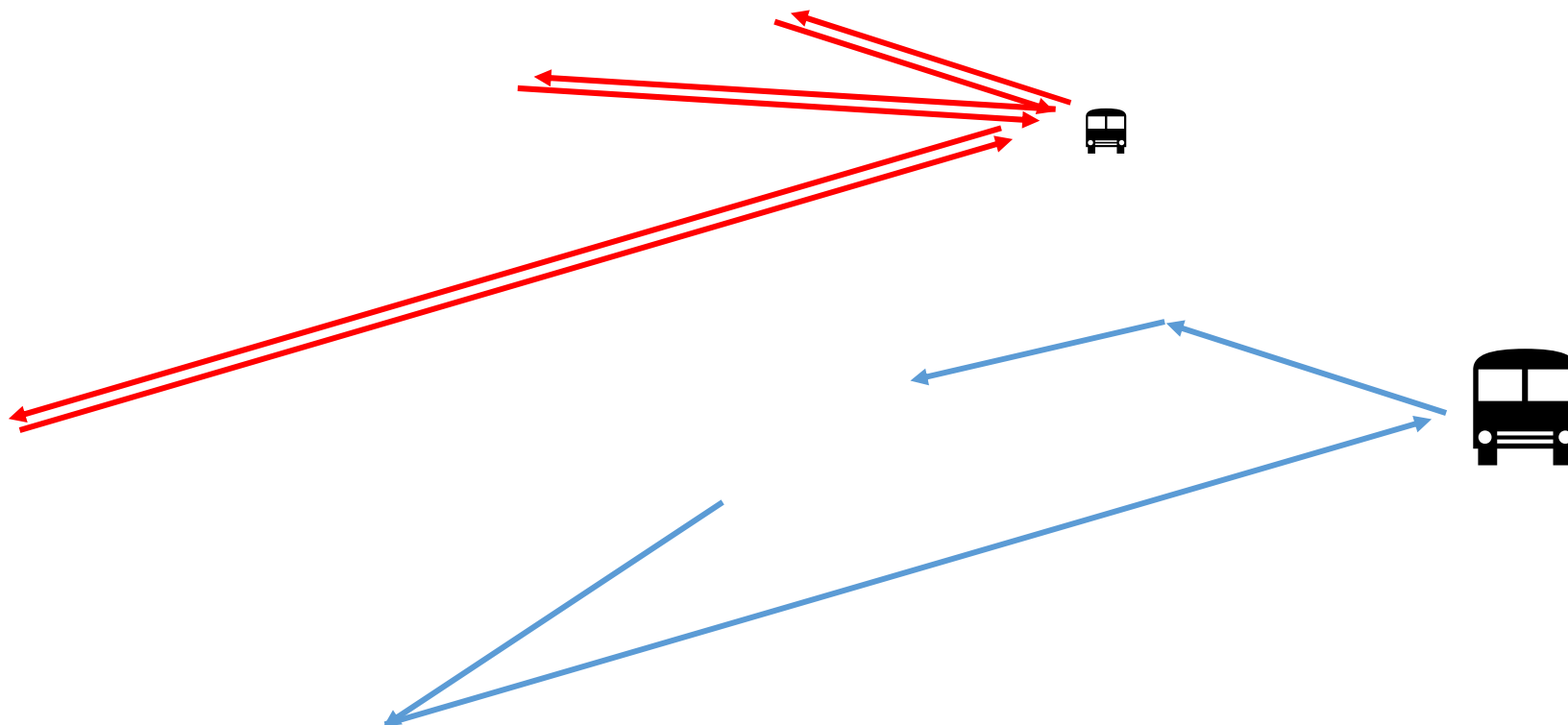
$2 + 2 + 4 + 4 + 8 + 8 = 28$
because each sub goal
should reduce difference



Means-end analysis allows
temporary goal state evasion:

$$2 + 2 + 4 + 8 = 16$$

Note: The choice of a problem-solving strategy takes effect on the technology you buy / develop / apply. In our case, difference reduction requires a smaller lorry but wear-and-tear is bigger (makes more miles); means-end analysis demands a larger lorry to transport all goods at once!!



Fetch robotics: Difference reduction or Means-end analysis?



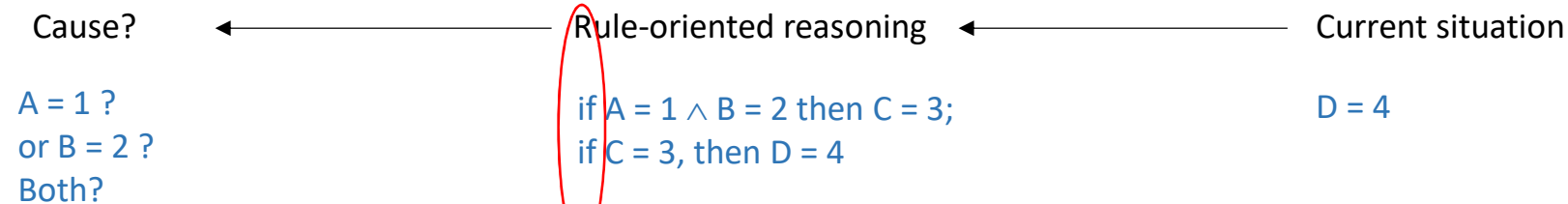
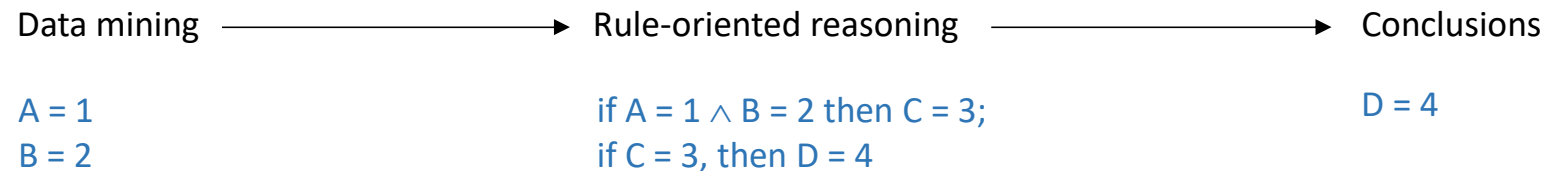
Difference reduction:
Means-end analysis:

more wear-and-tear
bigger robots / platforms

Forward and backward reasoning are recursive inference principles. Used to solve well-defined problems, which explicitly have formal limitations (chess, maths)

lonely = 1; socially isolated = 2; need for relationship = 3; social robot = 4; social media = 5; dating site = 6

Forward reasoning



Backward reasoning

Depends on formal relationship between variables. If there are more explanations for the same phenomenon, see next slide...

Forward and backward reasoning are recursive inference principles. Used to solve well-defined problems, which explicitly have formal limitations (chess, maths)



Your PC ran into a problem and needs to restart. We're just collecting some error info, and then we'll restart for you.

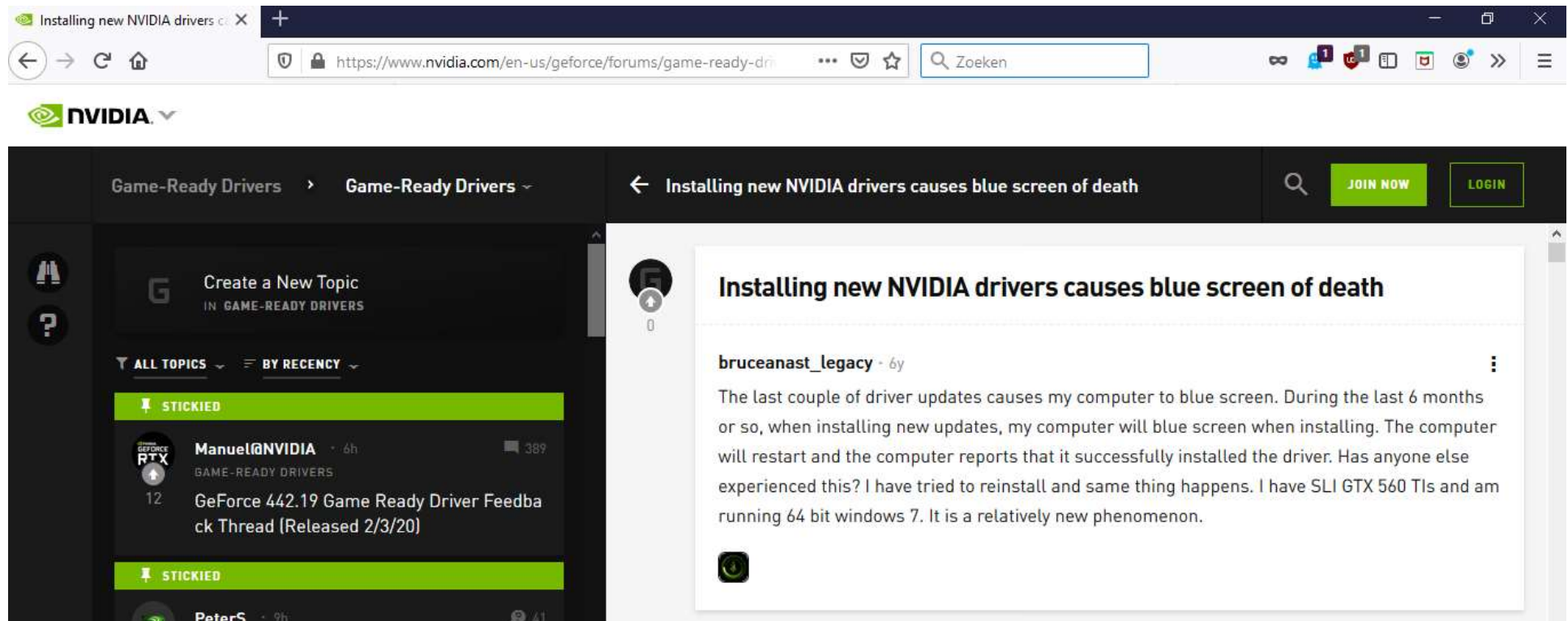
42% complete



For more information about this issue and possible fixes, visit
<https://www.windows.com/stopcode>

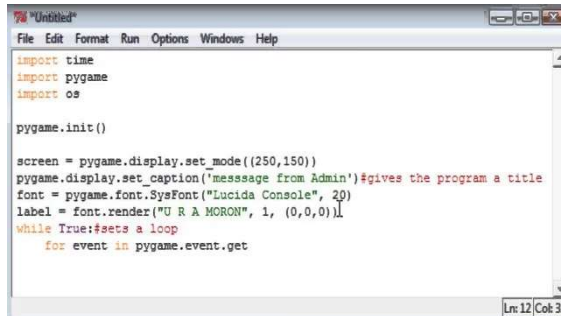
If you call a support person, give them this info:
Stop code: PAGE_FAULT_IN_NONPAGED_AREA
What failed: nvlddmkm.sys

If it is a known problem...
(I just installed that new NVIDIA driver)



... you are allowed to reason forward and in this case, backward
(diagnostics)

If the problem is unknown...
(more explanations for the same phenomenon)

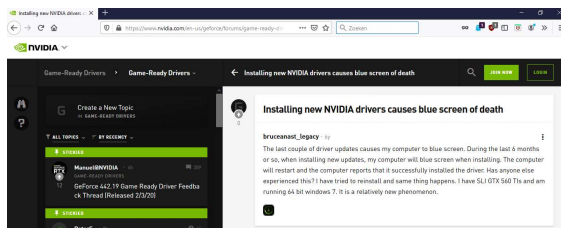


```
import time
import pygame
import os

pygame.init()

screen = pygame.display.set_mode((250,150))
pygame.display.set_caption('message from Admin')#gives the program a title
font = pygame.font.SysFont("Lucida Console", 20)
label = font.render("U R A MORON", 1, (0,0,0))
while True:#sets a loop
    for event in pygame.event.get
```

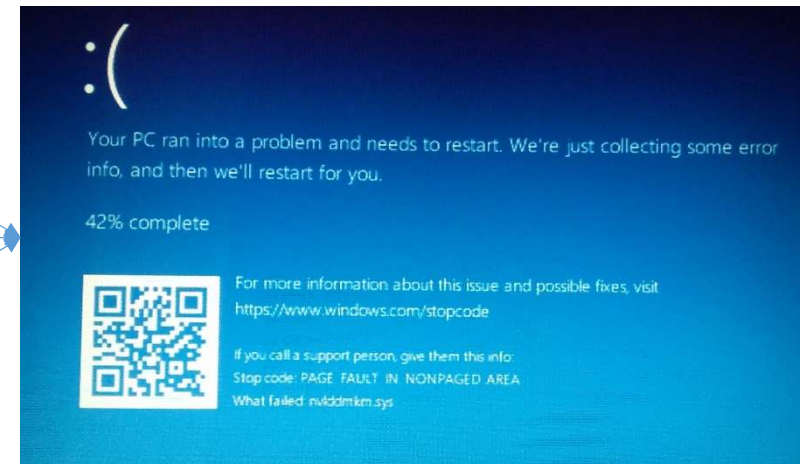
"I was coding a virus"



"Installed NVIDIA driver"



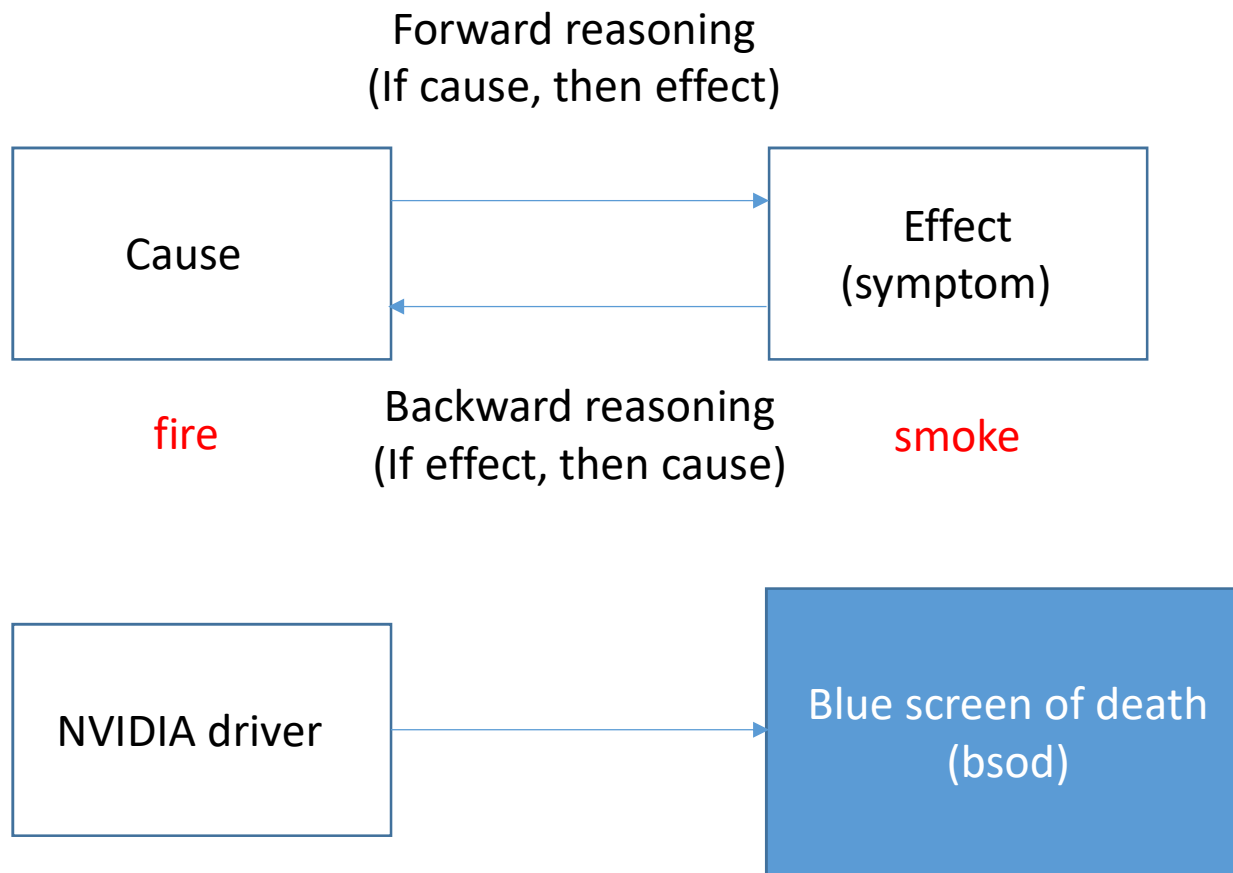
"I gamed so hard,
I overheated my CPU"



... you are NEITHER allowed to reason forward NOR backward
without empirical validation, excluding all causes but 1

Logic reasoning

Only allowed for known problems
(must have 1 to 1 relationship)
Else, guesstimate and test empirically



HCI: Q&A, break

Johan F. HOORN and Jeff TANG

Next: Analogy and creative problem solving



Analogical reasoning looks for relations between features and categories that are similar to relations between other features and categories

UNIX manual: “The clock signal is the heartbeat of your computer”

Features: clock signal = heartbeat

Analogical reasoning looks for relations between features and categories that are similar to relations between other features and categories

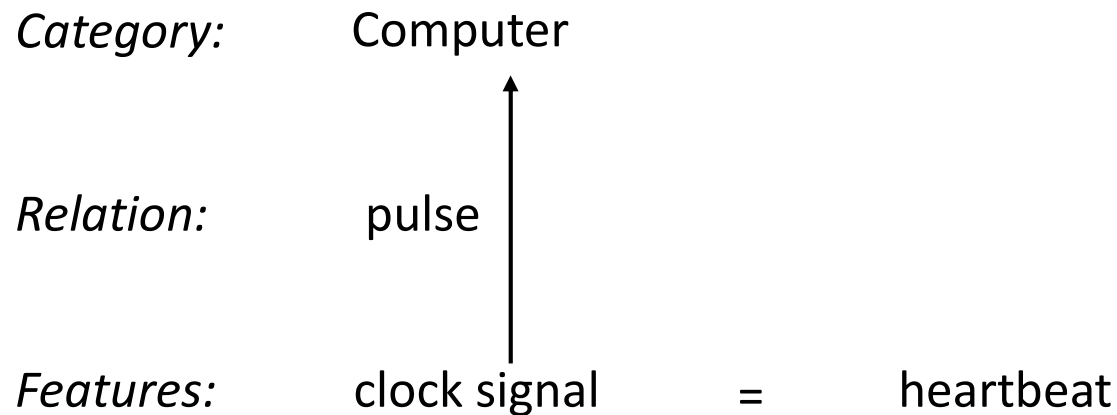
UNIX manual: “The clock signal is the heartbeat of your computer”

Category: Computer

Features: clock signal = heartbeat

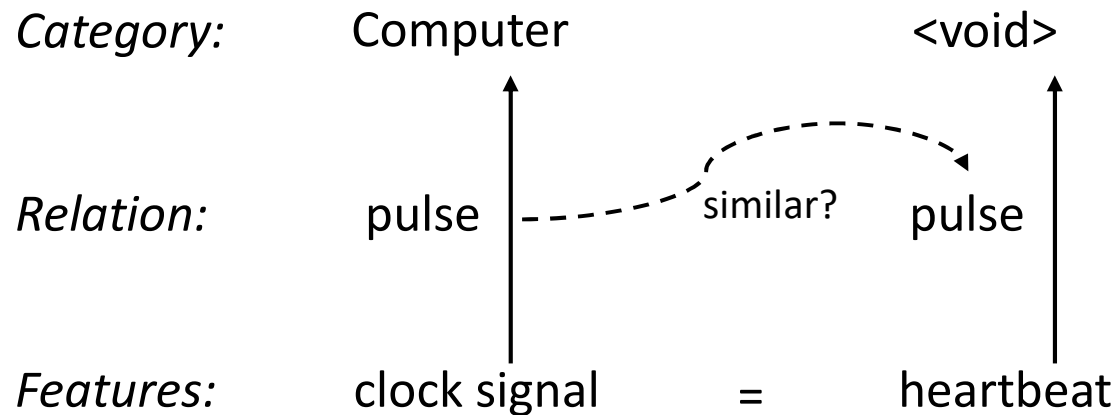
Analogical reasoning looks for relations between features and categories that are similar to relations between other features and categories

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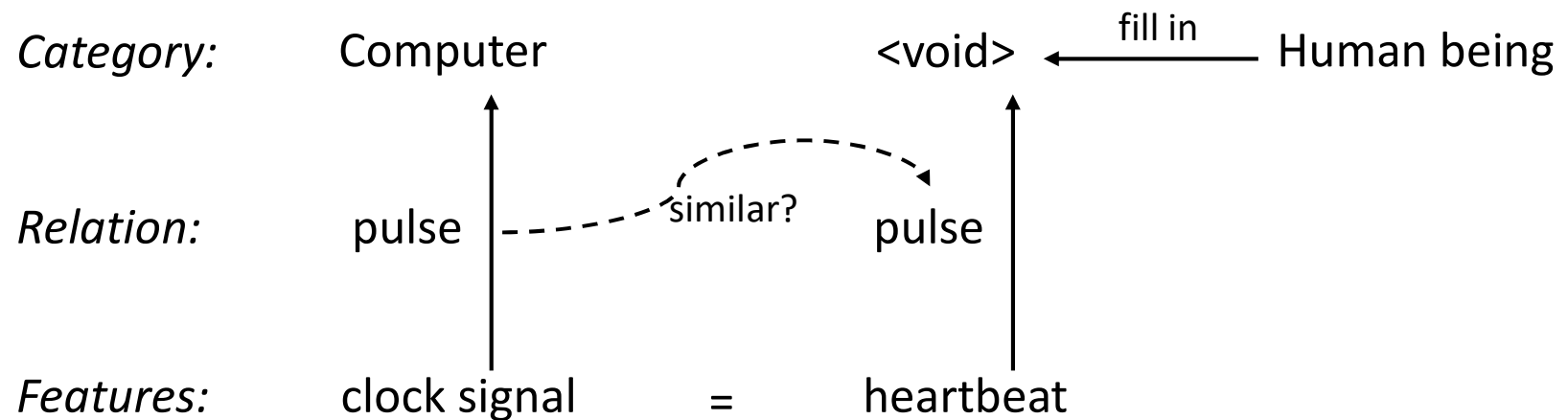
Analogical reasoning looks for relations between features and categories that are similar to relations between other features and categories

UNIX manual: “The clock signal is the heartbeat of your computer”



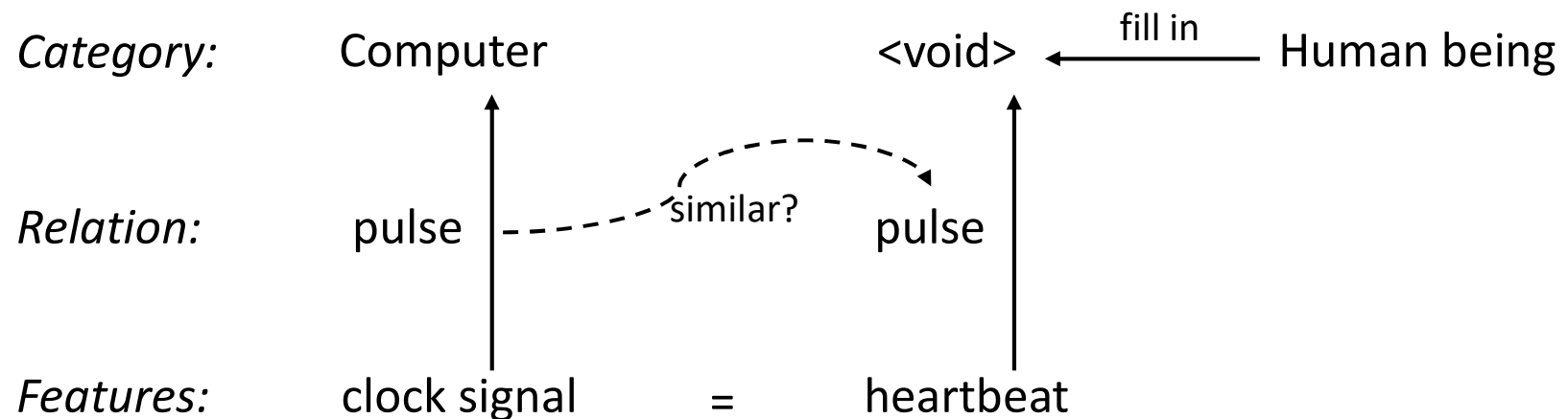
Analogical reasoning looks for relations between features and categories that are similar to relations between other features and categories

UNIX manual: “The clock signal is the heartbeat of your computer”



Analogical reasoning looks for relations between features and categories that are similar to relations between other features and categories

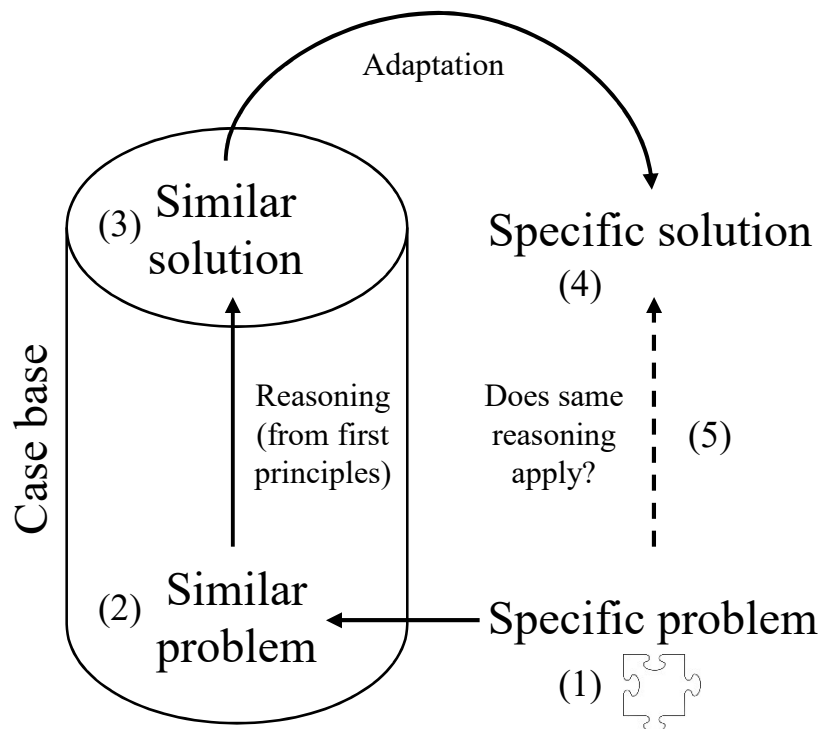
UNIX manual: “The clock signal is the heartbeat of your computer”



Clock signal : computer :: heartbeat : human being

Analogical reasoning is halfway intelligent and creative problem solving: It has a reasoning and an associative side

Case-based reasoning is a type of analogical reasoning



- 1) Axe imbalance and inefficiency
- 2) Who in nature works like an axe? (association)
- 3) How does he do it?
- 4) Adapt woodpecker neck form to fit my camp axe
- 5) It is probably for the same reason that it works so well (energy absorption)

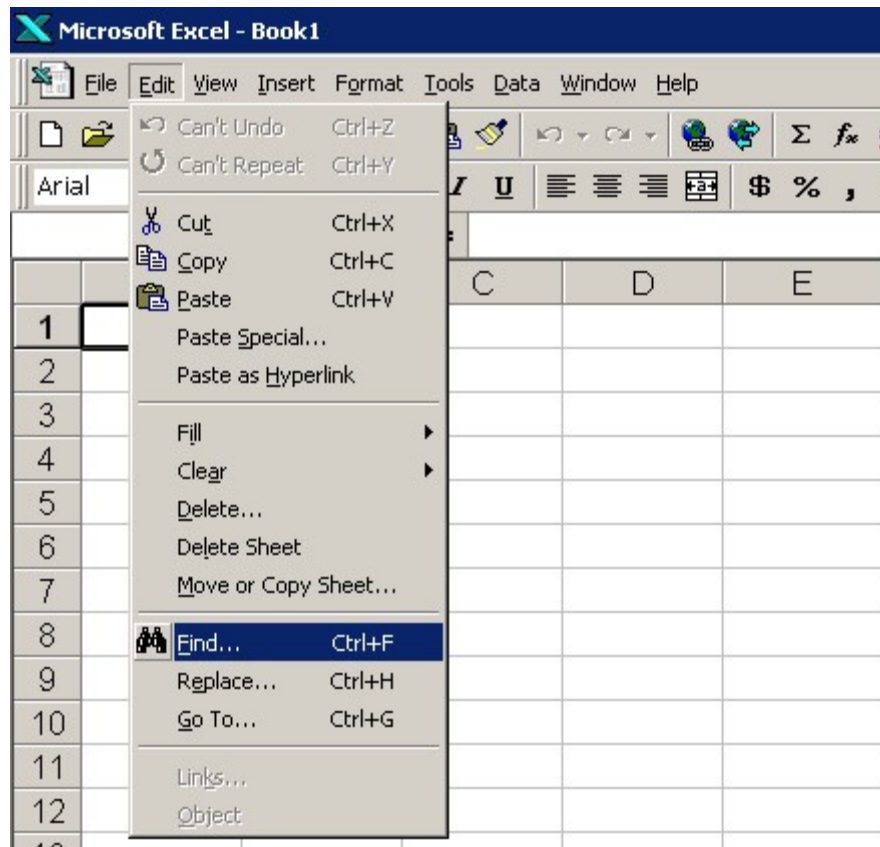
The creative combination is in the abstraction of finding similar/same principles in otherwise unrelated domains



Woodpecker-inspired camp axe for more balanced swing and more efficient blow

Old solution : old problem :: adapted solution : new problem

Random trial-and-error is a low level problem solving strategy but commonly found with non-expert users

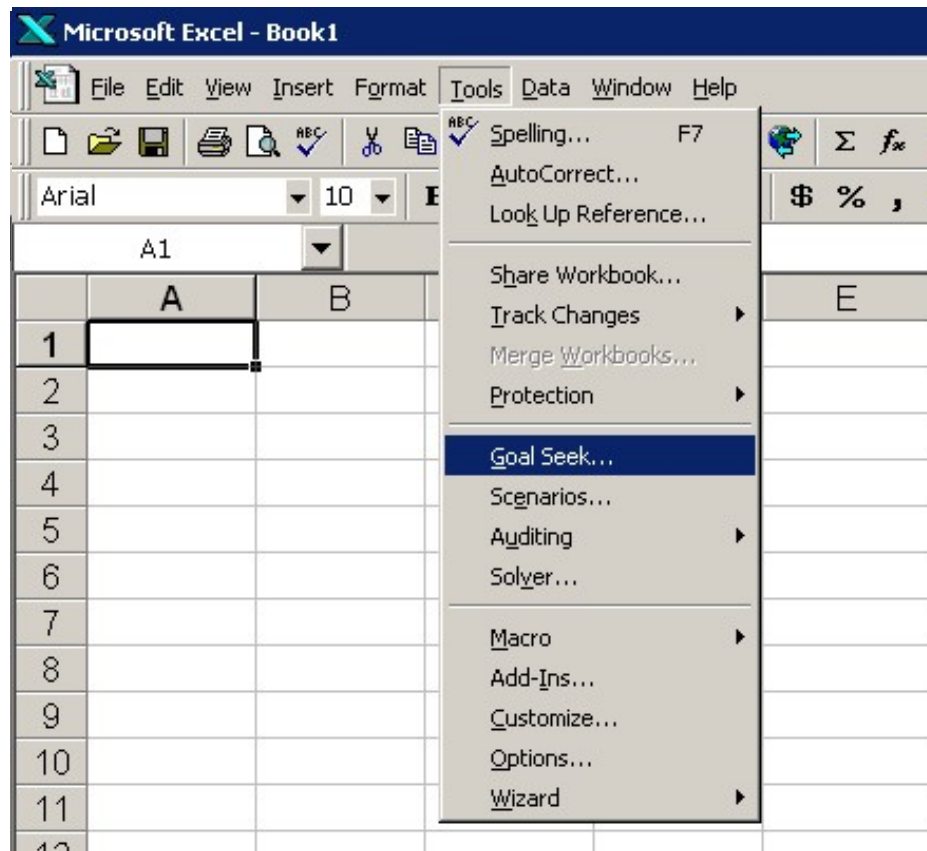


Novices follow first association.

Their problem: Find (Ctrl F) is not related in meaning to Edit so they start clicking some tabs

Violates the demand in intelligent problem-solving of systematic analysis and planned action. Yet, in creative problem-solving, serendipitous findings may occur

Random trial-and-error is a low level problem solving strategy but commonly found with non-expert users



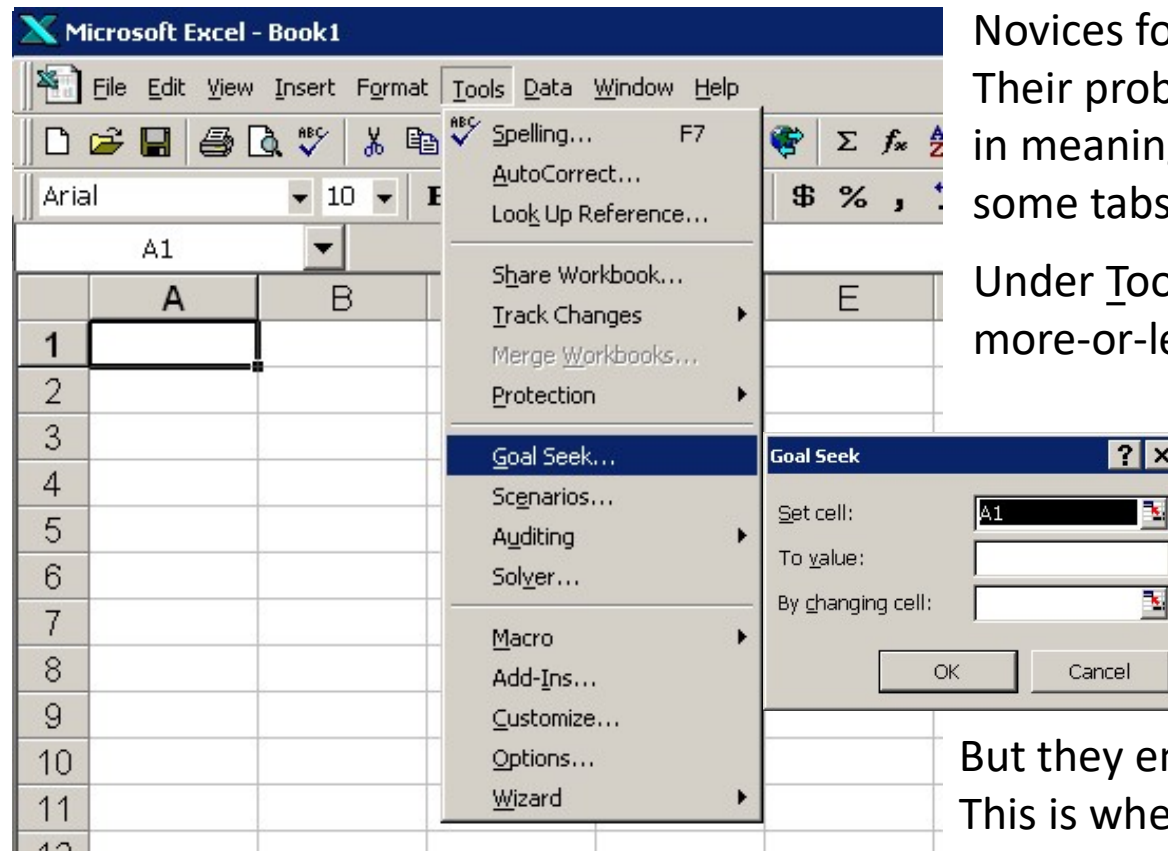
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Under Tools, Goal Seek... describes more-or-less what they want to do

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Novices follow first association.

Their problem: Find (Ctrl F) is not related in meaning to Edit so they start clicking some tabs

Under Tools, Goal Seek... describes more-or-less what they want to do

But they end up setting cells to values
This is where the wild clicking commences...

Violates the demand in intelligent problem-solving of systematic analysis and planned action. In creative problem-solving, serendipitous findings may occur

Problem solving strategies

Intelligence

Rule following
Difference reduction

Means-end analysis
Forward and backward reasoning

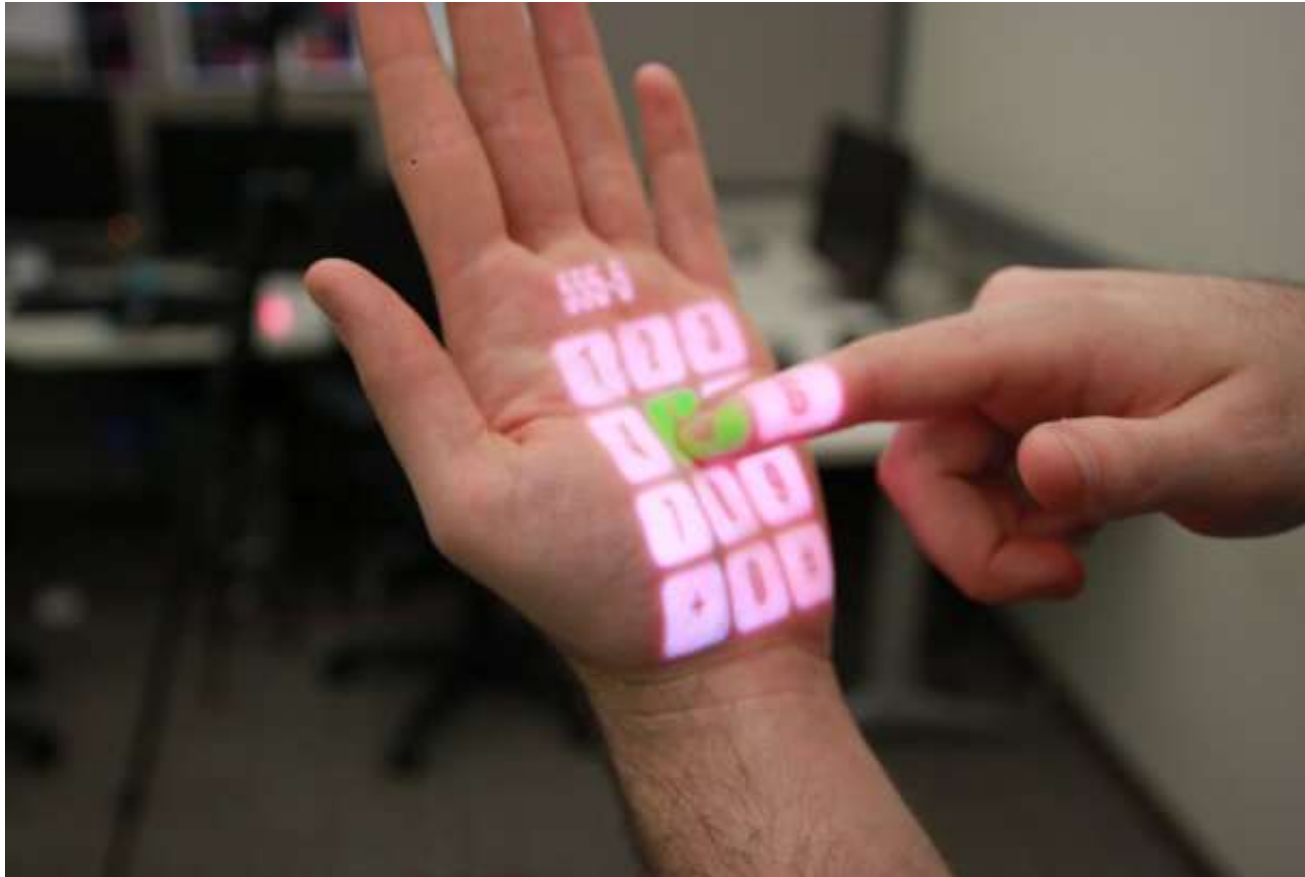
Creativity

Analogy
Random trial and error

Creative problem solving

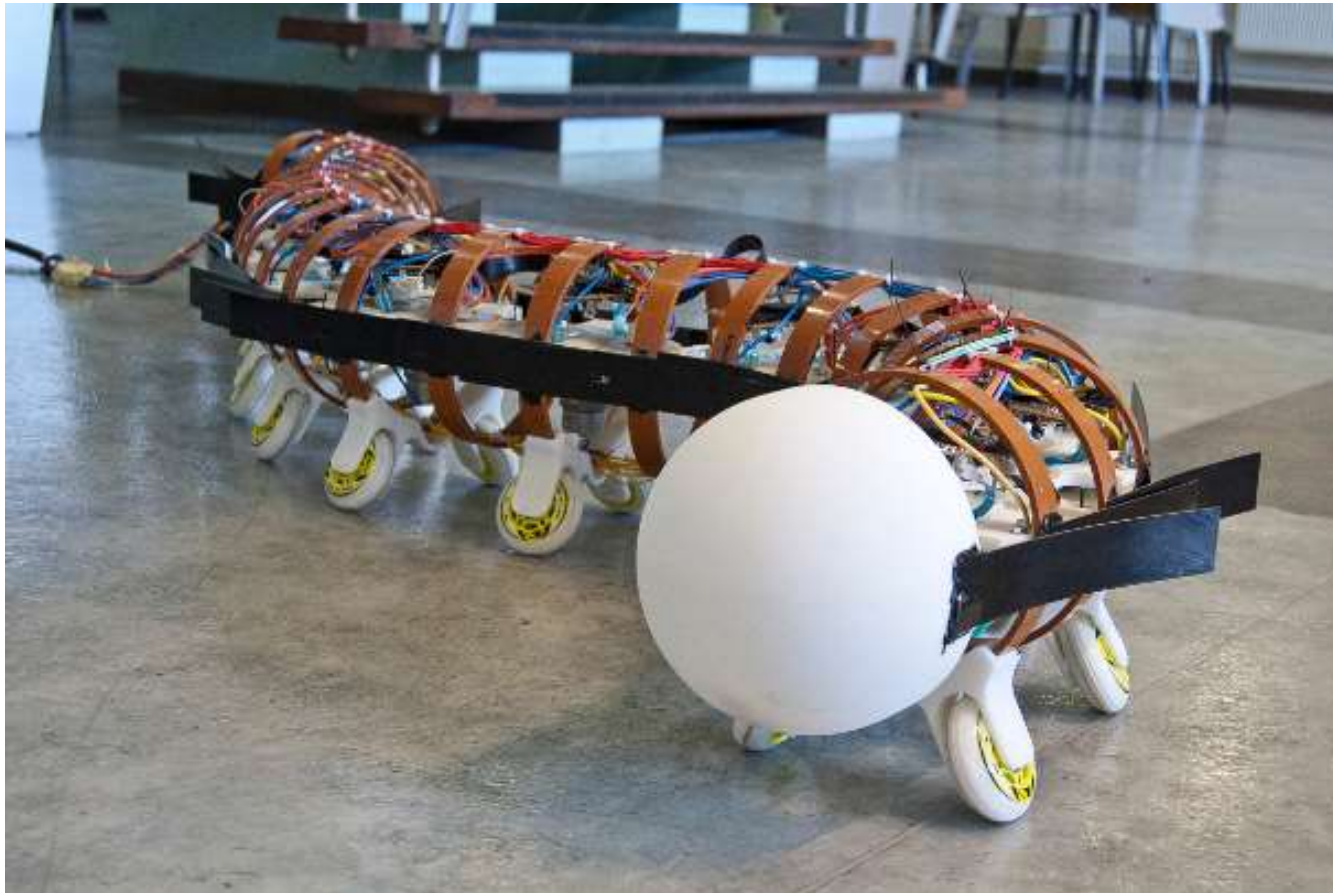
May lead to innovation

Creative problem solving: give a different function to the (standard) means that are available to solving the problem



OmniTouch augmented reality. Turns any surface - from a table to a hand - into a touchscreen. Wearable device that projects a graphical user interface, such as a computer desktop or a virtual keypad, onto any surface. Users are able to interact with these virtual interfaces using their fingers, which are tracked by sensors built into OmniTouch, in the same way they would use a touchscreen (Nick Heath, 2011).

Creative problem solving: The fusion of two worlds (e.g., biology (worm) and engineering (rescue robot))



University of Leeds (Jordan Boyle). Robot movement based on locomotion of real-life nematode worms. Possible applications: Searching for survivors in collapsed buildings or navigating gaps in damaged structures to deliver aid to individuals trapped in disaster zones (Nick Heath, 2011).

Creative problem solving: The fusion of two worlds (interface design and needle work)

Like a thimble. Solved through creative design



Fat-finger syndrome solved

Creative problem solving: Going beyond the obvious
(e.g., using the backside of a PC)



Crowd PC (Philipp Schaake). Runner-up prize in the Lifebook category of the Fujitsu Design Award 2011. Can be configured to resemble a notebook or slate, and can be tweaked so that the screen and keyboard are separate. The device's modifiable nature also allows it to be operated via the back cover when closed (Nick Heath, 2011).

Next Step: Creativity and Deep Learning (DL)

DL is a type of computer program (AI) from the family of neural networks

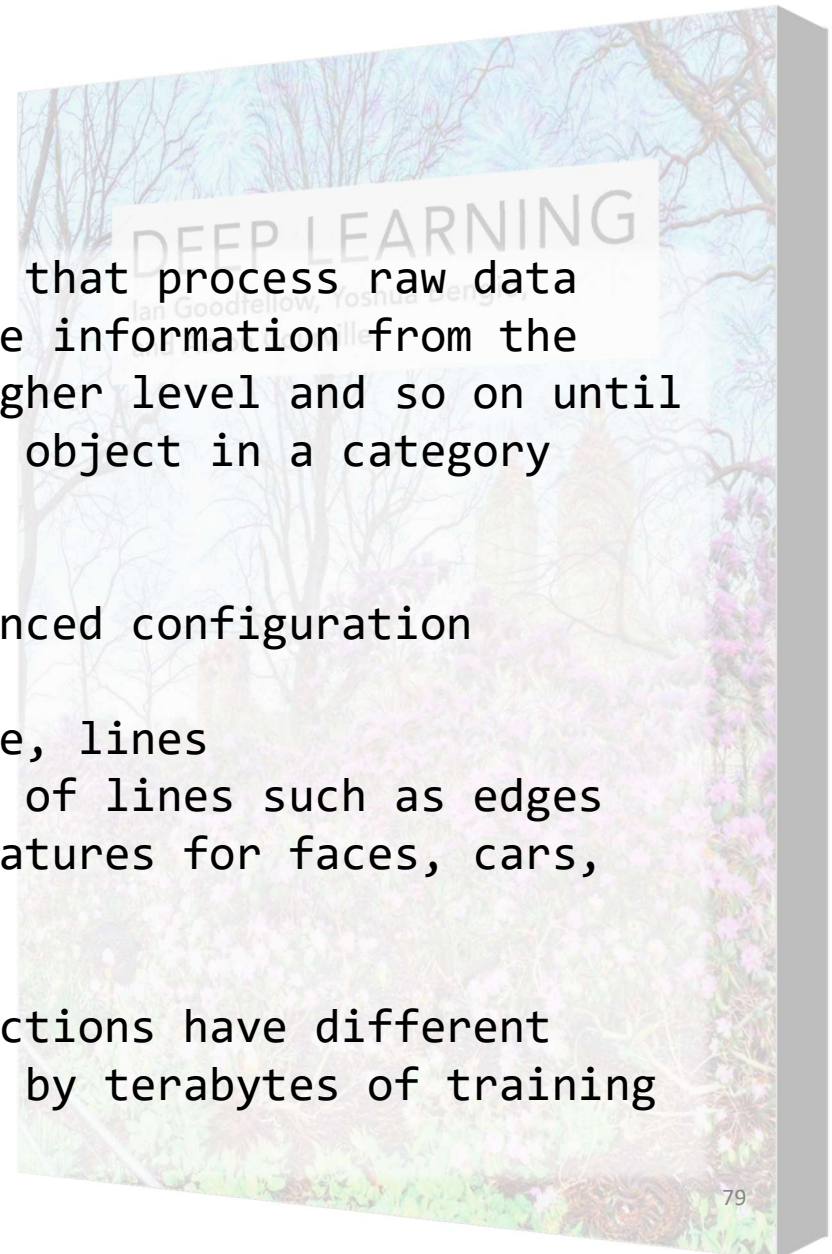
It has 10 to 20 layers of 'cells' that process raw data (e.g., one pixel). Cells integrate information from the bottom layers up into the next higher level and so on until they reach criterion and place an object in a category based on its features

DL is signal detection in an advanced configuration

Bottom layers detect, for instance, lines

Second layers search combinations of lines such as edges
Edges together make up typical features for faces, cars, elephants, chairs, etc.

All cells are connected but connections have different weights. Tweaking weights is done by terabytes of training data



Next Step: Creativity and Deep Learning (DL)

Which is the odd one out?



Weights indicate the relative importance of each cell

During training, the system gets thousands of pictures with the correct output it should produce

All weights are wrong but from the difference with the desired output the system adapts the weights to come closer (not this edge but that)

Through many many iterations, the weights are finetuned such that they produce 'fish' even in cases that were not in the training set. At that point, a robot with DL installed learns that you are a human and not a gorilla

(Courtesy GooglePhotos)

Next Step: Creativity and Deep Learning (DL)

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DIGITS

Google Mistakenly Tags Black People as 'Gorillas,' Showing Limits of Algorithms

By **ALISTAIR BARR**
Jul 1, 2015 3:40 pm ET

Google is a leader in artificial intelligence and machine learning. But the company's computers still have a lot to learn, judging by a major blunder by its Photos app this week.

The app tagged two black people as "Gorillas," according to Jacky Alciné, a Web developer who spotted the error and tweeted a photo of it.

Recommended Videos

1. Film Clip: 'Lost in Paris'
2. Film Clip: 'Cars 3'

Through many many iterations, the weights are finetuned such that they produce 'fish' even in cases that were not in the training set. At that point, a robot with DL installed learns that you are a human and not a gorilla

(Courtesy GooglePhotos)

Next Step: Creativity and Deep Learning (DL)

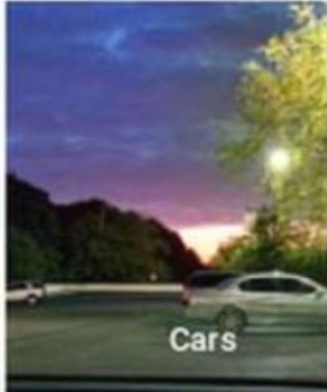
 **diri noir avec banan** @jackyalcine · Jun 28
Google Photos, y'all [redacted] up. My friend's not a gorilla.



Skyscrapers



Airplanes



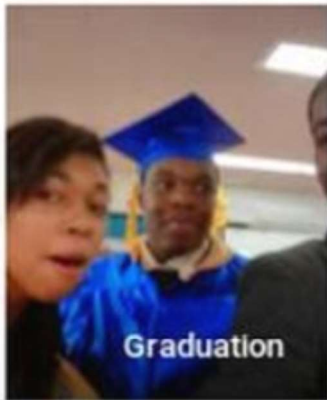
Cars



Bikes



Gorillas



Graduation

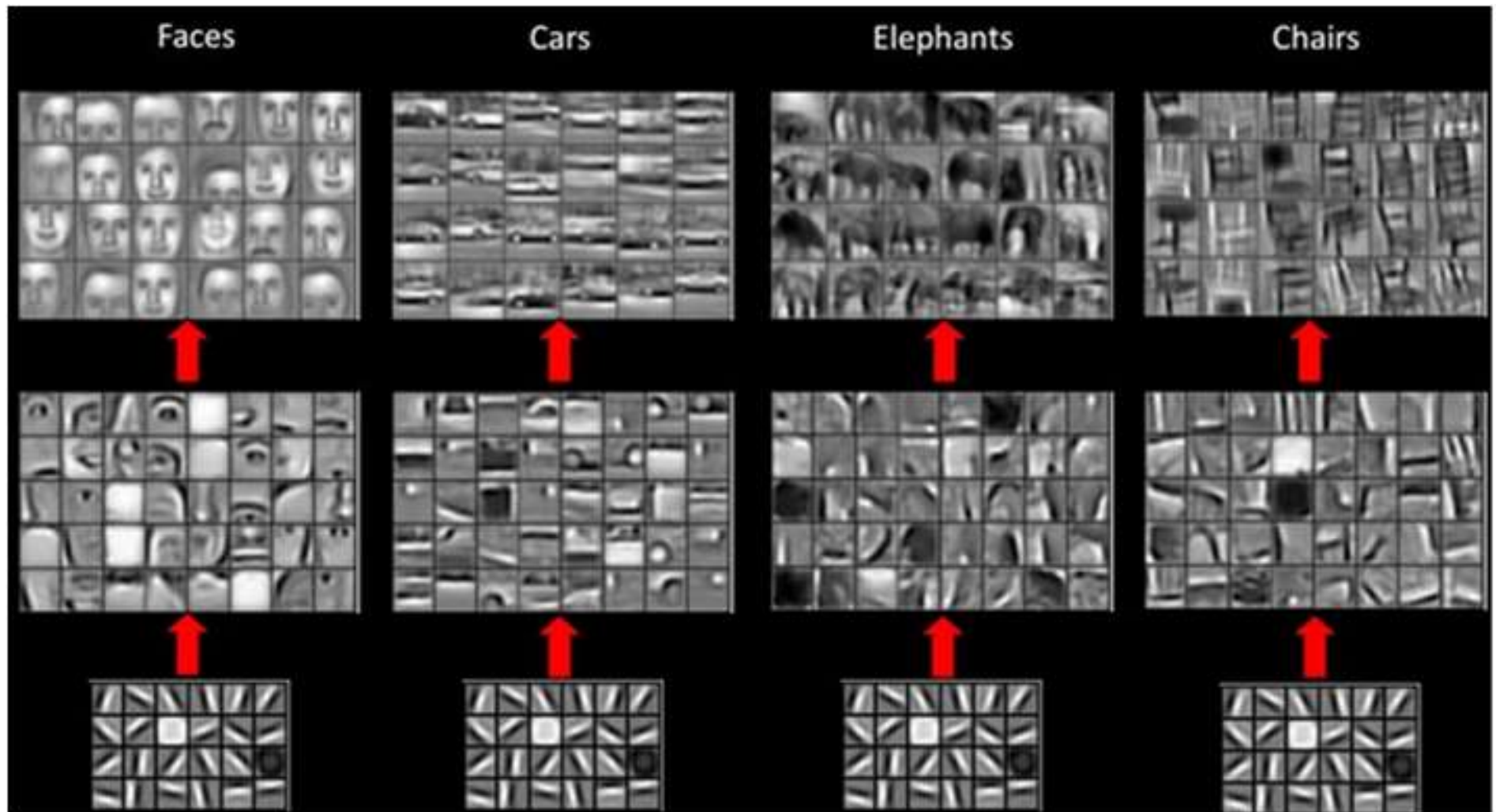
RETWEETS 1 031 FAVORITES 513



Why error is only error in a rigid system and not in creativity

The case of mistakes in Deep Learning

Categorization from the bottom features up

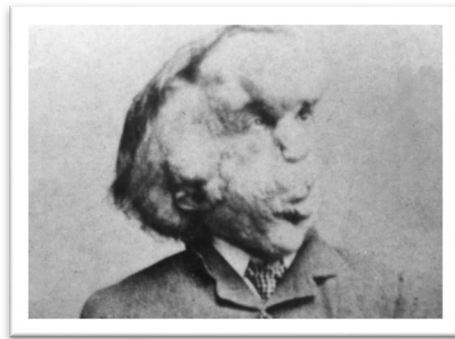


Why error is only error in a rigid system and not in creativity

The case of mistakes in Deep Learning

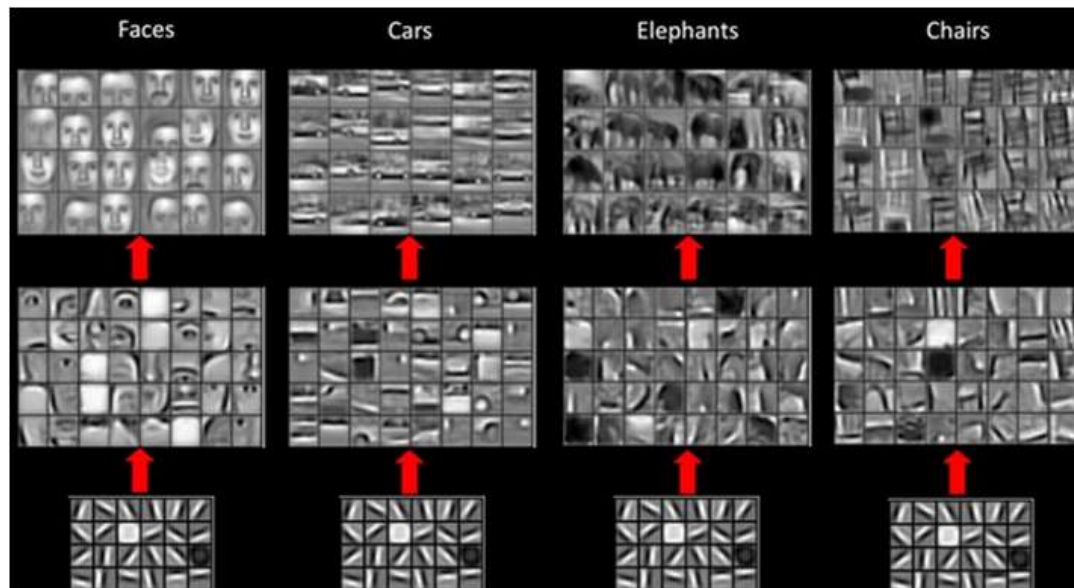
Categorization from the bottom features up
Features from the 'wrong' category may match on an abstract level

Elephant face?
Chair face?
Elephant chair?
Car chair!



Joseph Merrick
The Elephant Man

But DL may not be
that wrong:
Hybrids exist – more
than stiff taxonomists
wish for



Maximo Riera's Elephant Chair

Why error is only error in a rigid system and not in creativity

The case of mistakes in Deep Learning

Categorization from the bottom features up

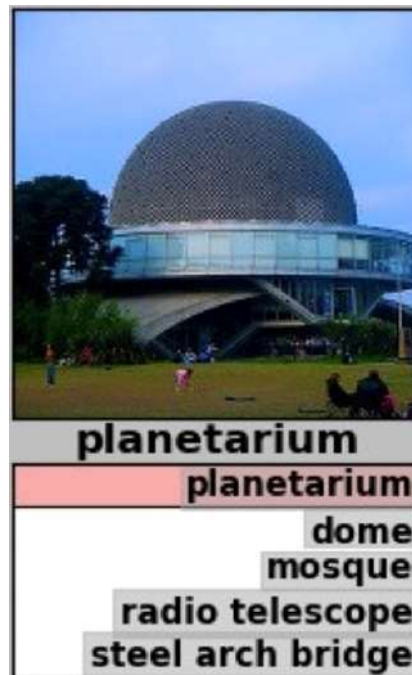
Creativity is matching features from the 'wrong' category on an abstract level

Mistake?



Machine

Mistake?



Machine

Not a mistake?

Why isn't this a pile of vegetables?



Arcimboldo

Human

Why might not these be great design ideas?



+

pencil sharpener

=

Interior Flaner Shop

★★★★★ 4.49 (5,778)

Shop Information

Contact Shop

Categories

« All Categories

Home & Office Supplies

Office Stationeries & Supplies

Writing Instruments

Pencil Sharpeners

Classic cameras! In fact pencil cutting ♪ camera pencil sharpner.

Kikkerland camera pencil sharpener (DTL)

KIKKERLAND

P 10X POINTS Hurry! 9 days left! Ends 17/04/2017 15:59 JST Buy it and earn 230 points!

Details

PayPal OK Details

€ 20.17 (¥ 2,376)

Actual price may vary according to exchange rates. Payment will be charged in yen.

[Estimate Shipping Fees](#)

★★★★★ 4.20 (5) [See Product Reviews](#)

Quantity:

* Please note 1 Non-mail flights

Add to Cart

This page is a simplified version of an original Japanese page. [See the original page](#)

Kikkerland camera Pencil Sharpener

Classic cameras! In fact pencil cutting ♪ camera pencil sharpner.

Kikkerland

Toy manufacturer founded in New York in 1992. Netherlands-born Jan van der lande started business from one vase was designed by Amsterdam-born friend.

Now also including the Netherlands, works of designers from around the world, especially in "daily life you can enjoy" engaged in product development with humor that stimulates the curiosity as a theme and design.

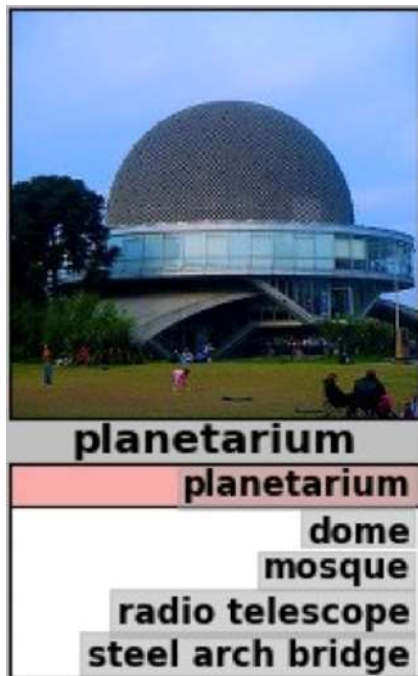
Lots of product introduction, announced Rainbow maker reflects prisms of rainbow-colored room attracting attention around the world

camera
pencil sharpner



Recommended for those who want to combine relaxing rooms!

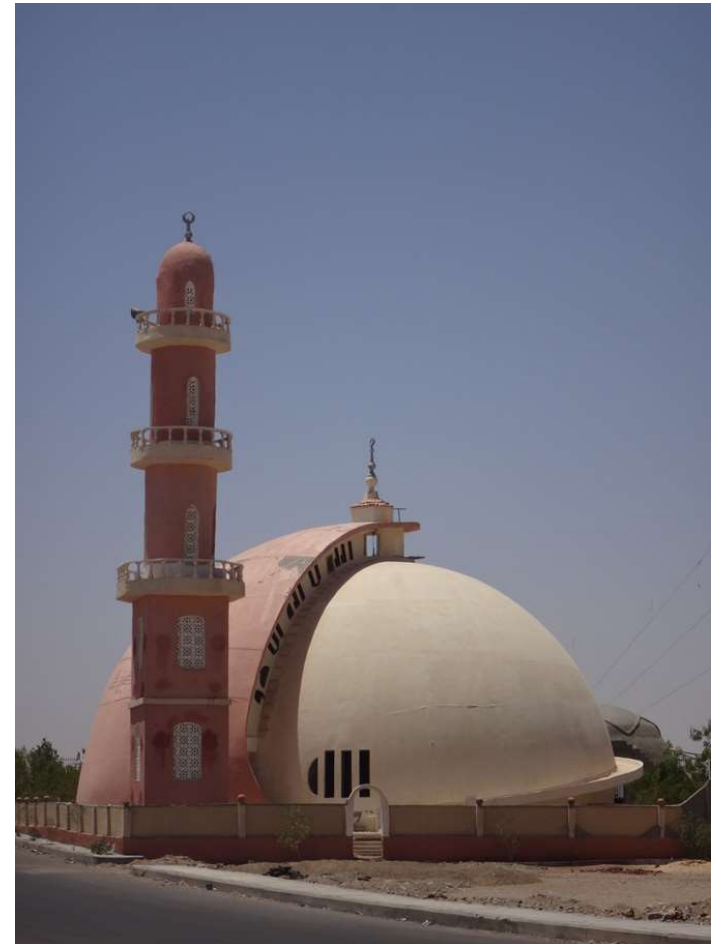
Why might not these be great design ideas?



+

mosque

=





Home Attraction ▾ Accommodation ▾ Restaurant ▾ Shopping ▾ Acti



National Planetarium Kuala Lumpur

Location: Kuala Lumpur

Location Info

Address: No. 53, Jalan Perdana, 50480 Kuala Lumpur, Malaysia

Office Number: +603 2273 4301 / +603 2271 4303

Opening Hours: 08:00am - 05:00pm (Monday-Friday)

Location

The National Planetarium is located conveniently inside the [Lake Gardens Kuala Lumpur](#). It was back in 1993 when the planetarium was first opened to the public. The planetarium has been intelligently designed and structured to mimic a mosque with a blue dome. In order to locate it in the large Lake Gardens is easy. The National Planetarium sits in between the Kuala Lumpur Bird Park and the National Museum.

“The planetarium has been intelligently designed and structured to mimic a mosque with a blue dome.”

HCI: Q&A, break

Johan F. HOORN and Jeff TANG



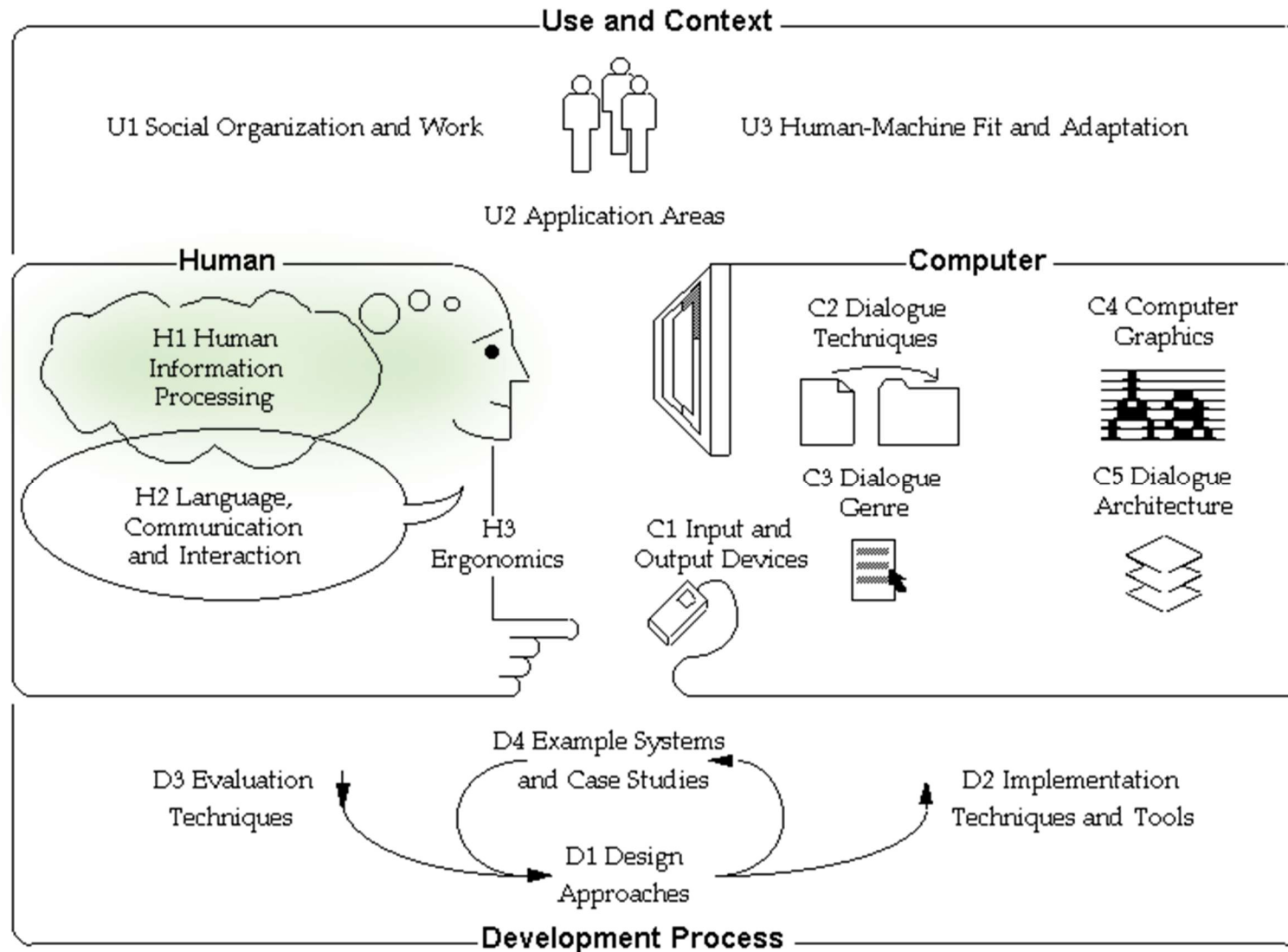
Human-Computer Interaction

Johan F. HOORN and Jeff TANG

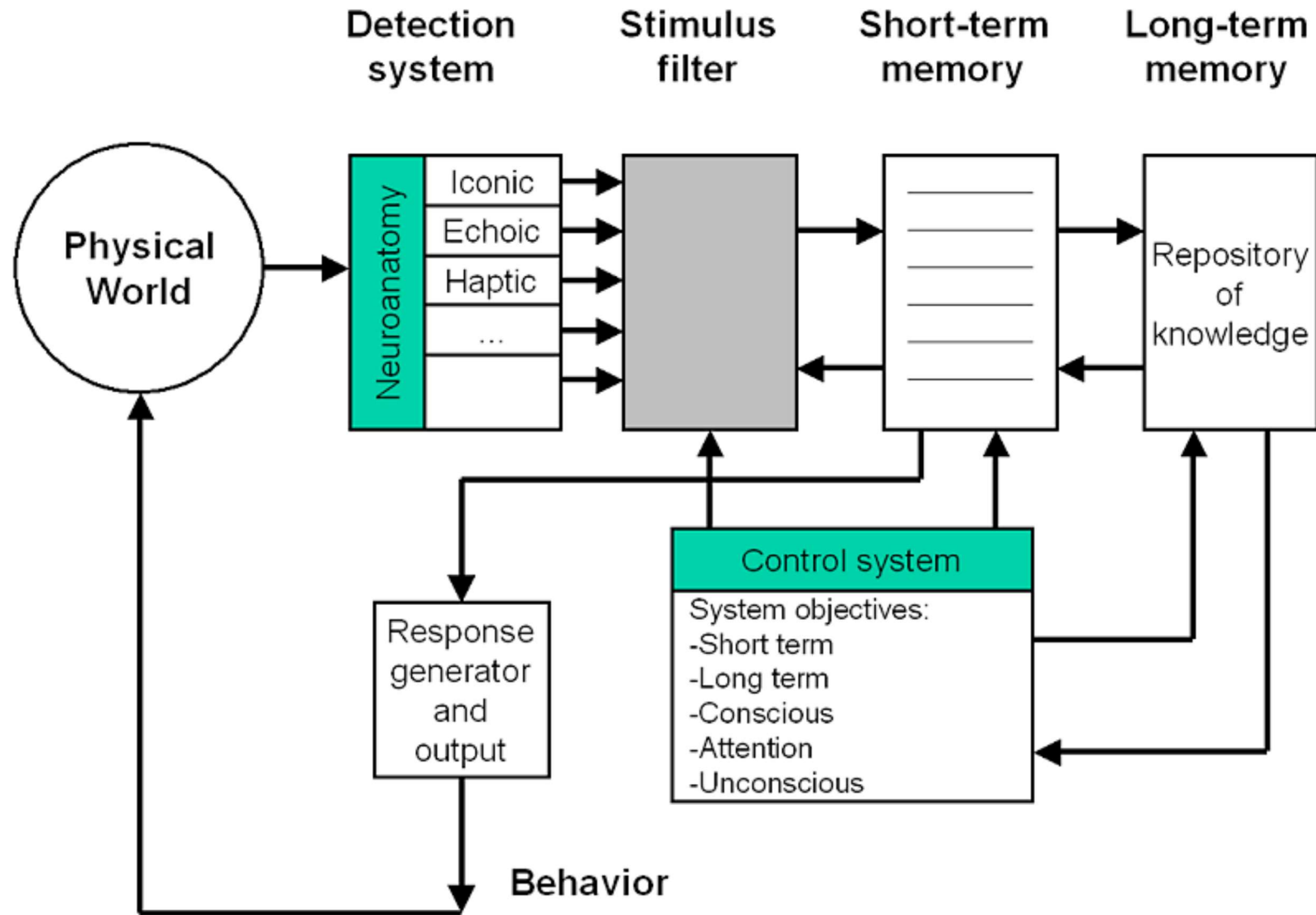
Human Information Processing, Part 2



Overview: Where were we?



Remember? The Human Information Processor



Problem representation

How you represent a problem (e.g., text or image) affects its solution

Text works better
to explain
abstractions:
'Communism'
'Democracy'

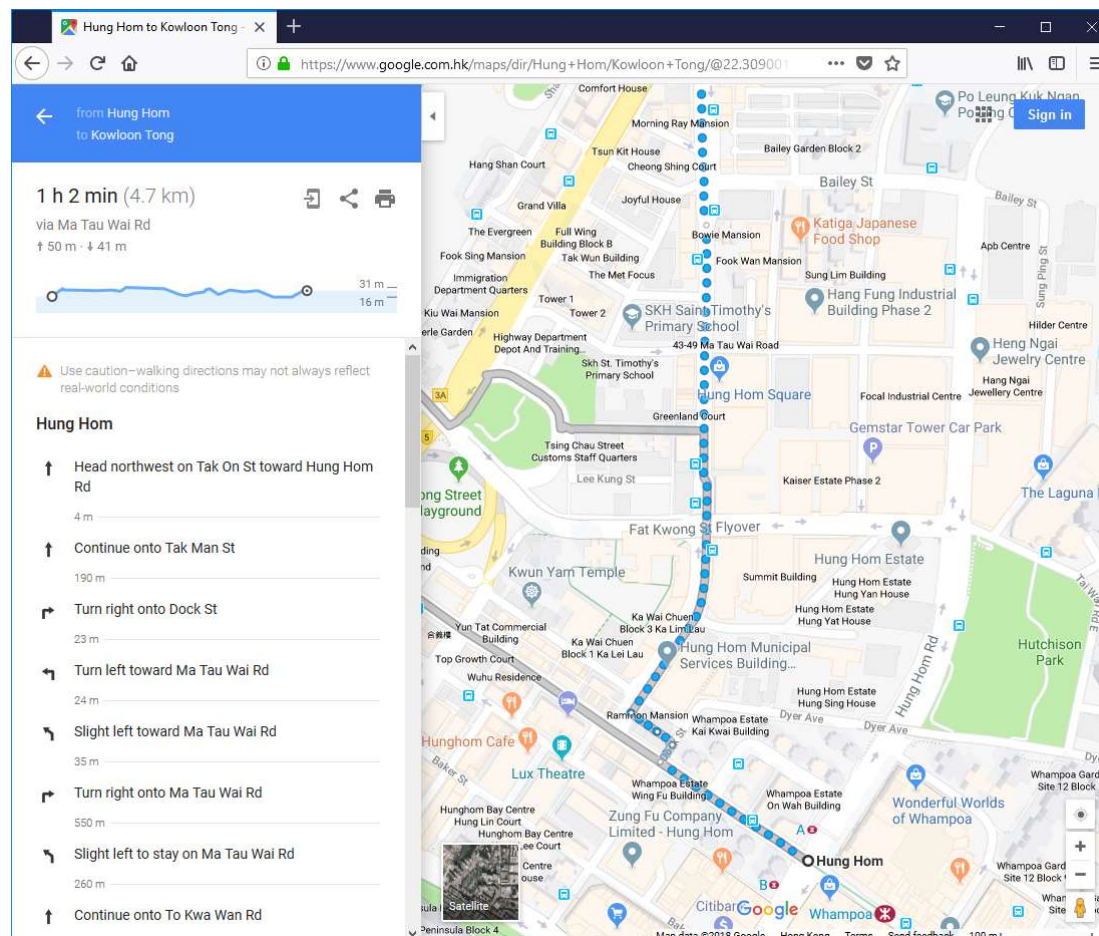


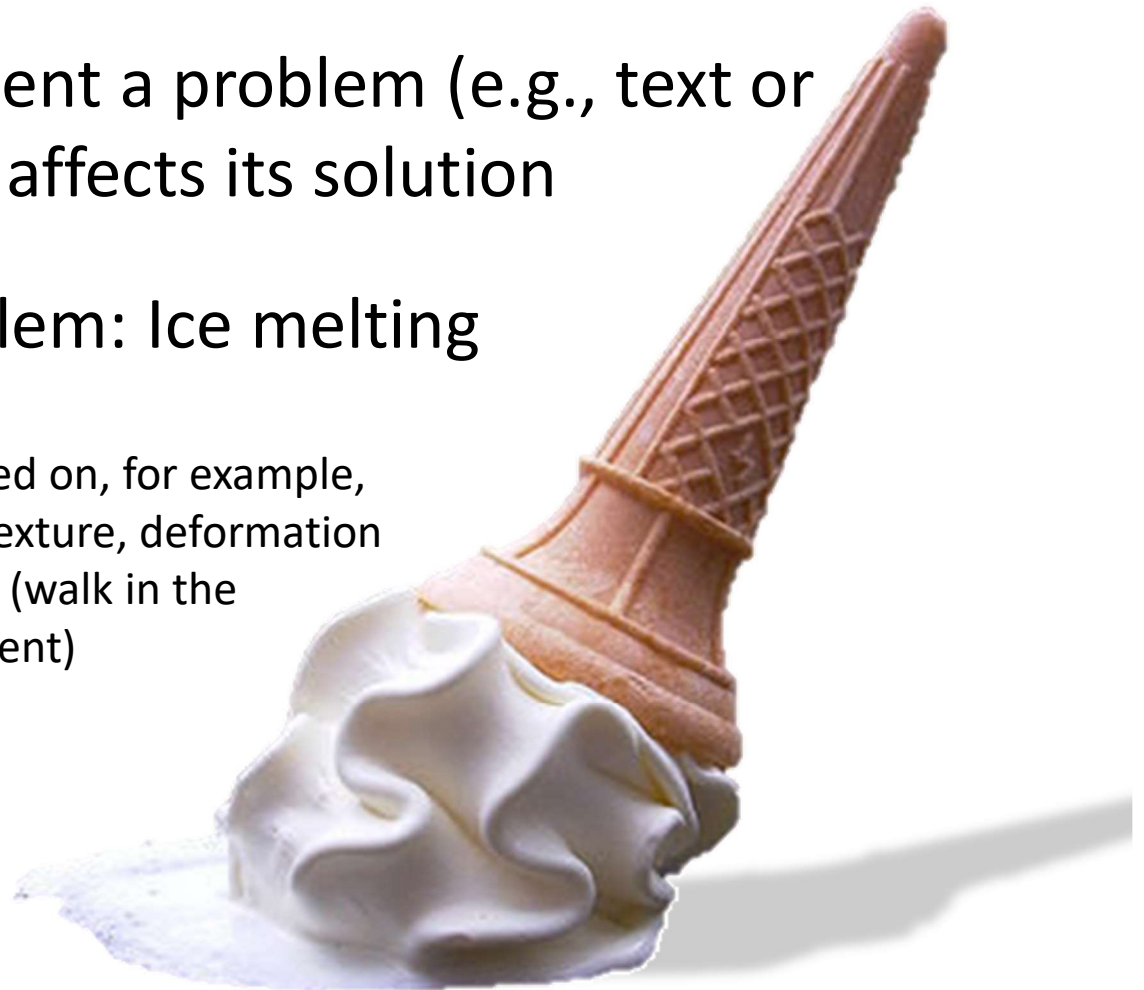
Image works
better for spatial
orientation and
localization. Where
is what?

Problem representation

How you represent a problem (e.g., text or image) affects its solution

Problem: Ice melting

Solutions become focused on, for example, color changes, shapes, texture, deformation of solids, sun, and stains (walk in the shadow, napkins, detergent)

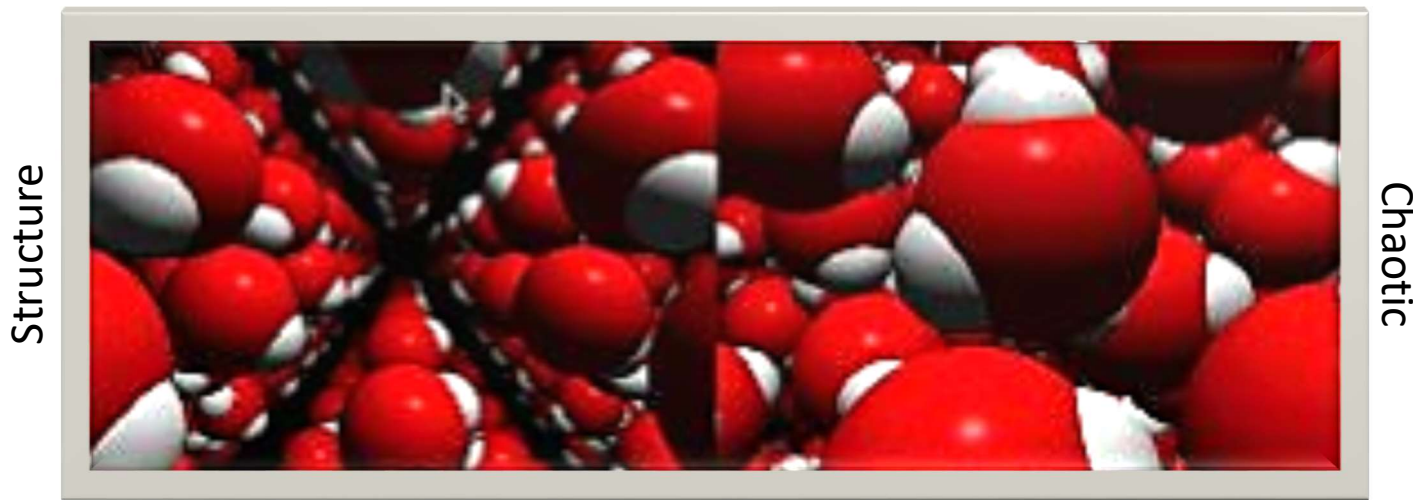


Surface level
Phenomenology

Problem representation

How you represent a problem (e.g., text or image) affects its solution

Problem: Ice melting



Solutions directed to fixing loss of structure without creating gritty ice crystals and managing processes and mechanisms of, for instance, atoms and ions (e.g., add emulsifiers, more sodium chloride)

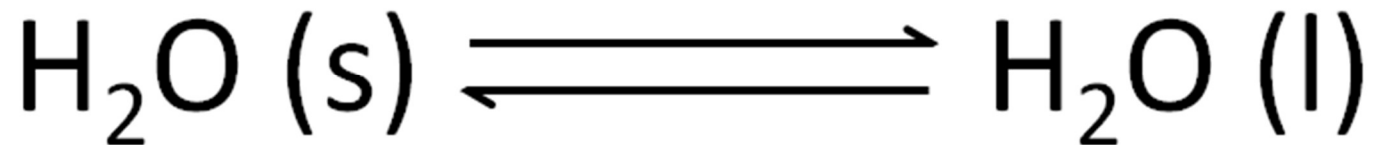
Structural level

Problem representation

3

How you represent a problem (e.g., text or image) affects its solution

Problem: Ice melting



Solutions derived from logics, reasoning, abstraction.

Protein BsIA binds with N_2 and O_2 (air), H_2O , and $-\text{COOH}$ (component of fat)

Symbolic level

Global warming solved: Scientists find ingredient that stops ice cream melting





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THE ESCAPIST PORTAL > THE ESCAPIST FORUMS > THE NEWS ROOM

Global Warming Solved: Scientists Find Ingredient That Stops Ice Cream Melting

THREAD INDEXSEARCH FORUMSCONTACT MODERATORSPOSTING GUIDELINES

PatrickJS

Muckraker





Posts: 260
Joined: 8 Jun 2015

1 POSTED: 1 September 2015 6:43 am

Global Warming Solved: Scientists Find Ingredient That Stops Ice Cream Melting

British scientists have isolated an ingredient that can keep ice cream from melting in the summer heat.

As the [U.N. convenes a closed-door meeting on climate change](#) this week, British scientists are one step ahead of the game on averting the worst kind of heat-related catastrophe: next summer, [your ice cream cone won't melt](#).

[Food researchers](#), from the Universities of Dundee and Edinburgh, have isolated a naturally occurring protein, BslA, that binds the air, fat and water in your triple scoop. In the end, this severely limits how quickly your frozen treat is going to melt.

The protein also keeps the ice cream frozen for longer, preventing gritty ice crystals from forming when you've kept the stuff in your freezer too long.

If you are interested in more nutritious food science news, you might like to [sample some bacon-flavored seaweed](#).

Cait MacPhee, a professor of [ice cream science](#) at Edinburgh and leader of the study, told the Telegraph, "we're excited by the potential this new ingredient has for improving ice cream, both for consumers and for manufacturers."

Personally, I'm not sure what the big deal is here. People have told me of this supposed 'melting,' but I've eaten three bowls of ice cream since I started writing this article and haven't noticed a thing.

Now, if only someone could invent ice cream that doesn't fall out of the cone just after you've paid for it...

Source: [Telegraph.co.uk](#)

[Permalink](#)



Image Source: McGill University

2 POSTED: 1 September 2015 7:10 am

Problem representation

How you represent a problem affects its solution,
ALSO IN USER INTERFACES!

Symbolic level
emphasises
form transition from
solid to liquid matter
as logic reasoning



Visible laboratory level
emphasises e.g., color
changes, formation of
solids, and boiling

Molecular level
(Quicktime animation)
emphasises changes in structure and processes of,
for instance, atoms and ions

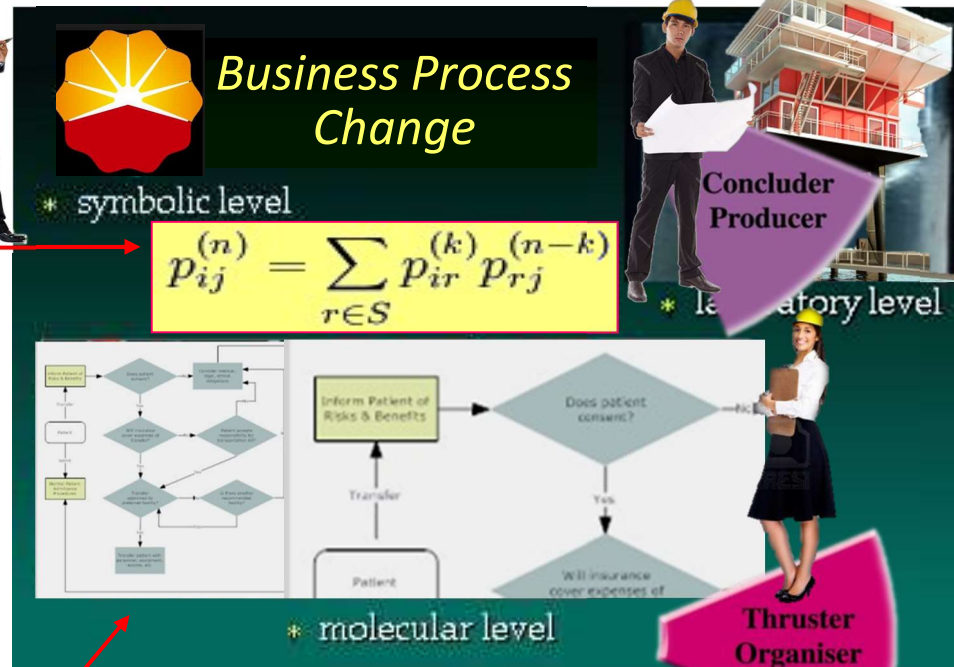
VisChem <http://www.uws.edu.au/science/Staff/rtasker/papers/images/interface.gif>

Change one level
and see how the
other levels change
as well. Insight!

Make multiple representations, each addressing another aspect

Problem: I want to solve this optimization problem

Symbolic level
emphasises
transition from
input to output
as logic reasoning



Visible laboratory level
emphasises,
for example,
type of workforce,
unforeseen mishaps,
work culture

Structural level
(Quicktime animation)

emphasises
changes in structure and processes of,
e.g., work units and their actions

So to improve the production, three sub-goals
took three different representations

As a designer, your clients will represent their problems in one specific way
Draw in more information than the problem representation that your client offers!

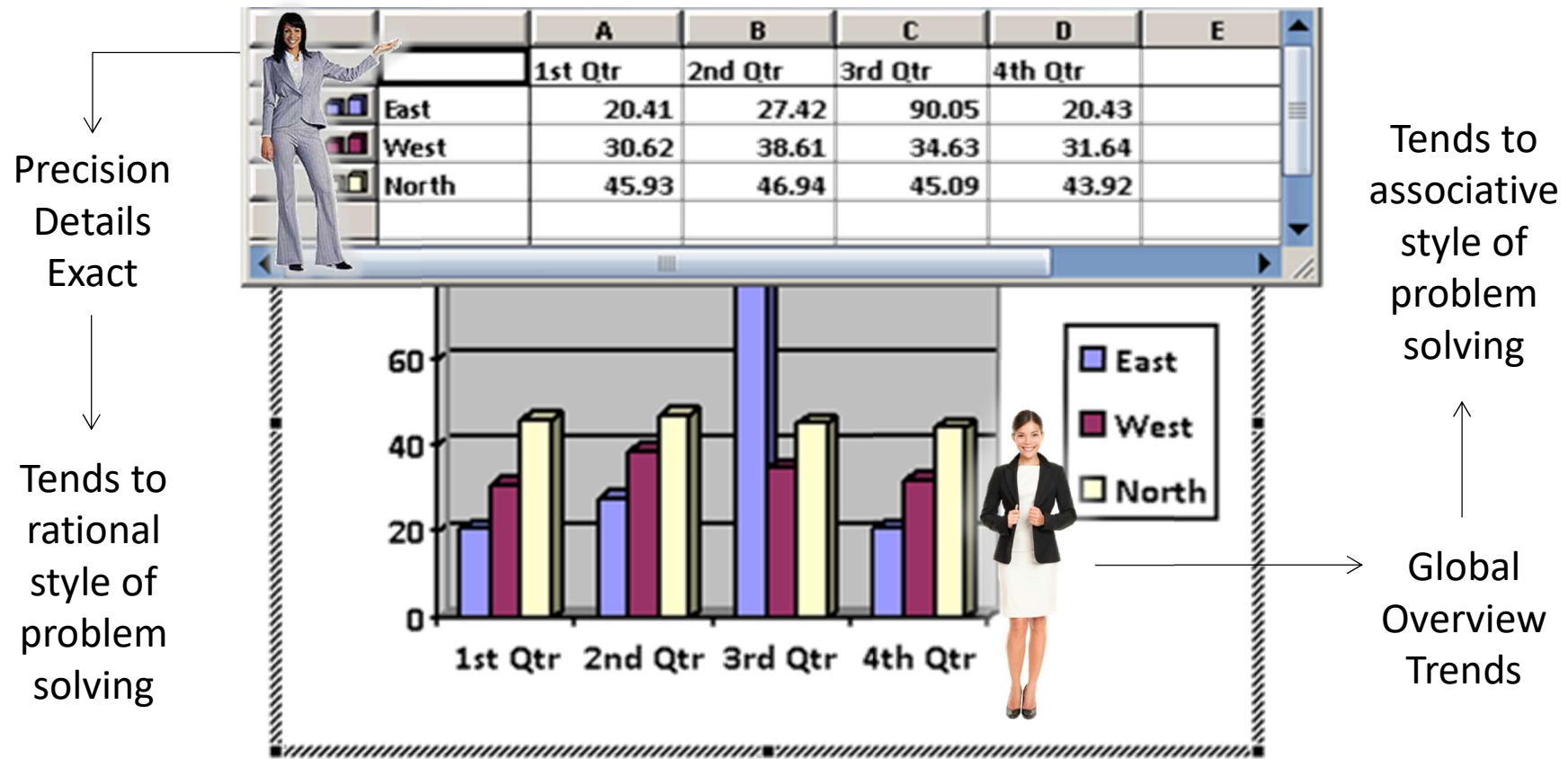
You copied everything from the Internet
We have a legal issue here

Let's try a new business model



In sum, how you represent a problem to your client (or your client to you), begs its solution

Each level of aggregating the data (cf. goal-decomposition) requires its own form of representation (cf. design)



Microsoft Word chart editor

Different mind sets solve problems differently

Who is your client (and who are you)?

The logic rationalist

~~COMP~~
Feeling happy in closed systems,
like a fish in a tank



The associative empiricist

~~SD~~
Feeling happy in open systems,
like a bird in a tree

Our students

The associative empiricist

Feeling happy in open systems

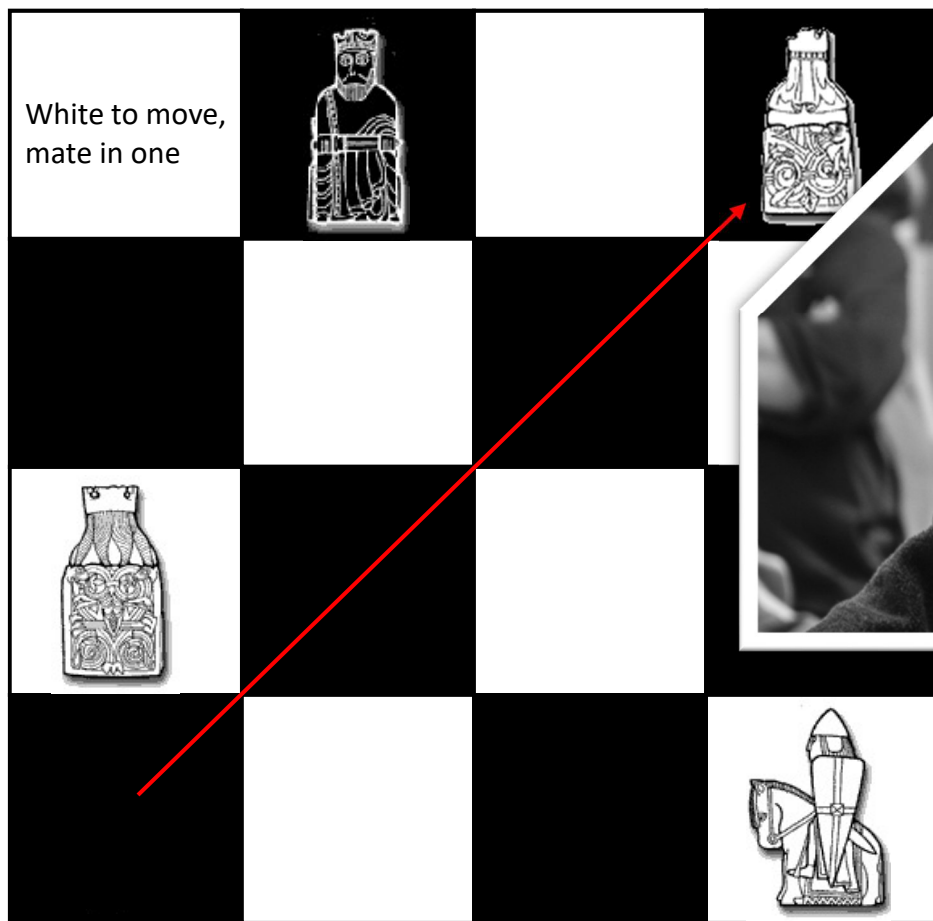


The logic rationalist

Feeling happy in closed systems



Structured decisions usually relate to well-defined problems and a more precise problem state



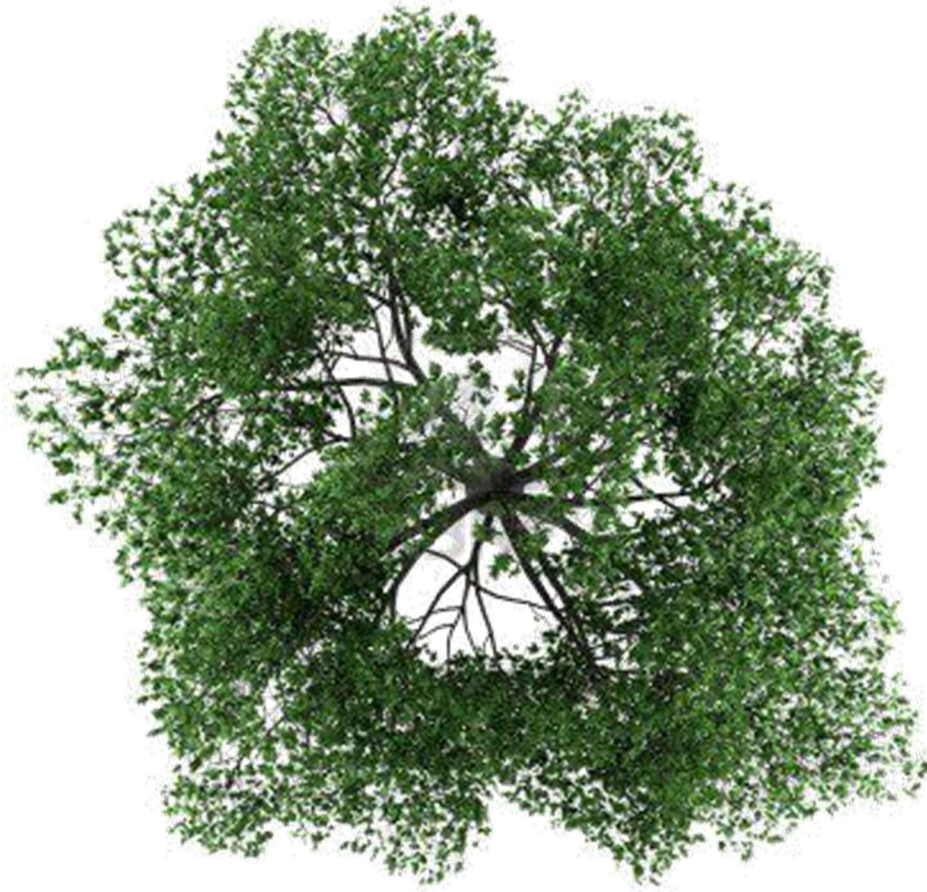
Unstructured decisions usually relate to ill-defined problems and a more vague problem state



What are you drawing?

Don't know.
I'm drawing.

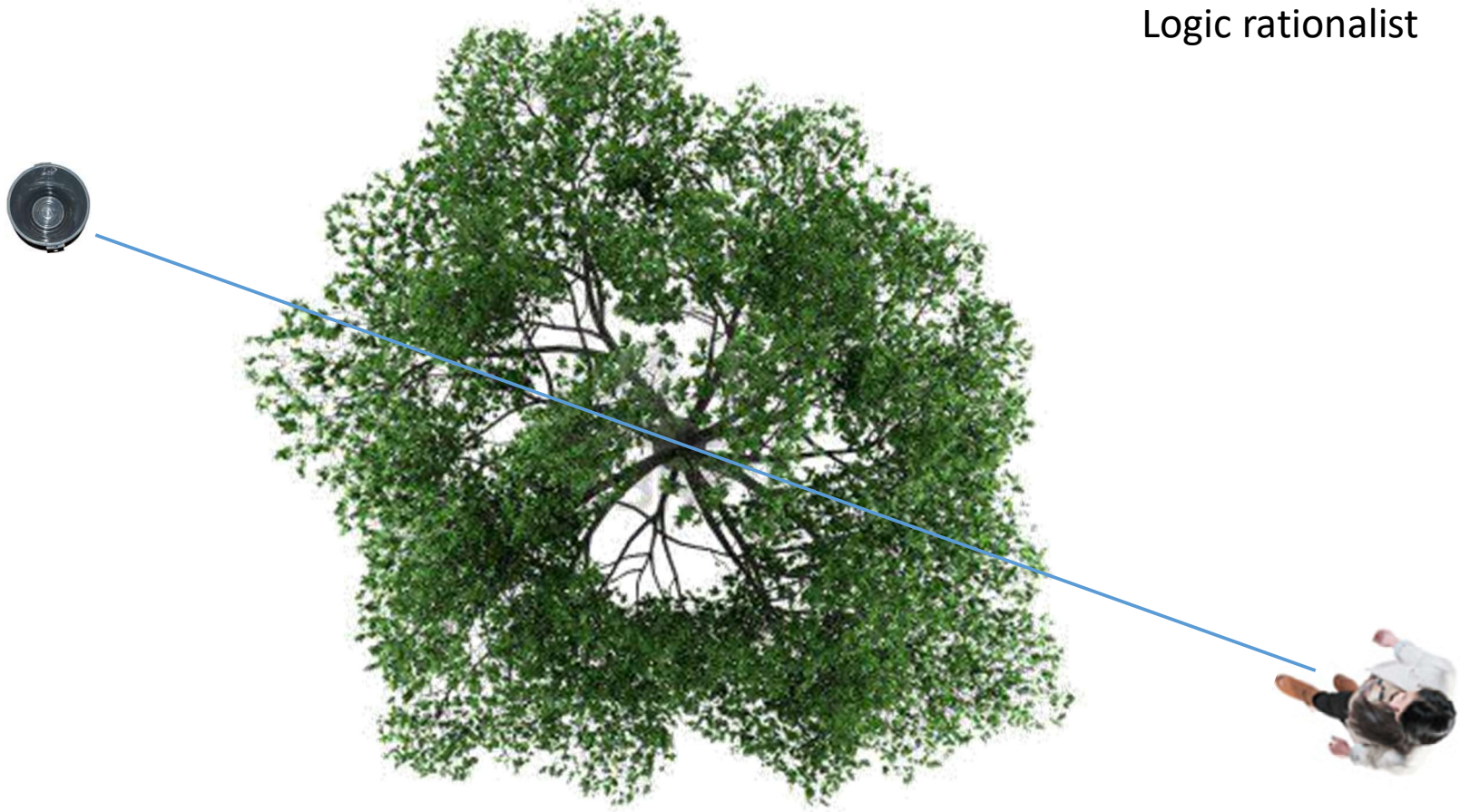
Tree - bin - me



Position yourself such that the bin, the tree, and you form a straight line

Tree - bin - me

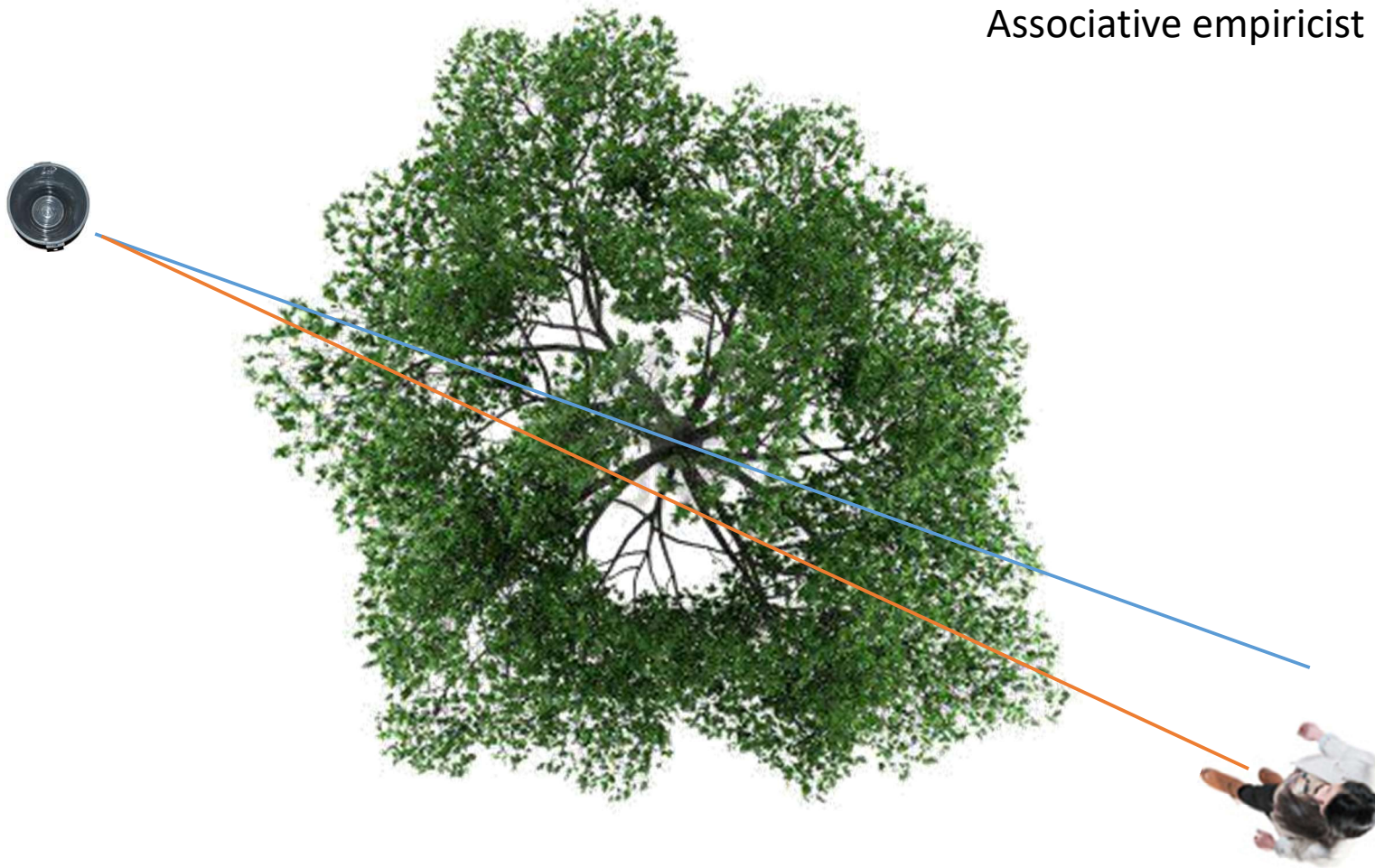
Logic rationalist



Position yourself such that the bin, the tree, and you form a straight line

Tree - bin - me

Associative empiricist



Position yourself such that the bin, the tree, and you form a straight line

Tree - bin - me

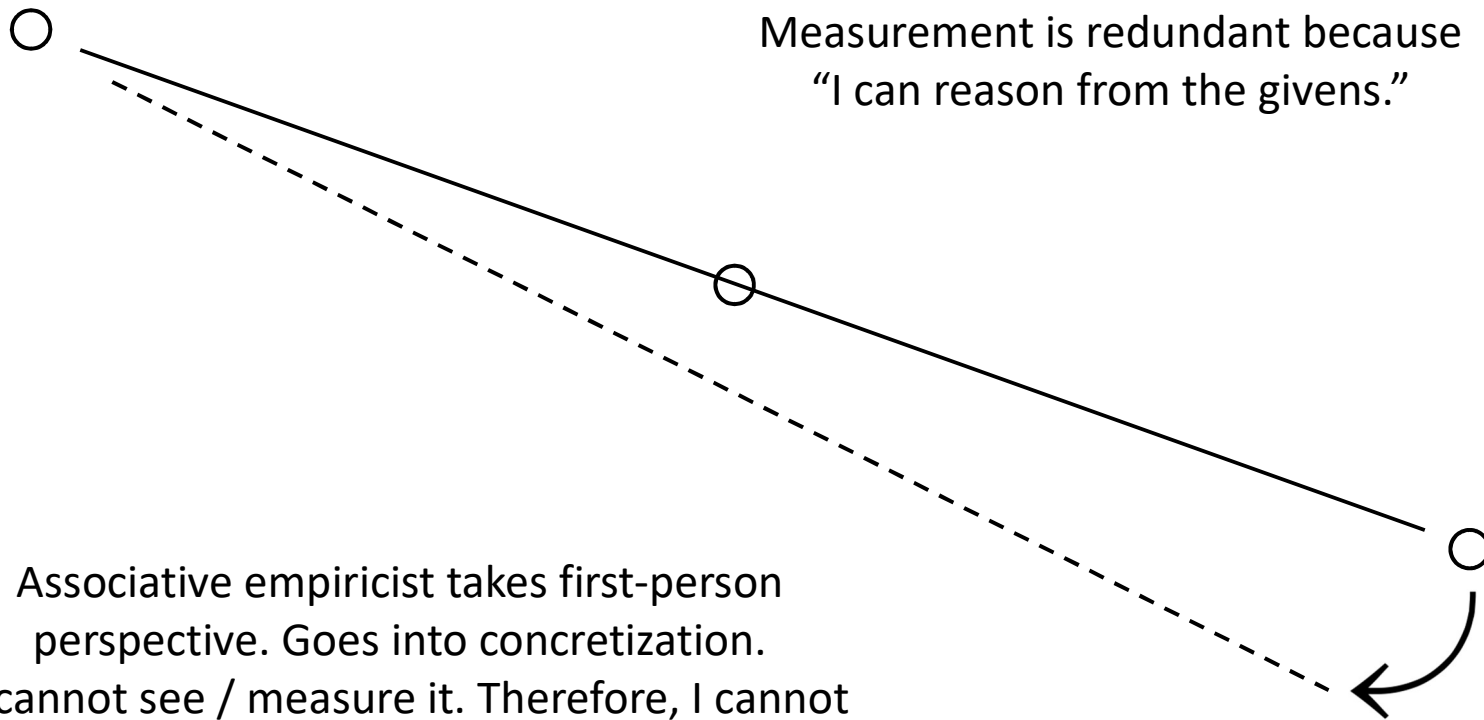
Logic rationalist takes person-independent overview.

Sees him/herself as a point in space.

Abstracts from the real world.

Measurement is redundant because

“I can reason from the givens.”



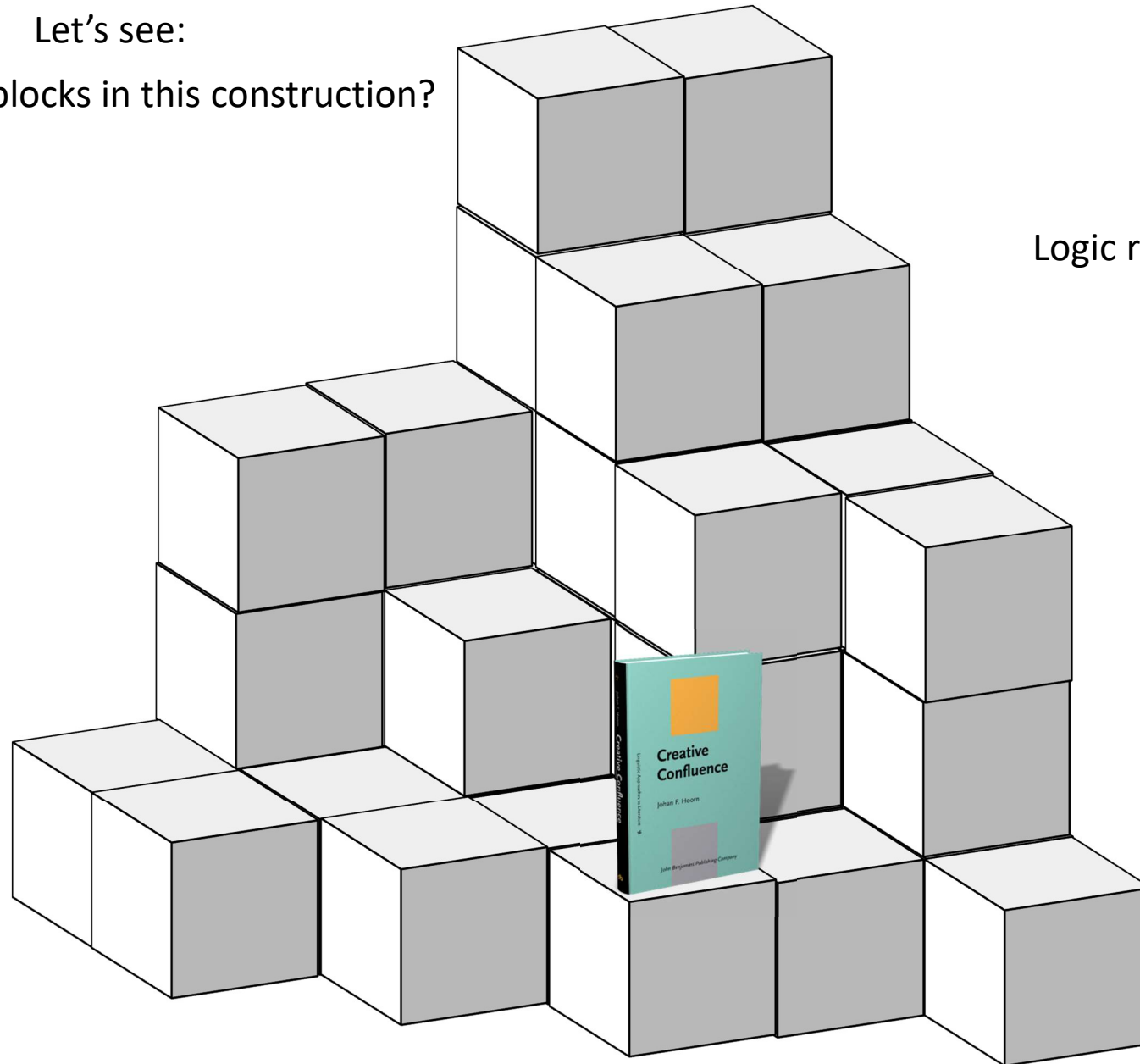
Associative empiricist takes first-person perspective. Goes into concretization.

“I cannot see / measure it. Therefore, I cannot answer the question.”

By now you probably think that
associative empiricists
are pretty stupid

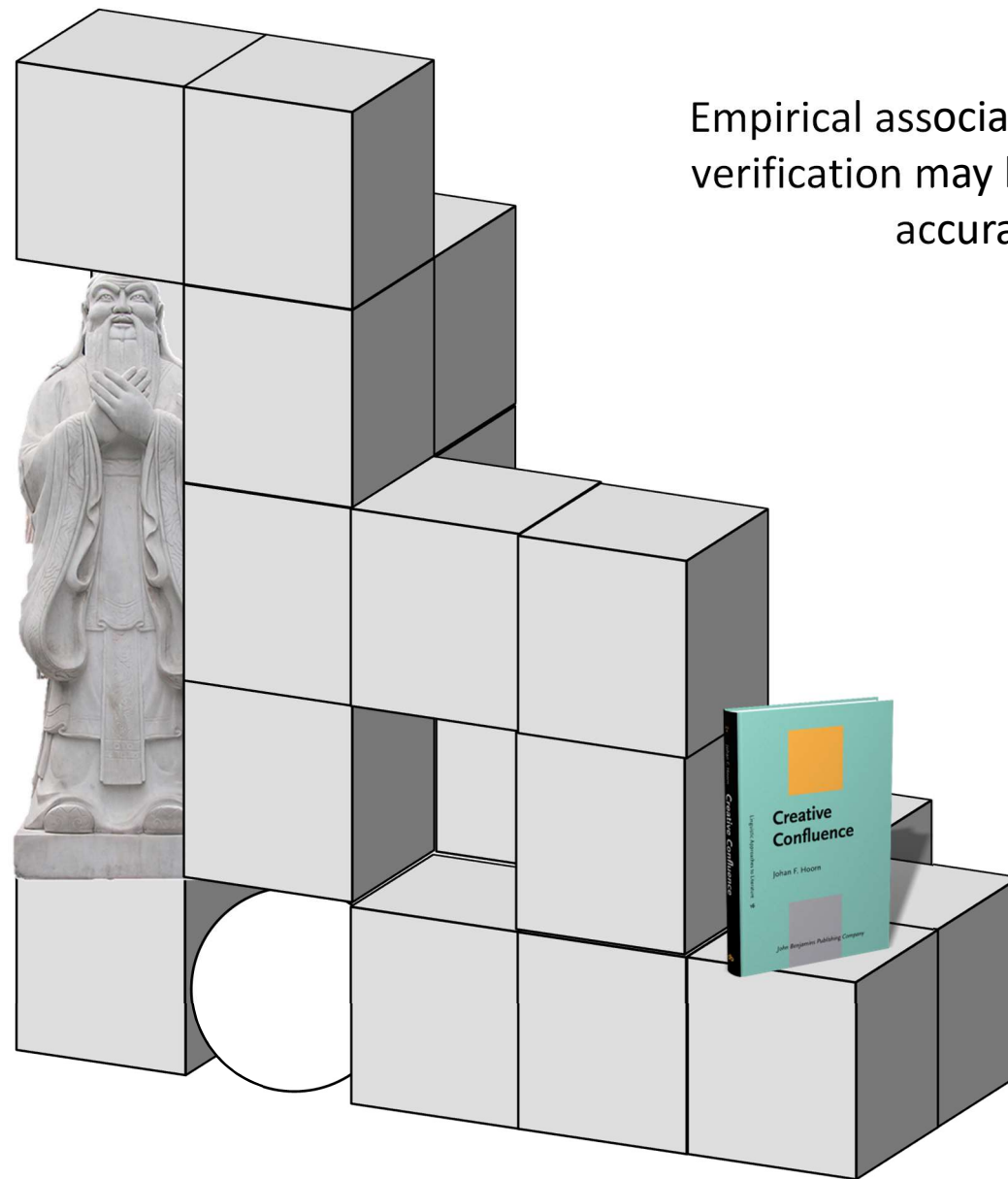


Let's see:
How many blocks in this construction?



Logic rationalism

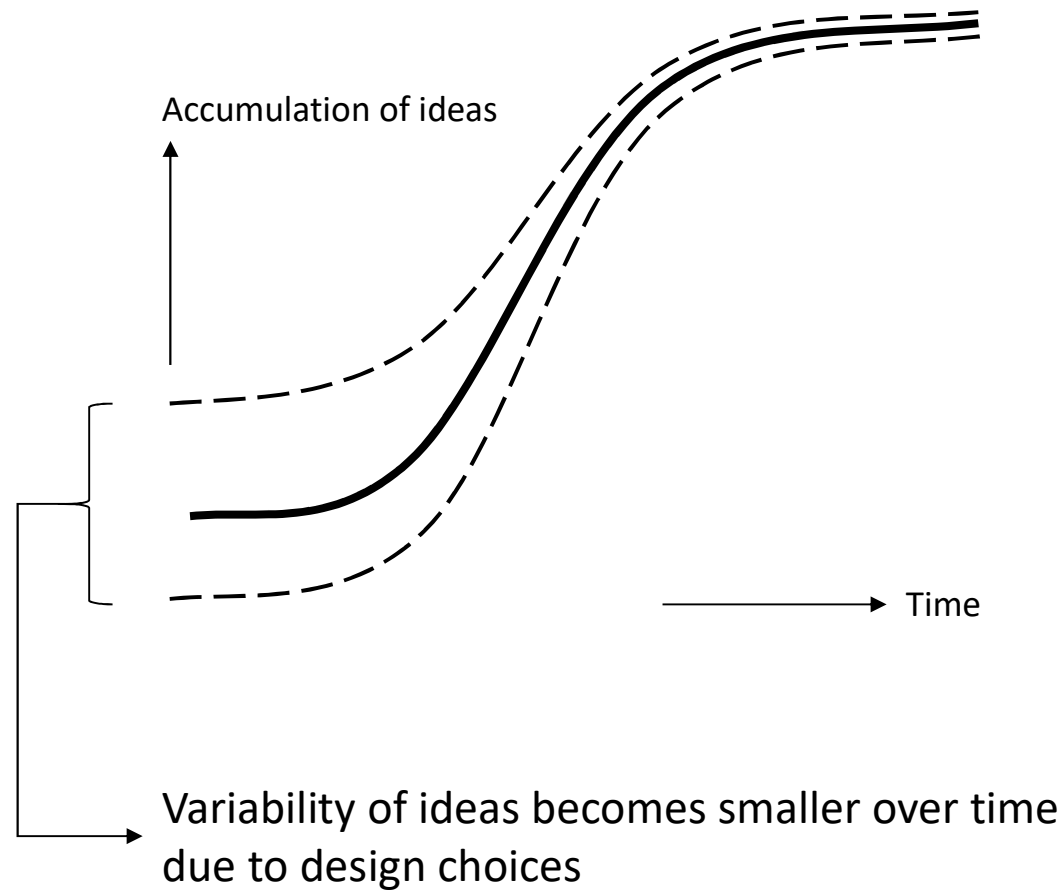
Hoorn, J. F. (2014). *Creative confluence* (p. 77). Philadelphia, PA: John Benjamins.



Empirical association and then
verification may lead to better
accuracy

Our students working together

Product life-cycle follows the Innovation S-curve



HCI: Q&A, break

Johan F. HOORN and Jeff TANG

Next: Response, motor output, Fitts' Law

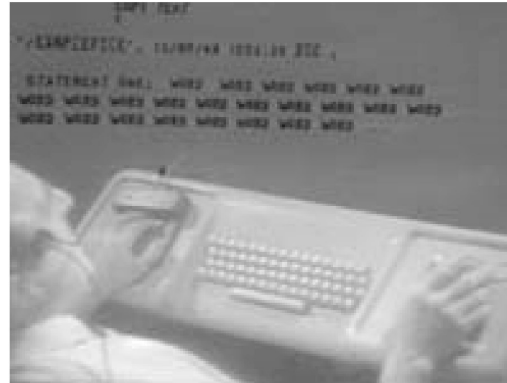


Output: response

❑ Motor

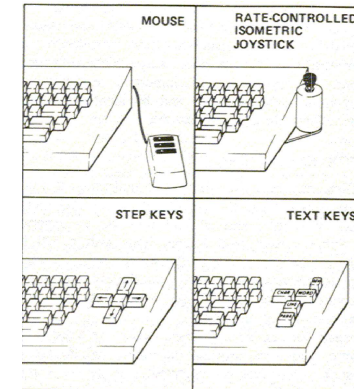
- Speed-accuracy trade-off
 - Fitts' law
 - Shannon's theory
- Performance evaluation
 - Index of Difficulty
 - Index of Performance

Problem solving

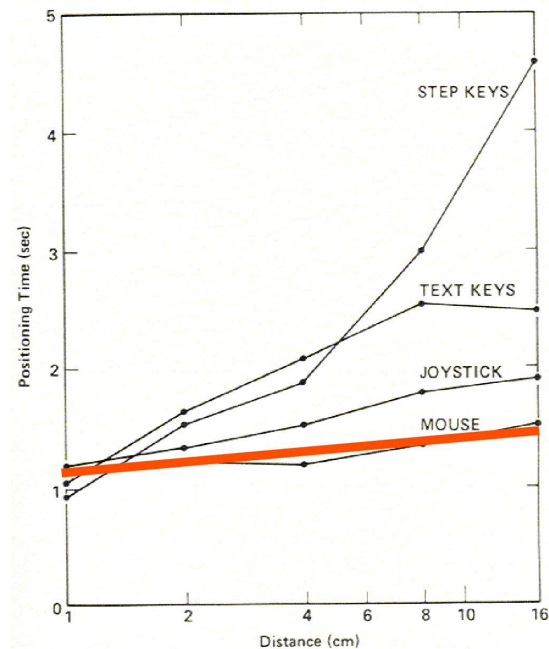


Engelbart: Which is fastest?

Multiple solutions



User testing (performance)



User testing (evaluation)

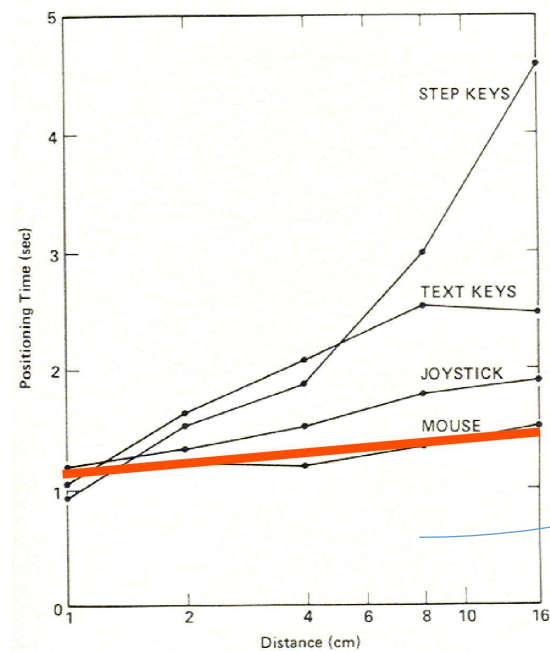




The 'Dislike' Button is finally coming

It looks like it's finally here, the Dislike button for Facebook. CEO Mark Zuckerberg has for many years said they wouldn't add a "dislike" button, but in a Question and Answer session Tuesday, Zuckerberg let the world know they will soon be adding one.

"I think people have asked about the dislike button for many years. Today is a special day because today is the day I can say we're working on it and shipping it," said Zuckerberg.

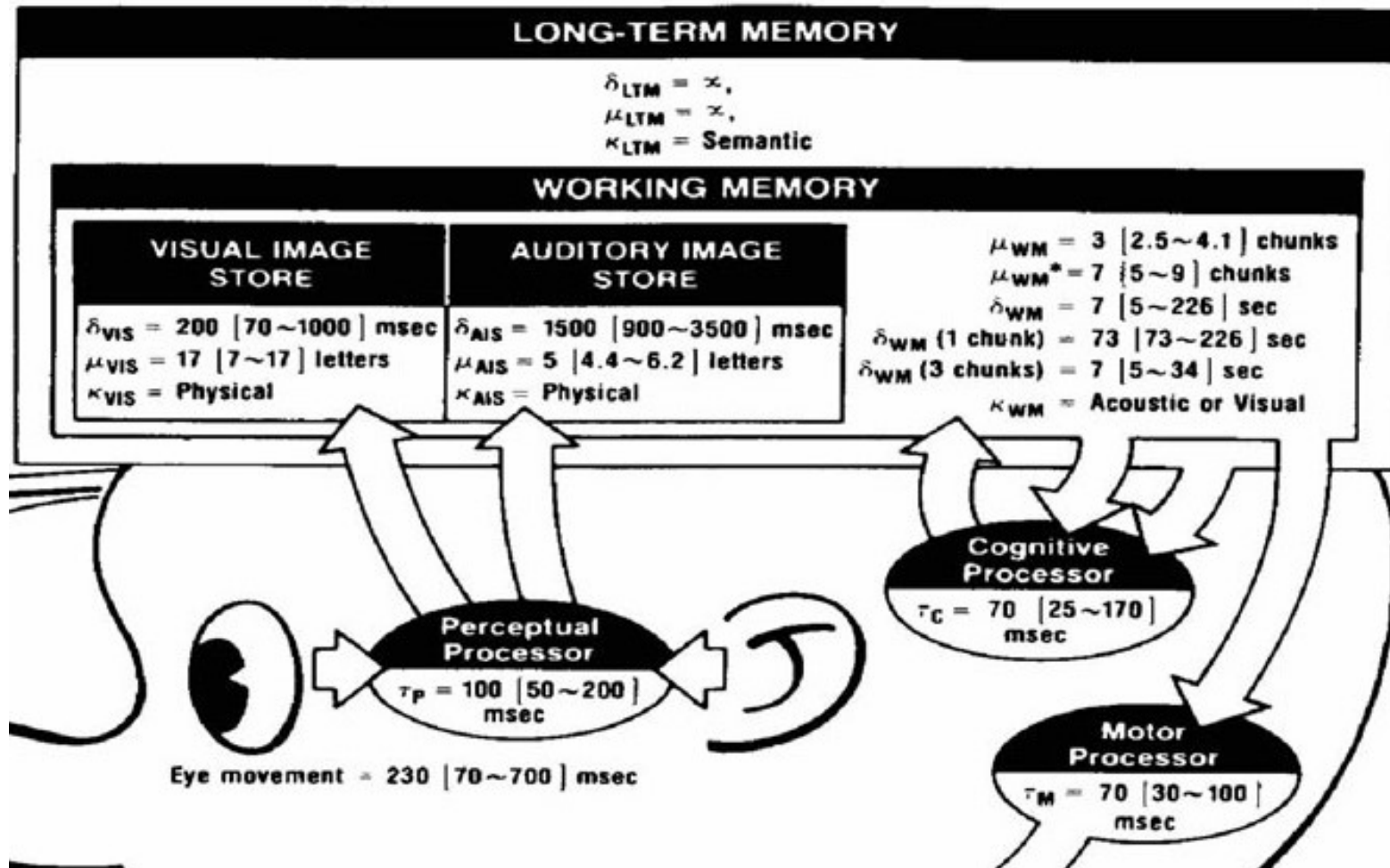


Attitude

Time scale of human behavior

10^7 (months) 10^6 (weeks) 10^5 (days)	Social	Social Behavior
10^4 (hours) 10^3 10^2 (minutes)	Rational	Adaptive Behavior
10^1 10^0 (seconds) 10^{-1}	Cognitive	Immediate Behavior
10^{-2} 10^{-3} (milliseconds) 10^{-4}	Biological	

Performance



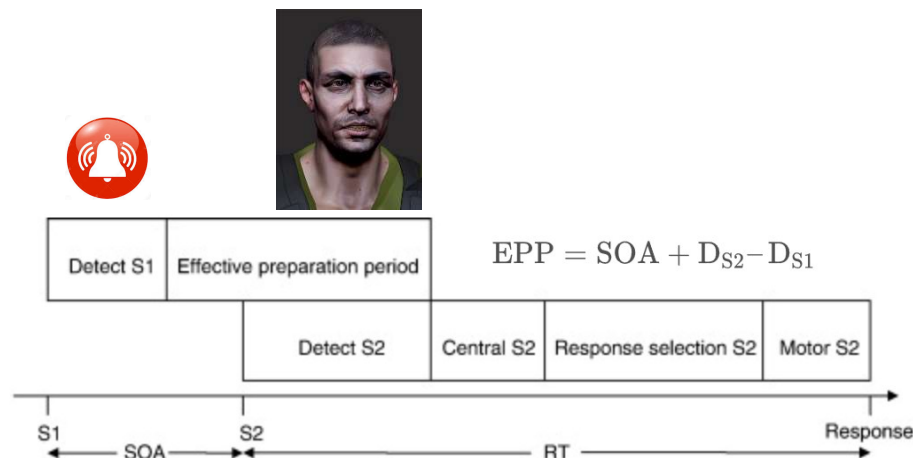
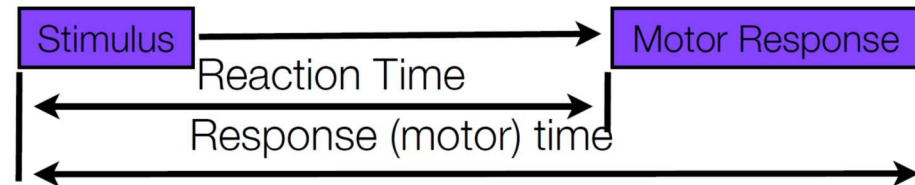
Output: response time

❑ Motor

Time taken to respond to stimulus (speed): reaction time + movement time

- Movement time is dependent on physical characteristics of subject: e.g. age, fitness level
- Reaction time is dependent on stimulus type:

- Visual: ~200 ms
- Auditory: ~150 ms
- Pain: ~700 ms



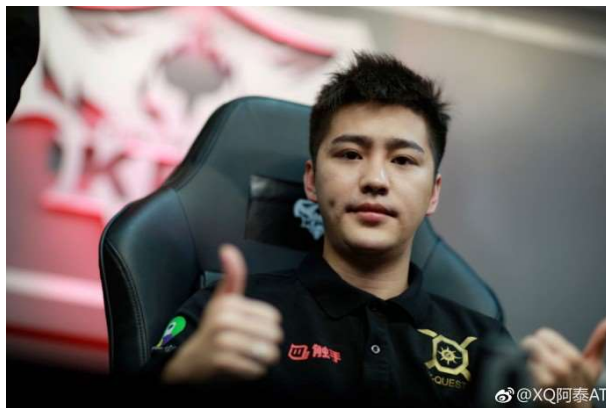
Trigger-happy game research.
See Lecture 01:
Use and Context

Output: accuracy of response

❑ Motor

- Another measure of motor skill is accuracy
- There is a trade-off between speed and accuracy:
 - Faster is sloppier (inaccurate)
 - Slower is precise (accurate)
- However, whether speed of reaction results in reduced accuracy depends on the task and the user

That is why you do training: fast + accurate



AT, professional gamer, *Honor of Kings*



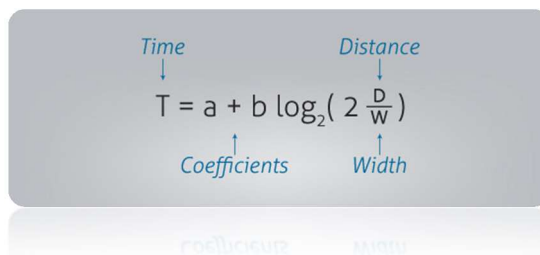
Fafa Yim, Logitech pro-team captain, *League of Legends*
Faster than AT with more target hits

Output: speed vs accuracy

❑ Motor

Fitts' Law: speed vs accuracy

- Established in 1954 by Prof. Paul Morris Fitts
- Lieutenant-colonel of the US Air Force
- Head of the US Air Force Human Engineering Division
- Later professor at Ohio State U and U of Michigan
- A model of human movement
- Predicts the time required to rapidly move from a starting position to a final target area
- Models the act of pointing
- In the real world (e.g., with a hand or finger, or even eye tracking)
- On computers (e.g., with a mouse or a joystick)



The diagram shows the Fitts' Law equation $T = a + b \log_2 \left(2 \frac{D}{W} \right)$ inside a grey rounded rectangle. Labels with arrows point to parts of the equation: 'Time' points to 'T', 'Coefficients' points to 'a' and 'b', 'Distance' points to 'D', and 'Width' points to 'W'.

The wider an interface widget and the closer it is to the pointer, the easier to hit it

(Lecture 01: Use and Context)

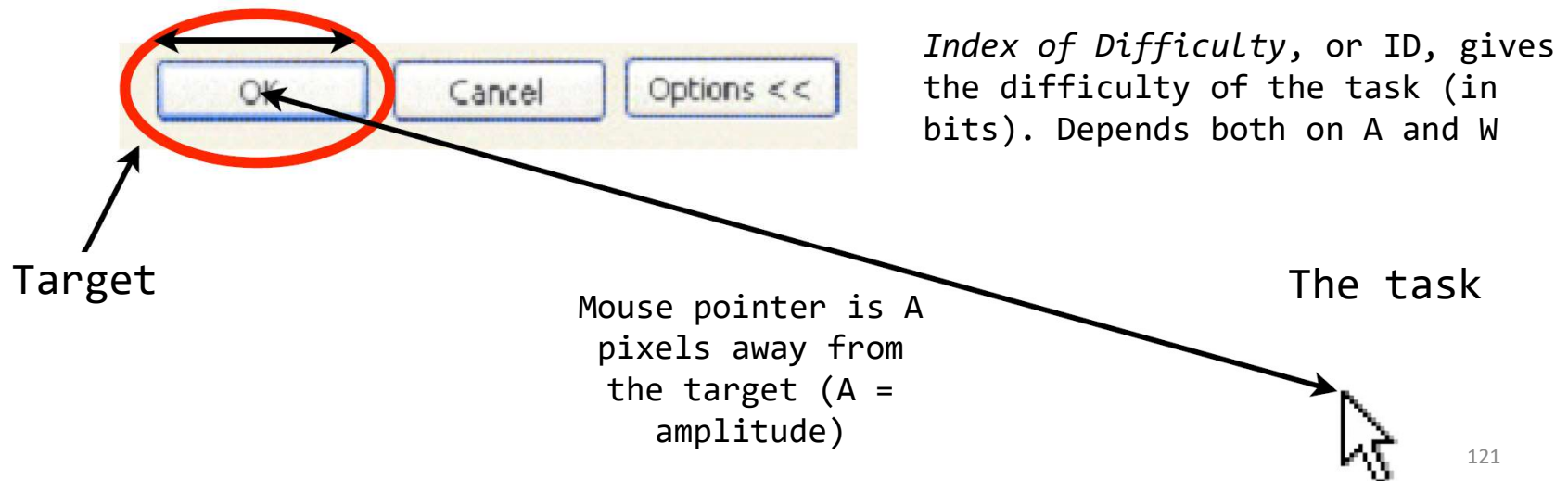
Output: speed vs accuracy

□ Motor

Fitts' Law: speed vs accuracy

- Model of human movement
- Based on Shannon's theory of information systems
- Human motor system modeled as channel used to transmit information
- Task has a certain difficulty (measured by number of bits)
- Channel has a certain bandwidth (measured by seconds/bit)

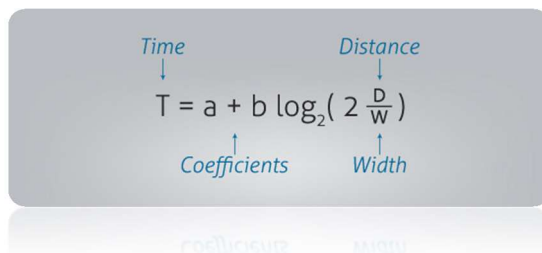
Target is W pixels wide (W = tolerance)



Output: speed vs accuracy

Index of Difficulty

- Movement to target requires more time when
- Distance to target (A) increases
- Error Tolerance of target (W) decreases
- That is, when target is farther away or of smaller size
- Movement to target requires less time when
- Distance to target (A) decreases
- Error Tolerance of target (W) increases
- That is, with a closer target or one of larger size
- Originally: $I_D = \log_2(2A/W)$
- Shannon-style reformulation:
- $I_D = \log_2(A/W + 1)$, or $I_D = \log_2(A/W + 0.5)$ (when $I_D < 3$ bits)



The diagram shows the equation $T = a + b \log_2(2 \frac{D}{W})$ inside a light blue rounded rectangle. Arrows point from labels to parts of the equation: 'Time' points to 'T', 'Coefficients' points to 'a' and 'b', 'Distance' points to 'D', and 'Width' points to 'W'.

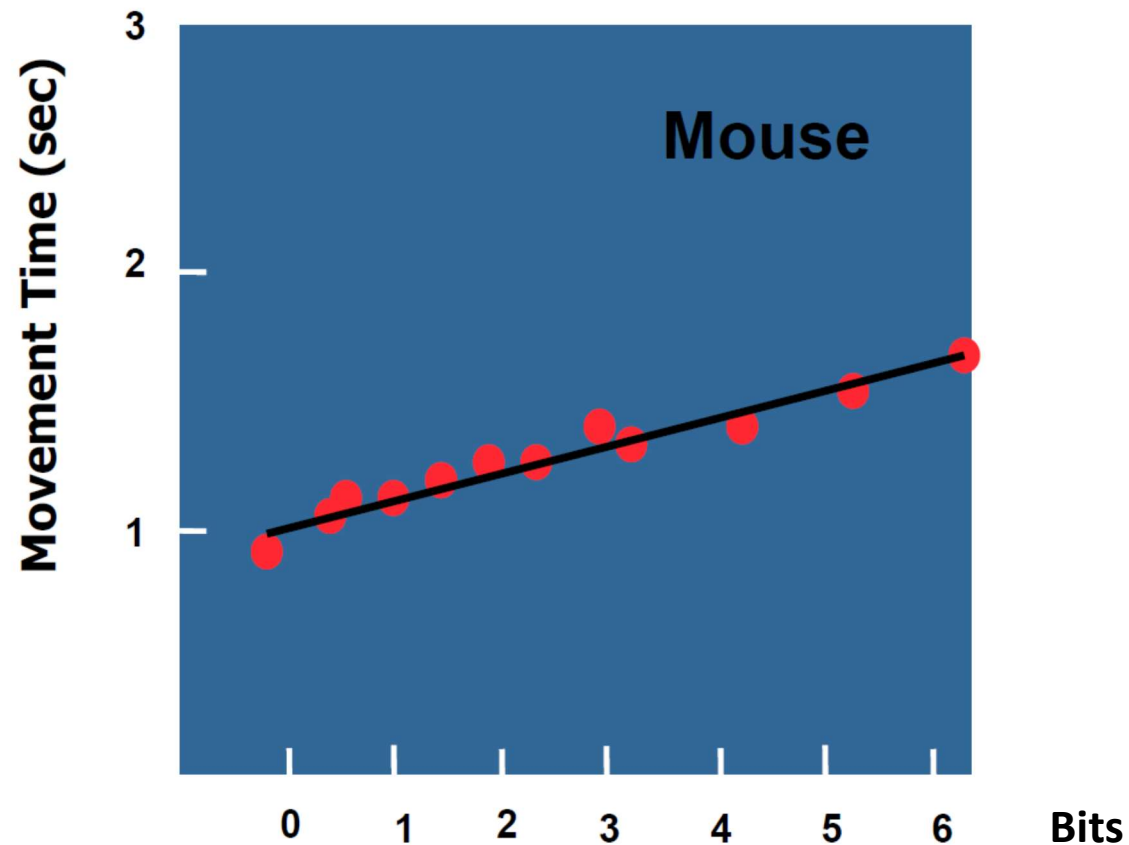
$$T = a + b \log_2(2 \frac{D}{W})$$

The wider an interface widget and the closer it is to the pointer, the easier to hit it

(Lecture 01: Use and Context)

Output: speed vs accuracy

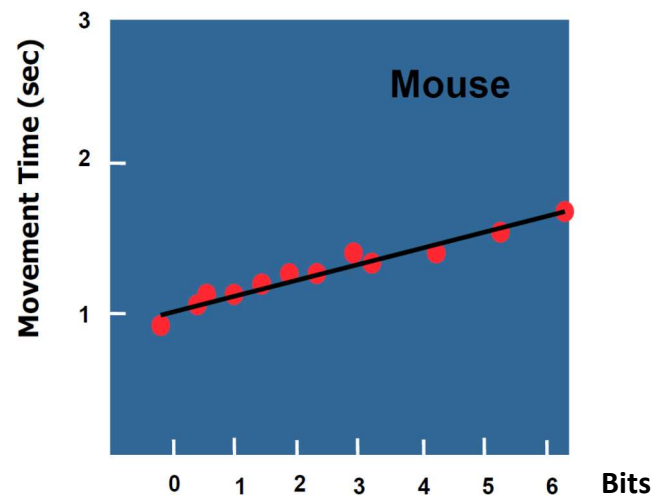
The channel is characteristic of the modality of the interaction



Output: speed vs accuracy

Putting everything together:

- Time T_{pos} needed to complete the task is linear with respect to ID
- $T_{pos} = a + b \cdot ID$
- a and b are empirically determined through experimental data
- a = y-intercept
- b = slope (secs/bit)
- T_{pos} = time taken (seconds)



Output: speed vs accuracy

Evaluating a device

Index of Difficulty depends on the task, not the device

The regression constants a and b determine how fast the task is done (or how fast the device processes the task)

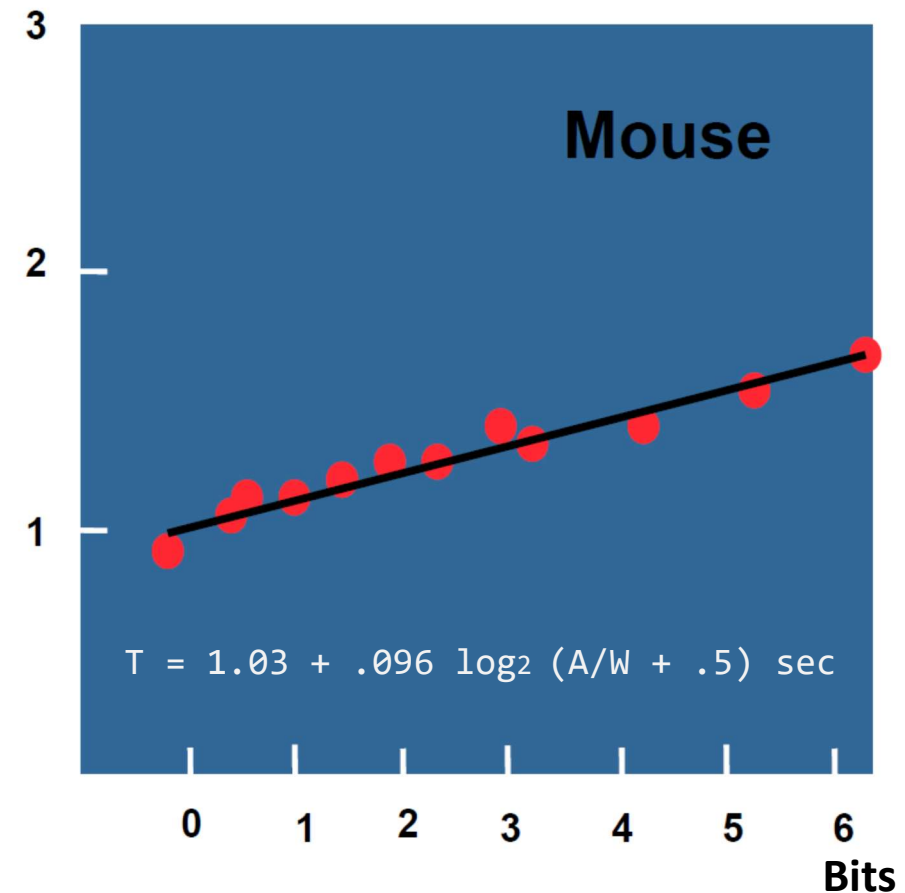
The *Index of Performance* IP (or *throughput* TP) defines how quickly the task can be finished

$$IP = 1/b$$

Aside: the mouse is very close to being an optimal device
(hard to do better than the mouse)

$$I_D = \log_2(\text{Dist}/\text{Size} + .5)$$

Time



Virtual Tutor Summary



Human Information Processing is different from computation because perception, memory, problem solving, and (motor) responses are less 'stable' but also more adaptive

All the visual, textual, sound, and touch input become an integrated whole and recede in memory, sometimes in a single cell (e.g., faces and places)

Memory is not fixed but changes over time (neurons change their connections). Memory is context-sensitive

Problem-solving skills may differ. For deterministic problems, you need reasoning (e.g., means-end analysis). For underdetermined problems, you need creative solutions

Virtual Tutor Summary



When you design a device, you need to reckon with speed-accuracy trade-offs. That is, trained users can be highly efficient, combining high speed with great accuracy (cf. pro-team gamers)

It also means that untrained users (novices, first-time users, older adults) may not be as advanced as you

User testing of speed and accuracy tells you how your system or device does in terms of user performance. Don't forget to ask for appreciation as well (like-dislike)

Fitts' law as derived from Shannon's Theory of Information Systems is your means to determine user performance of motor output

Exam Sample Item

Category “Knowledge”

When can you consider a problem to be ‘well-defined’?

A When goal decomposition allows to break down a complex problem into smaller ones by determining sub goals

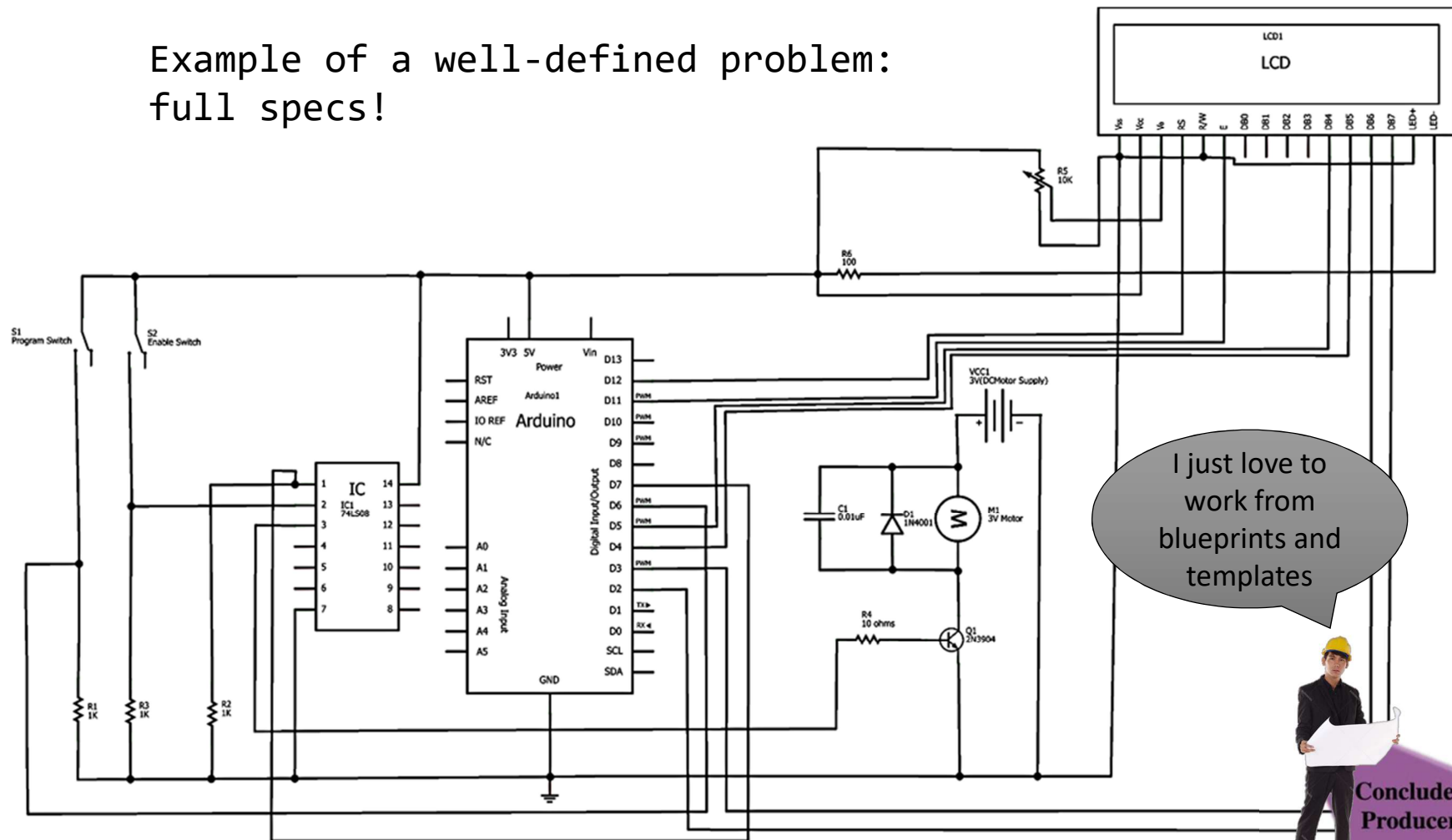
B When structured decisions are made in a problem state that has multiple representations



C With full specification of the provided data, actions, strategies, and goal states, which can be represented in a problem space

D When at the structural level, the focus lies on changes in structure and processes such as work units and actions

Example of a well-defined problem:
full specs!



I just love to
work from
blueprints and
templates



Concluder
Producer

In *Arduino Electronics Blueprints*, Don Wilcher explains how a programmable logic controller (PLC) is used to operate various electronic and electromechanical devices that are wired to I/O wiring modules (<https://hub.packtpub.com/programmable-dc-motor-controller-lcd/>)

References and recommended reading

- Anderson, J. R. (1995). *Cognitive Psychology and its Implications*. New York, NY, USA: W. H. Freeman.
- Entwistle, N. J. (2013). *Styles of Learning and teaching: An integrated outline of educational psychology for students, teachers and lecturers*. New York: Routledge.
- Felder, R. M. (1988). Learning and teaching styles in engineering education. *Engineering Education*, 78(7), 674-681.
- Hoorn, J. F. (Ed.) (2013). *Organic creativity and the physics within*. Philadelphia, PA: John Benjamins.
Available from <http://www.jbe-platform.com/content/books/9789027212030>
- Hoorn, J. F. (2014). *Creative confluence*. Philadelphia, PA: John Benjamins. Chapter 3.
- Kahneman, D., Knetsch, J. L., & Thaler, R. (1986). Fairness as a constraint on profit seeking: Entitlements in the market. *American Economic Review*, 76, 728-741.
- Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. *Econometrica*, 47, 263-291.
- Newell, A., & Simon, H. A. (1972). *Human Problem solving*. Englewood Cliffs, N.J., USA: Prentice-Hall.
- Schneider, S. L., & Lopes, L. L. (1986). Reflection in preferences under risk: Who and when may suggest why. *Journal of Experimental Psychology: Human Perception and Performance*, 12, 535-548.
- Schneider, S. L. (1992). Framing and conflict: Aspiration level contingency, the status quo, and current theories of risky choice. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 18, 1040-1057.
- Simon, H. A. (1960). *The New Science of Management Decision*. New York, NY, USA: Harper and Row.
- Tversky, A., & Kahneman, D. (1981). The framing of decisions and the psychology of choice. *Science*, 211, 453-458.